

Strategic Multinational Location Choice with Endogenous Pricing*

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Abstract

Multinational firms choose portfolios of locations that jointly serve foreign markets, while rivals' locations and endogenous prices shape each location's return. We study the resulting finite-player portfolio game. Downstream pricing and network effects can violate the ordering conditions exploited by order-based methods in the quantitative trade literature. We develop an exact best-response algorithm that discards portfolios using payoff bounds for each portfolio size, together with conditions for equilibrium uniqueness and convergence. In a calibration to five cloud providers' observed network across 42 countries, the equilibrium with endogenous Atkeson–Burstein markups has 124 active locations, 19 more than under constant CES markups, and aggregate provider profit net of fixed costs is 35.6 percent lower under CES. Holding rivals' repricing fixed changes the equilibrium AB network. Raising the fixed cost of operating EU locations reduces deployment in both networks, with fewer exits under AB pricing at a moderate increase and more at a large one. Country-by-country entry rules can select different networks and market shares even when profits are close to the exact optimum.

Keywords: strategic multinational location choice, portfolio equilibrium, combinatorial discrete choice, exact portfolio best response, variable markups

JEL Codes: F12, F23, L86, C72, C62, L13

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1. INTRODUCTION

Firms often choose networks of activities rather than isolated actions. A multinational selects production and sourcing locations, an exporter chooses a set of destination markets, and an infrastructure provider builds a network of facilities. The return to one link can depend on the firm’s other links and on the portfolios chosen by its rivals. The economic action is then a complete portfolio.

Quantitative trade models treat this choice in two useful ways. In canonical export models, a firm enters a destination when its profit there exceeds the fixed cost of serving it (Chaney, 2008; Dickstein and Morales, 2018). This market-by-market rule is exact under the model’s separable payoff structure. Other models make the complete set of links an economic object: prior export destinations affect subsequent entry costs and export decisions are interdependent across markets (Morales et al., 2019; Alfaro-Ureña et al., 2025), sourcing locations jointly determine input costs (Antràs et al., 2017), and multinational locations jointly determine market access (Tintelnot, 2017; Castro-Vincenzi, 2024; Arkolakis et al., 2023; Antràs et al., 2026). These models show why portfolio dependence matters and develop exact methods for the single-firm choice problem. Recent work also studies where a multiproduct firm produces and sells each model when choices can be complements or substitutes (Castro-Vincenzi et al., 2025).

Our question is what changes when complete portfolios are strategic. If AWS opens a region in Korea, Microsoft and Google may change their networks. Their responses alter prices and market shares and can feed back into AWS’s original entry incentive. We study the resulting finite-player portfolio equilibrium, its uniqueness, and its computation. We then use the exact response as a benchmark for asking when a simpler country-by-country entry rule remains valid.

We formulate a Bayesian game in which each firm observes private link costs and chooses a complete feasible portfolio. Downstream demand and pricing determine the operating payoff for every joint portfolio. For each realized cost vector, an exact response selects the profit-maximizing complete portfolio.

Existing methods reduce this search using additional order structure, and each gives up one of the two margins our setting requires. Jia (2008) solves a strategic chain-entry game, so oligopoly is within reach, but the lattice argument requires supermodular struc-

ture. Arkolakis et al. (2023) dispense with supermodularity through a weaker ordering, but that ordering need not survive downstream price adjustment, and their framework supplies no game-level uniqueness or convergence. Antràs et al. (2017) treat a single firm, and Castro-Vincenzi et al. (2025) allows both complements and substitutes but requires their pairwise signs to be known in a single-agent, monopolistically competitive problem. Our entry game needs both margins at once: strategic rivals with downstream oligopoly pricing, and no usable order structure. Indeed, the Jia and AES orderings fail in our calibrated response problems under both variable and constant markups. We therefore use branch and bound to compute each firm’s exact best response without one-link local search or monotone elimination.

The exact response also yields a useful link-level threshold. Fix one candidate link and hold the firm’s other private link costs fixed. Compare the best complete portfolio containing the link with the best complete portfolio excluding it. The difference between their optimized values determines the highest private cost at which the firm includes the link. We call this threshold the link’s inclusion cutoff. Because the two optimizing portfolios can differ at several other links, the cutoff generally depends on the firm’s other costs and on rivals’ portfolio distributions. It represents the exact portfolio response rather than an independent link decision.

Changes in rivals’ portfolio distributions shift these contingent inclusion boundaries. We collect these shifts in a nonnegative interaction matrix K . The entry K_{fg} bounds how a change in firm g ’s portfolio distribution moves firm f ’s inclusion cutoffs. The construction converts payoff-gap shifts into private-cost units, so K and the private-cost dispersion σ_ε are measured on the same scale. The spectral radius $\rho(K)$ summarizes how direct and indirect responses amplify around the game. Greater private-cost dispersion smooths portfolio choice and weakens the response to a shifted cutoff. When $\sigma_\varepsilon > \rho(K)$, the distributional best-response map is a contraction: the exact portfolio Bayesian Nash equilibrium (BNE) is unique and iteration converges from any initial distribution.¹

A separate result covers the other end of the noise scale. Suppose the mixed extension of the complete-information game has a unique equilibrium and that equilibrium is a strict

¹Related contraction arguments compare private heterogeneity with the strength of strategic interaction in Bayesian games (Mason and Valentinyi, 2007; Melo, 2022). Our setting differs because each firm chooses a complete portfolio under a low-dimensional vector of link-cost shocks, and the interaction matrix is built from link-level cutoff shifts rather than scaling with the number of feasible portfolios.

pure profile. Under a boundary condition on the cost distribution, the BNE is then unique for sufficiently small private-cost noise and converges to that profile. The cloud application uses the zero-noise case: we solve its complete-information equilibrium directly and verify every provider’s equilibrium portfolio against unrestricted deviations.²

Exact responses still require a combinatorial search. Our branch-and-bound algorithm uses portfolio-size-specific profit bounds to eliminate groups of portfolios that cannot improve on the current best choice. When the search exhausts the remaining branches, no omitted portfolio can be more profitable. The search problem is NP-hard even under constant markups, so no generally fast exact rule should be expected; in practice the number of expanded branches is governed by how many portfolios come close to the optimum, and the calibrated searches stay small (Appendix C).

We use the exact response to assess country-by-country entry rules. When firms may choose any subset of candidate countries, such a rule reproduces the exact portfolio for every deterministic cost vector if and only if operating value is additive across links. In the additive case the inclusion cutoff collapses to the market-entry threshold of canonical export models (Chaney, 2008; Dickstein and Morales, 2018); the contingent cutoff is its generalization to interdependent portfolios. When additivity fails, we fit an additive anchor to the exact payoff table, and the anchor’s worst-case error bounds the profit loss from the simpler rule. The country-by-country rule built from the anchor still selects the exact portfolio when this error is small relative to the payoff gap between the best and second-best portfolios.

We specialize the framework to a Fréchet–constant-elasticity-of-substitution (CES) environment with Atkeson–Burstein (AB) markups that vary with market shares (Atkeson and Burstein, 2008; Edmond et al., 2015). A location changes service costs, accessibility, equilibrium shares, prices, and the value of other locations. Full repricing can therefore make an additional location profitable at one network and unprofitable after the network expands. We observe reversals in both directions in the calibrated AB and CES response problems, so neither the lattice ordering used by Jia nor either monotone ordering used by Arkolakis et al. (2023) applies. The failure persists under CES because service-cost and market-access gains still depend nonlinearly on the entire active network.

²The calibration targets a single cross-section, which does not identify σ_ε ; the application therefore solves and verifies the complete-information game directly, and the finite-noise theory serves the model’s extension.

The application studies five cloud providers choosing regions in 42 countries. A region supplies digital services across borders and changes latency, compliance, and serviceability in several markets. Launch histories are consistent with providers adjusting locations in multi-country waves: 79 percent of the country entries we date over 2007–2024 occur in years in which the provider enters more than one country (Figure 1). We calibrate provider- and country-level cost shifters to observed deployment counts after fixing the demand elasticities and gravity-score coefficients and selecting the accessibility curvature from a reported grid.

Our main counterfactual asks how market-share-dependent markups shape the equilibrium international network. Holding demand, geography, accessibility, service costs, and fixed costs fixed, we replace AB pricing with the constant CES markup and recompute the strategic portfolio equilibrium. The equilibrium with AB markups contains 124 active provider-country locations, 19 more than under CES; the CES network is a strict subset of the AB network. Changing the pricing rule leaves aggregate provider profit net of fixed costs 35.6 percent below the AB outcome after providers reoptimize their networks.

A second counterfactual isolates rivals’ strategic repricing. Providers still optimize over complete portfolios, but they evaluate each deviation without allowing rivals to adjust their prices. This restriction leaves the CES network unchanged but selects four additional locations under AB pricing.

We also compare exact responses with rules that evaluate countries separately on the same payoffs. The simpler rules select different international networks and move provider shares within markets. When providers can reconsider their full candidate sets, the simpler rules also admit profitable deviations involving several countries at once. Similar aggregate link counts and profits close to the exact optimum therefore need not imply the same network.

A set of policy experiments shows why these modeling margins matter beyond the baseline. The first experiment captures the compliance costs that European data-protection and digital-sovereignty rules impose on operating local data centers (Ferracane et al., 2020; Demirer et al., 2024; Manganelli and Schnurr, 2024): a moderate increase in the fixed cost of operating each EU location removes 10 locations under AB pricing but 13 under CES, and a large increase reverses the ordering, with 26 exits under AB against 20 under CES. Provider profit net of fixed costs rises in both cases as competition softens. A second ex-

periment separates costs from access in China, motivated by the Great Firewall and app restrictions (Sun, 2024; Sun and Trefler, 2023): lower China-related costs alone leave the network unchanged, while deployment changes when feasible sets change, with broad Alibaba entry when its candidate set expands. Both experiments are computed on the exact benchmark equilibria.

The paper connects three literatures. The first develops tractable portfolio methods under order structure (Jia, 2008; Antràs et al., 2017; Arkolakis et al., 2023; Castro-Vincenzi et al., 2025), including fulfillment and multi-plant networks chosen by a single firm (Houde et al., 2023; Oberfield et al., 2024); we compute exact best responses with no order restrictions in a finite-player game with equilibrium repricing. Relative to strategic product and location models (Holmes, 2011; Wollmann, 2018; Fan and Yang, 2020), including joint assortment-and-pricing competition (Draganska et al., 2009) and revealed-preference chain networks (Ellickson et al., 2013), we provide an exact best-response method that does not rely on one-link local search. Relative to market-by-market trade models, we characterize when the separable representation remains exact and how errors matter when it does not. Cardinality relaxations and exact assortment methods have established precedents in operations research (Feldman and Topaloglu, 2015; Leitner et al., 2024; Zhang et al., 2025); our computational contribution adapts this logic to a firm’s portfolio serving many markets, with multiple aggregates, rival portfolio states, and a downstream pricing equilibrium.

The rest of the paper proceeds as follows. Section 2 defines exact portfolio equilibrium, gives its uniqueness conditions, and asks when country-by-country choice recovers the exact portfolio response. Section 3 specializes the pricing and interaction bounds to AB demand. Section 4 presents the cloud calibration, counterfactuals, and response comparisons. Section 5 concludes.

2. EXACT PORTFOLIO EQUILIBRIUM

A firm chooses a complete portfolio after observing its vector of private link costs. That portfolio determines the downstream pricing game, whose solution determines operating profit. We call the response exact because every realized cost vector leads the firm to optimize globally over its feasible portfolios. Across cost realizations, these choices generate

a distribution over complete portfolios that preserves dependence among the firm's links.

This section first defines the portfolio game and its equilibrium. It then shows how a global response can be represented by link-level cutoffs after conditioning on the firm's other costs. Those cutoffs provide the sensitivity measure used in the large-noise uniqueness result; a separate argument covers the small-noise limit. The section closes with the computation of global portfolio responses and the conditions under which country-by-country choice recovers them.

2.1 Portfolio game and equilibrium

There are F firms. Firm f has a finite set of candidate links $\mathcal{A}_f$ and a nonempty feasible family $\mathcal{X}_f \subseteq 2^{\mathcal{A}_f}$. For every pure portfolio profile $S = (S_1, \dots, S_F)$, first solve the downstream game and record the resulting operating profits. If that game has multiple equilibria, a fixed rule $p^\dagger(S)$ selects one continuation outcome. The selected payoff, net of deterministic link costs, is

$$r_f(S) = \Pi_f(S; p^\dagger(S)) - \sum_{j \in S_f} \phi_{fj} w_{fj}. \quad (2.1)$$

Here $w_{fj} > 0$ is a link-specific cost loading and ϕ_{fj} is the deterministic component of link j 's entry cost, so $\phi_{fj} w_{fj}$ is the link's deterministic cost in payoff units; the private cost component enters through the same loading in (2.2) below. The collection of these portfolio-contingent profits is the reduced payoff table used in the entry game. A unique pricing equilibrium is sufficient to construct it, but the exact equilibrium result below also permits a fixed selection from multiple pricing equilibria.

Firm f observes the vector of private link costs $\varepsilon_f = (\varepsilon_{fj})_{j \in \mathcal{A}_f}$. Its payoff is

$$u_f(S; \varepsilon_f) = r_f(S) - \sigma_\varepsilon \sum_{j \in S_f} w_{fj} \varepsilon_{fj}, \quad w_{fj} > 0. \quad (2.2)$$

Throughout, $\sigma_\varepsilon > 0$. The positive loading w_{fj} converts the standardized private-cost shock into payoff units and allows its magnitude to differ across links.

Assumption 1 (Private link costs). Private types are independent across firms. Within a firm, the coordinates may be dependent. For every j , the conditional density of ε_{fj} given

$\varepsilon_{f,-j}$ is bounded by $\bar{g}_{fj} < \infty$ almost surely. Independent normal or logistic link shocks satisfy this condition. The small-noise result additionally imposes the local boundary condition in Appendix A.3.

Let $Q_f \in \Delta(\mathcal{X}_f)$ be firm f 's portfolio distribution and write $Q_{-f} = \bigotimes_{g \neq f} Q_g$. Its interim portfolio value is

$$V_f(S_f; Q_{-f}) = \sum_{S_{-f}} r_f(S_f, S_{-f}) Q_{-f}(S_{-f}). \quad (2.3)$$

Each firm observes its own cost vector but not those of its rivals. A realized cost vector maps into one complete portfolio, while the distribution of cost vectors induces Q_f . Thus Q_f does not impose independent link choices: it tracks how globally optimal portfolios vary across the firm's private-cost realizations. Independence across firms gives the product distribution Q_{-f} ; costs and link choices within a firm may remain dependent. A firm evaluates a candidate portfolio by averaging its reduced payoff over this distribution of rival networks.

The type-specific global best response and the distribution it induces are

$$b_f(\varepsilon_f; Q_{-f}) \in \arg \max_{S_f \in \mathcal{X}_f} \left\{ V_f(S_f; Q_{-f}) - \sigma_\varepsilon \sum_{j \in S_f} w_{fj} \varepsilon_{fj} \right\}, \quad (2.4)$$

$$\mathcal{B}_f(Q_{-f})(S_f) = \Pr\{b_f(\varepsilon_f; Q_{-f}) = S_f\}. \quad (2.5)$$

Definition 1 (Exact portfolio BNE). An exact portfolio Bayesian Nash equilibrium is a profile $Q^* = (Q_1^*, \dots, Q_F^*)$ satisfying

$$Q_f^* = \mathcal{B}_f(Q_{-f}^*), \quad f = 1, \dots, F.$$

Every realized type therefore chooses a globally optimal portfolio in $\mathcal{X}_f$.

Proposition 1 (Existence). *Suppose the continuation selection in (2.1) is fixed independently of entry-cost types and beliefs, all selected payoffs are finite, types are independent across firms, and each within-firm type vector has an absolutely continuous joint distribution. Then an exact portfolio BNE exists.*

Appendix A.1 gives the proof. With finite action sets and absolute continuity, the

global best response is unique for almost every type. The resulting distributional best-response map is continuous on $\prod_f \Delta(\mathcal{X}_f)$. Brouwer's theorem then applies.

2.2 Contingent link thresholds

Although the firm solves a global combinatorial problem, each link has a threshold representation after conditioning on the firm's other link shocks. Fix those other costs and solve two complete-portfolio problems: the best feasible portfolio containing link j , and the best feasible portfolio excluding it. The value of the first option falls with j 's own cost, while the second does not. The cost at which the two optimized values cross is the link's inclusion cutoff.

Fix rival beliefs Q_{-f} and write $v_f(S_f) = V_f(S_f; Q_{-f})$ for the resulting deterministic interim portfolio value. For a link j , define the best values among portfolios that include and exclude j :

$$V_{fj}^{\text{in}} = \max_{\substack{S_f \in \mathcal{X}_f \\ j \in S_f}} \left\{ v_f(S_f) - \sigma_\varepsilon \sum_{k \in S_f \setminus \{j\}} w_{fk} \varepsilon_{fk} \right\},$$

$$V_{fj}^{\text{out}} = \max_{\substack{S_f \in \mathcal{X}_f \\ j \notin S_f}} \left\{ v_f(S_f) - \sigma_\varepsilon \sum_{k \in S_f} w_{fk} \varepsilon_{fk} \right\}.$$

The comparison portfolios may contain different other links, so the cutoff is contingent rather than a separate optimization for link j .³

Lemma 1 (Contingent inclusion cutoff). *Except on a probability-zero tie set, the globally optimal portfolio contains link j if and only if*

$$\varepsilon_{fj} < \bar{\varepsilon}_{fj}(\varepsilon_{f,-j}; v_f) \equiv \frac{V_{fj}^{\text{in}} - V_{fj}^{\text{out}}}{\sigma_\varepsilon w_{fj}}. \quad (2.6)$$

Example 1: A two-link portfolio. Suppose all four portfolios $\emptyset$, $\{a\}$, $\{b\}$, and $\{a, b\}$ are feasible, and write their deterministic values as $v_\emptyset$, v_a , v_b , and v_{ab} . Conditional on ε_b , the

³Links that are required in every feasible portfolio or excluded from every feasible portfolio are omitted because they are not choice margins.

cutoff for link a is

$$\bar{\varepsilon}_a(\varepsilon_b) = \frac{\overbrace{\max\{v_a, v_{ab} - \sigma_\varepsilon w_b \varepsilon_b\}}^{\text{best portfolio with } a} - \overbrace{\max\{v_\emptyset, v_b - \sigma_\varepsilon w_b \varepsilon_b\}}^{\text{best portfolio without } a}}{\sigma_\varepsilon w_a}. \quad (2.7)$$

The best portfolio containing a can switch between $\{a\}$ and $\{a, b\}$, while the best portfolio excluding a can switch between $\emptyset$ and $\{b\}$. The cutoff therefore depends on the other link's realized cost even though the firm still chooses one globally optimal portfolio. If $v_{ab} - v_a = v_b - v_\emptyset$, the two maximization terms move together and the cutoff reduces to the separable value $(v_a - v_\emptyset)/(\sigma_\varepsilon w_a)$. Outside that special case, the four joint portfolio probabilities contain information that the two marginal entry rates do not.

For example, interpret a as entry into Korea and b as entry into Japan. If $v_{ab} - v_b > v_a - v_\emptyset$, operating in Japan raises the value of adding Korea, so the two locations are complements. The reverse inequality makes them substitutes. Only equality makes Korea's cutoff invariant to the realized cost of the Japanese link.

Own costs and portfolio order. Two monotonicity concepts are distinct. Lemma 1 uses only own-cost monotonicity: raising the cost of the Korean link cannot make a portfolio containing Korea more attractive relative to one excluding it. Order-based portfolio algorithms require a stronger property governing how Korea's marginal value changes as Japan or another link is added. Jia (2008) uses supermodularity to bracket a global response, while Arkolakis et al. (2023) require a weaker one-way ordering of marginal-value signs across nested portfolios. Our equilibrium argument uses only own-cost monotonicity, and the general portfolio algorithm does not impose the stronger portfolio ordering. Computing an inclusion cutoff still requires the two global optimizations that define V_{fj}^{in} and V_{fj}^{out} . The cutoff is useful for measuring equilibrium sensitivity, but it is not by itself a shortcut for the combinatorial response.

Jia's equilibrium iteration further uses strategic substitutability across chains. If one chain's portfolio is ordered from small to large and its rival's from large to small, both best responses are increasing in that product order, which permits monotone iteration. The exact BNE argument below does not require an ordering of portfolios that makes every cross-firm response monotone.

Suppose deterministic portfolio values change because rivals' behavior changes, so each link has an old and a new cutoff. Compare the two environments using the same realization of private costs. A link changes status only when its cost shock lies between the old and new cutoffs. Conditional on $\varepsilon_{f,-j}$, this event has probability at most $\bar{g}_{fj}$ times the cutoff movement. Integrating over the other shocks and taking a union across links bounds how much the induced distribution over complete portfolios can change, without imposing independence among link choices. This comparison drives the uniqueness argument below.

2.3 Large-noise uniqueness

Consider moving rival g from one portfolio to another while holding the remaining rivals fixed. For each candidate link of firm f , ask how much this change can move an include-versus-exclude payoff gap. Only firms whose private costs lie near the displaced boundary can change their portfolios. The density weight converts boundary movement into a bound on affected probability mass, and summing across links gives the bilateral interaction K_{fg} .

Formally, common shifts in all portfolio values leave choice unchanged, while shifts in include-versus-exclude gaps matter. The factor $1/\sigma_\varepsilon$ enters separately in Theorem 1.

To see where that factor enters, hold $\varepsilon_{f,-j}$ fixed and let h denote a change in firm f 's deterministic portfolio-value table. Applying the fact that a maximum can move by no more than its most affected alternative, separately to the best include and best exclude portfolios, makes common shifts of h cancel, and the cutoff movement obeys

$$\left| \bar{\varepsilon}_{fj}(v_f + h) - \bar{\varepsilon}_{fj}(v_f) \right| \leq \frac{1}{\sigma_\varepsilon w_{fj}} \max_{\substack{S_f \ni j \\ T_f \not\ni j}} |h(S_f) - h(T_f)|. \quad (2.8)$$

Multiplying by the conditional-density bound $\bar{g}_{fj}$ converts this cutoff movement into the probability that link j changes status. Summing across links gives a weighted measure of payoff-boundary movement; it is a seminorm because a common shift to every portfolio value receives value zero.

For a function $h : \mathcal{X}_f \rightarrow \mathbb{R}$, define the link-boundary sensitivity measure (formally, a

seminorm)

$$\|h\|_{\partial f} = \sum_{j \in \mathcal{A}_f} \underbrace{\frac{\bar{g}_{fj}}{w_{fj}}}_{\text{choice-sensitivity weight}} \underbrace{\max_{\substack{S_f \ni j \\ T_f \not\ni j}} |h(S_f) - h(T_f)|}_{\text{largest include-exclude gap shift}}. \quad (2.9)$$

Up to the factor $1/\sigma_\varepsilon$, $\|h\|_{\partial f}$ bounds the total probability, summed over links, that perturbing firm f 's payoff table by h changes a link's inclusion status.

For rival- g portfolios A, B and a portfolio profile $U = S_{-(f,g)}$, let

$$\Delta r_{A,B,U}^{fg}(S_f) = \underbrace{r_f(S_f, A, U)}_{g \text{ plays } A} - \underbrace{r_f(S_f, B, U)}_{g \text{ plays } B}.$$

This difference is the perturbation that firm f experiences when rival g switches its portfolio from B to A while the remaining rivals stay at U . Define the nonnegative interaction matrix K by

$$K_{fg} = \max_{A,B,U} \|\Delta r_{A,B,U}^{fg}\|_{\partial f} \quad (g \neq f), \quad K_{ff} = 0. \quad (2.10)$$

The entry K_{fg} bounds the worst-case change in firm f 's weighted payoff gaps between portfolios that include and exclude a link. The weights are the conditional-density bounds divided by the link cost loadings. The comparison changes firm g 's portfolio and maximizes over the remaining rivals. Choice-probability sensitivity is $K_{fg}/\sigma_\varepsilon$; common payoff levels cancel. The matrix is a worst-case global envelope, not a local derivative evaluated at equilibrium.

Throughout, $d_{\text{TV}}(P, Q)$ denotes the total variation distance: the largest difference in the probability that P and Q assign to the same event. On a finite support it equals $\frac{1}{2} \sum_x |P(x) - Q(x)|$, half the summed absolute differences of the point probabilities.

The matrix K collects bilateral portfolio sensitivities. Its spectral radius $\rho(K)$, the largest absolute value of its eigenvalues, captures their amplification around feedback cycles. With two firms, $\rho(K) = \sqrt{K_{12}K_{21}}$: a strong response in one direction can coexist with uniqueness when the reverse response is weak. In larger games, the spectral radius similarly aggregates direct and indirect feedback rather than selecting the largest bilateral effect.

Theorem 1 (Exact large-noise uniqueness). *Under the assumptions in Section 2.1, the distri-*

butional best responses satisfy

$$d_{\text{TV}}\left(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})\right) \leq \frac{1}{\sigma_\varepsilon} \sum_{g \neq f} K_{fg} d_{\text{TV}}(Q_g, Q'_g). \quad (2.11)$$

Consequently, if

$$\sigma_\varepsilon > \rho(K), \quad (2.12)$$

the exact portfolio BNE is unique. Simultaneous distributional best-response iteration converges globally to that equilibrium.

In economic terms, private link-cost heterogeneity must be large enough relative to cross-firm payoff feedback that a change in portfolio beliefs shrinks as it passes through successive best responses.

Because $\rho(K)$ never exceeds the largest row sum, the theorem's condition is easier to satisfy than the simple row-sum test $\max_f \sum_{g \neq f} K_{fg} < \sigma_\varepsilon$: uniqueness can survive one large bilateral response as long as it is only weakly fed back through the rest of the system. Greater dispersion does not change the deterministic payoff effects in K . It reduces the probability mass of types whose costs lie near a moving boundary, so $\sigma_\varepsilon > \rho(K)$ admits positive firm weights under which best response is a contraction. Within the same interaction bounds, no re-weighting of firms in the metric delivers a weaker condition, and re-scaling any firm's units leaves $\rho(K)$ unchanged (Proposition A.1 in Appendix A).

The proof has two steps. First, payoff perturbations move contingent link boundaries, and holding private-cost realizations fixed converts those movements into total-variation changes. Second, swapping one rival's distribution at a time telescopes these single-rival bounds into the matrix inequality. A suitable positive weighting of firm-level distances then makes the map a contraction whenever $\rho(K) < \sigma_\varepsilon$. Appendix A.2 gives the formal proof.

The argument applies to any finite reduced payoff table and feasible portfolio family, including cases with changing cross-difference signs. The bound adds one term per candidate link. Evaluating K and global best responses remains sensitive to the structure of the feasible family.

Remark 1 (Separable threshold model as a special case). Suppose $\mathcal{X}_f = 2^{\mathcal{A}_f}$ and the interim

value separates across own links:

$$V_f(S_f; Q_{-f}) = C_f(Q_{-f}) + \sum_{j \in S_f} v_{fj}(Q_{-f}).$$

Then the include and exclude problems in Lemma 1 share the same optimization over all links other than j , so $V_{fj}^{\text{in}} - V_{fj}^{\text{out}} = v_{fj}(Q_{-f})$. The inclusion cutoff therefore reduces to

$$\varepsilon_{fj} < \frac{v_{fj}(Q_{-f})}{\sigma_\varepsilon w_{fj}},$$

which is independent of $\varepsilon_{f,-j}$. If own link shocks are also independent, each link is then included independently with its own probability, so Q_f is fully described by one inclusion probability per link, and the equilibrium fixed point is over these probabilities rather than over portfolio distributions. Thus the separable threshold model is the product-form special case of the exact portfolio equilibrium.

The contingent-boundary argument uses conditioning and bounded conditional densities.⁴ Here the argument yields a distributional best-response contraction for arbitrary deterministic portfolio payoffs and feasible families.

For independent standard normal shocks, $\bar{g}_{fj} = 1/\sqrt{2\pi}$. With heterogeneous positive scales σ_{fj} , the corresponding probability-sensitivity coefficient becomes $\bar{g}_{fj}/(\sigma_{fj}w_{fj})$ in place of $\bar{g}_{fj}/(\sigma_\varepsilon w_{fj})$; the maintained loading $w_{fj} > 0$ and the same spectral argument apply.

2.4 The small-noise limit

Theorem 1 uses dispersion to smooth strategic responses. At the other end of the noise scale, suppose the complete-information game has a unique Nash equilibrium in the full mixed-strategy space and that this equilibrium is a strict pure profile. If private shocks satisfy the local boundary condition, noisy best responses become nearly optimal in the complete-information game, so every small-noise BNE lies near the pure profile. Strictness gives a positive payoff gap there, and the local boundary bound makes best response a

⁴Related winner-gap anti-concentration results for discrete optimization appear in Beier and Vöcking (2006).

contraction. The BNE is therefore unique for sufficiently small σ_ε and converges to the complete-information profile.

Proposition 2 (Small-noise uniqueness). *Suppose the complete-information portfolio game has a unique Nash equilibrium in the full mixed-strategy space and that this equilibrium is a strict pure profile S^* . Suppose the private shock vectors satisfy the local boundary condition in Appendix A.3: near a strict optimum, the probability that the payoff gap between a firm’s best and second-best portfolios falls below t is at most proportional to t , with a coefficient that vanishes as $\sigma_\varepsilon \downarrow 0$. Then there is $\underline{\sigma} > 0$ such that the exact portfolio BNE is unique whenever $0 < \sigma_\varepsilon < \underline{\sigma}$. Its portfolio distributions converge to the point masses on S^* as $\sigma_\varepsilon \downarrow 0$.*

Independent standard normal or logistic link shocks satisfy this boundary condition. No moment conditions are required: the argument caps complete-information regret at the finite payoff-table range, so some infinite-mean shock distributions are admissible when they satisfy the stated tail-density condition. Appendix A.3 gives primitive conditions and proves the proposition.

The uniqueness results use different mechanisms. Large private-cost noise smooths best responses enough to deliver a global contraction. Small noise instead inherits local stability from a complete-information game that already has a unique strict equilibrium, including against mixed strategies. The results cover the two tails of the noise scale; they do not assert uniqueness at every intermediate noise level. The cloud application is the zero-noise case. It checks complete-information global responses directly rather than applying the finite-noise contraction.

2.5 Computing the exact equilibrium

With the Atkeson–Burstein (AB) pricing block developed in Section 3, computing the exact equilibrium has an inner and an outer stage. Given a private-cost realization and rivals’ portfolio distributions, the inner stage finds the firm’s globally optimal complete portfolio. At finite noise, integrating these choices across cost realizations produces the distributional best response; the outer stage iterates those distributions to a fixed point. In the complete-information application, the outer stage instead iterates realized global portfolio responses.

The main computational difficulty lies in the inner stage. When own-portfolio supermodularity or AES choice monotonicity is verified, the corresponding Jia or AES procedure can solve that stage exactly. Recomputing the AB pricing equilibrium at every evaluated portfolio does not guarantee either order condition; Online Appendix [OA.3.3](#) gives a three-link counterexample even though the downstream pricing equilibrium is unique.

Our general implementation therefore uses branch and bound rather than order-based link elimination. It divides the portfolio menu into groups of portfolios, computes an upper bound on the best payoff attainable within each group, and discards a group when that bound falls below the best payoff already found. For a type ε_f , write the objective as

$$\tilde{V}_f(T_f; \varepsilon_f, Q_{-f}) = V_f(T_f; Q_{-f}) - \sigma_\varepsilon \sum_{j \in T_f} w_{fj} \varepsilon_{fj}. \quad (2.13)$$

The first term is the type-independent interim value; only included links incur their realized cost shocks. A search node (S_f, U_f) records links already selected in S_f and links still undecided in U_f . Its feasible completions form the set $C_f(S_f, U_f)$, and a node bound is any value $B_f(S_f, U_f)$ that dominates the payoff of every completion:

$$C_f(S_f, U_f) = \{T_f \in \mathcal{X}_f : S_f \subseteq T_f \subseteq S_f \cup U_f\}, \quad B_f(S_f, U_f) \geq \max_{T_f \in C_f(S_f, U_f)} \tilde{V}_f(T_f; \varepsilon_f, Q_{-f}). \quad (2.14)$$

If $B_f(S_f, U_f)$ does not exceed the best payoff already found, no completion in that node can be optimal.

The useful bound retains one feature of the common portfolio: its number of additional links. Write the objective (2.13) equivalently as $\sum_r q_r \pi_{fr}(T_f) - \sum_{j \in T_f} c_{fj}$, grouping revenue against total link costs: r indexes a finite rival-portfolio state, $q_r \geq 0$ is the probability the rival portfolio distribution assigns to state r (a point mass on the realized rival profile in the complete-information application), and $c_{fj} = w_{fj}(\phi_{fj} + \sigma_\varepsilon \varepsilon_{fj})$ combines deterministic and realized private costs. Suppose the continuation payoff in state r is bounded by a coordinatewise increasing function of nonnegative additive statistics,

$$z_{r\ell}(T_f) = z_{r\ell}^0 + \sum_{j \in T_f} a_{r\ell j}, \quad a_{r\ell j} \geq 0, \quad \pi_{fr}(T_f) \leq \bar{\pi}_{fr}(z_{r1}(T_f), \dots, z_{rL}(T_f)).$$

For a completion with k additional links, let $a_{r\ell, (h)}^\downarrow(U_f)$ be the h th largest contribution

to statistic (r, ℓ) among undecided links, and let $c_{f,\ell(h)}^\uparrow(U_f)$ be the h th smallest link cost. Define

$$\begin{aligned}\bar{z}_{r\ell}(S_f, U_f, k) &= z_{r\ell}(S_f) + \sum_{h=1}^k a_{r\ell,(h)}^\downarrow(U_f), \\ \underline{C}_f(S_f, U_f, k) &= \sum_{j \in S_f} c_{fj} + \sum_{h=1}^k c_{f,\ell(h)}^\uparrow(U_f).\end{aligned}$$

Together they describe the best case for a completion with k additional links: every statistic is credited with its k largest undecided contributions, while only the k cheapest undecided costs are subtracted.

Proposition 3 (Cardinality-conditioned completion bound). *Under the conditions above,*

$$B_f^{\text{card}}(S_f, U_f) = \max_{0 \leq k \leq |U_f|} \left\{ \underbrace{\sum_r q_r \bar{\pi}_{fr}(\bar{z}_r(S_f, U_f, k))}_{\text{best-case revenue with } k \text{ additions}} - \underbrace{\underline{C}_f(S_f, U_f, k)}_{\text{cheapest } k\text{-addition cost}} \right\} \quad (2.15)$$

is an upper bound on every feasible completion of the node. The result applies to any finite feasible portfolio family; including cardinalities with no feasible completion only weakens the bound.

The relaxation allows different links to generate the most favorable statistic in different markets, so the resulting vector need not correspond to a feasible portfolio. What it preserves is a common completion size and the lowest total cost possible at that size. In Example 1, the $k = 1$ relaxation may credit Korea's most favorable statistic in one market and Japan's in another while subtracting only the cheaper link cost, a combination neither single-link portfolio attains, so the bound can only overstate the best feasible completion value. Appendix C proves the result, shows that it weakly improves on the maximal-portfolio envelope built from the same component bounds (the bound that credits every undecided link at once), and specializes the bound to the two pricing continuations used in the application of Section 4, CES and AB. The construction is related to assortment bounds that relax a common product set across customer types or nested choice components (Feldman and Topaloglu, 2015; Leitner et al., 2024; Zhang et al., 2025). Here the common object is a strategic location portfolio shared across markets and rival states, followed by a CES or AB pricing continuation.

Branch and bound and top- k envelopes are standard tools in discrete optimization. Our contribution is to establish the validity of the cardinality-conditioned bound for a common portfolio evaluated across multiple markets and rival states with an equilibrium pricing continuation, and to show that it weakly dominates the maximal-portfolio envelope built from the same component bounds. This construction supplies exact responses in the strategic environment where the Jia and AES order-based elimination steps can fail. The two approaches differ in what justifies discarding a portfolio: order-based elimination fixes a link permanently when its marginal value moves in one direction across nested portfolios, while the bound discards a branch when even its most favorable completion cannot beat the best portfolio already found, a test that remains valid when equilibrium repricing flips marginal-value signs across portfolios.

Appendix C links search effort to the tightness of the node bound and the number of near-optimal portfolios. Near-additivity with a clear fixed-size margin is one transparent sufficient case, while in the calibrated problems the cardinality bound itself discards almost the entire search tree. Because fixed-size choice is NP-hard even under constant markups (Appendix C.5), the relevant object is certified instance-specific search rather than a generally fast closed-form entry rule.

Online Appendix [OA.6](#) gives the search algorithm, numerical-error accounting, and performance results. Exhausting the search establishes global optimality over the discrete portfolio choice set under the supplied continuation values. CES has a closed-form continuation. AB recomputes its unique pricing equilibrium at every evaluated profile and carries the pricing residual into the payoff bound. A finite-noise computation additionally requires control of the integration error across private-cost types.

2.6 When is country-by-country choice exact?

The exact response treats the firm's network as one choice. A common simplification instead assigns each country a value and enters whenever that value exceeds the country's fixed cost, as in canonical destination-entry models (Chaney, 2008; Dickstein and Morales, 2018). This country-by-country rule is attractive because it replaces a search over portfolios with a collection of scalar comparisons. It is also a stronger restriction than the inclusion cutoff of Lemma 1: the cutoff for country j must remain unchanged when the firm's other realized link costs change.

The restriction has a sharp boundary. Suppose every subset of candidate links is feasible and private costs have rich support. The country-by-country rule reproduces the exact complete-portfolio response for every deterministic cost vector if and only if operating value is additive across links. To state this condition in the notation of Section 2, let $W_f(S_f; Q_{-f})$ denote interim operating value before deterministic and private link costs. Then there are coefficients $\alpha_f(Q_{-f})$ and $v_{fj}(Q_{-f})$ such that

$$W_f(S_f; Q_{-f}) = \alpha_f(Q_{-f}) + \sum_{j \in S_f} v_{fj}(Q_{-f}) \quad \text{for every } S_f \subseteq \mathcal{A}_f.$$

Subtracting deterministic link costs gives the interim value $V_f(S_f; Q_{-f})$ defined in (2.3); because those costs are additive, they do not change the condition. Appendix B proves the equivalence. The condition concerns the payoff representation, not the joint distribution of costs: within-firm shocks and the induced link choices may remain correlated.

The two-link example gives the intuition. Complementarity creates regions in which the firm compares the empty portfolio with both links, so the choice boundary depends on the sum of their costs ($w_a \varepsilon_a + w_b \varepsilon_b$). Substitution creates regions in which the firm compares one link with the other, so the boundary depends on their cost difference ($w_a \varepsilon_a - w_b \varepsilon_b$). A collection of country-specific cutoffs produces only axis-aligned boundaries, each fixing one ε_j alone, and therefore cannot reproduce either comparison for all cost vectors. Additivity is the case in which these portfolio interactions vanish and the inclusion cutoffs cease to depend on other realized costs.

Failure of additivity does not imply that the simpler rule selects the wrong portfolio in every state. For fixed rival beliefs, approximate the complete payoff table V_f by an additive anchor

$$M_f(S_f) = \alpha_f + \sum_{j \in S_f} v_{fj}, \quad \gamma_f = \max_{S_f} |V_f(S_f) - M_f(S_f)|,$$

where γ_f is the anchor's worst-case approximation error. We use two choices of the fitted values. The Shapley anchor sets v_{fj} to link j 's marginal contribution to V_f averaged across portfolio orders. The Chebyshev anchor instead chooses (α_f, v_{fj}) to minimize γ_f , the best additive approximation in the worst case. A rule that maximizes either anchor loses at most $2\gamma_f$ relative to the exact response. For the discrete choice itself, let Δ_f be the exact

payoff gap between the best and second-best portfolios. If

$$\Delta_f > 2\gamma_f,$$

the anchor and the exact problem select the same portfolio. Modularity is thus the condition for uniform exactness across cost vectors, whereas the decision margin determines whether a particular response is preserved.

At the complete-information limit, applying this comparison at every firm’s equilibrium response gives a direct equilibrium-preservation test. The cloud application uses this limit and compares exact and country-by-country responses on common payoff tables. Appendix B gives the formal decision-margin result and extends the analysis to finite private cost noise. Under the large-noise contraction condition, it also shows how a one-step approximation error is amplified through subsequent rounds of rival responses.⁵

3. AB SPECIALIZATION: PRICING AND PORTFOLIO INTERACTIONS

Section 2 establishes equilibrium results for a reduced payoff table $r_f(S)$ and an interaction matrix K . This section supplies those two objects in the AB environment used in the application. Solving the pricing game at each joint portfolio constructs $r_f(S)$. Bounding how one firm’s portfolio changes another firm’s fully repriced payoff constructs K . Portfolios affect service-cost and accessibility or quality aggregators, while AB pricing jointly determines equilibrium shares, prices, and markups.

Empirical evidence documents strong strategic complementarities in the pricing of large firms (Amiti et al., 2019). Recent work uses AB demand to quantify how firm shocks propagate through market shares and markups for a fixed set of firms (Burstein et al., 2025). Here the location portfolios that determine costs, shares, and markups are endogenous:

Portfolio $\rightarrow$ costs/access $\rightarrow$ pricing equilibrium $\rightarrow$ profit $\rightarrow$ portfolio response.

Pricing uniqueness makes the reduced payoff well defined at every activation profile. The

⁵This finite-noise propagation result extends the model. The cloud calculations instead solve the complete-information portfolio equilibrium directly.

portfolio-interaction bound then supplies the matrix in the uniqueness condition of Theorem 1. The general result therefore remains portable across downstream models, while the AB specialization completes the pricing and profit block used below.

3.1 Benchmark environment

There are F firms, indexed by f , and N markets, indexed by n . Firm f has a finite set of candidate sites $\mathcal{A}_f$. An activation set is denoted $\mathcal{I}_f \subseteq \mathcal{A}_f$; it is the portfolio S_f from Section 2, written in site notation because the pricing block tracks activated sites serving every market. The quantitative benchmark is a fixed-expenditure AB reduced form with $\sigma > \eta > 1$ and exogenous sectoral expenditure R_n . The parameter σ is the elasticity of substitution across firms within a market, while η is the upper-tier elasticity across sectors. As a firm's market share rises, its perceived demand elasticity falls from σ toward η ; equation (3.3) below displays the schedule. This specification isolates the markup channel used in the application. A fully nested-CES version additionally lets sectoral expenditure move with its price index.

Firm f 's marginal cost of serving market n from activation set $\mathcal{I}_f$ is

$$c_{nf}(\mathcal{I}_f) = \frac{\Gamma_\theta}{a_f} \Sigma_{fn}(\mathcal{I}_f)^{-1/\theta}, \quad \Sigma_{fn}(\mathcal{I}_f) = \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}. \quad (3.1)$$

Adding a site raises Σ_{fn} and therefore weakly lowers the cost of serving market n . Sites with low $w_i \tau_{in}$ make the largest contribution. Here Γ_θ is the Fréchet scale normalization; for unit-scale Fréchet draws it equals $\Gamma(1 + 1/\theta)$. The common factor can be absorbed into the productivity normalization. The term a_f is firm productivity, w_i is the input cost at site i , τ_{in} is the trade or access cost from site i to market n , and θ is the Fréchet dispersion parameter. When link j corresponds to site i , the Section 2 cost loading w_{fj} specializes to the site scale w_i used here.

The portfolio can also shift demand through a positive provider-market quality or accessibility term $\psi_{fn}(\mathcal{I}_f)$, constructed in Section 4.2 from the provider's network; environments without this channel set it to one. Define quality-adjusted price power and market share by

$$y_{nf} = \psi_{fn}(\mathcal{I}_f) p_{nf}^{1-\sigma}, \quad s_{nf} = \frac{y_{nf}}{\sum_g y_{ng}}. \quad (3.2)$$

Under the CES demand block, expenditure on firm f in market n is $R_n y_{nf} / \sum_g y_{ng}$, so s_{nf} is also firm f 's revenue share in market n . If $\mathcal{I}_f = \emptyset$, we set $c_{nf} = +\infty$ and $y_{nf} = s_{nf} = 0$, so firm f is absent from the pricing block in market n and earns no variable profit there.

Given active firms, prices follow the Atkeson–Burstein rule. In the fixed-expenditure block,

$$p_{nf} = \mu(s_{nf})c_{nf}, \quad \mu(s) = \frac{\mathcal{E}(s)}{\mathcal{E}(s) - 1}, \quad \frac{1}{\mathcal{E}(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad (3.3)$$

where s_{nf} is firm f 's revenue share in market n . Markups are bounded:

$$\mu(s) \in \left[\frac{\sigma}{\sigma - 1}, \frac{\eta}{\eta - 1} \right].$$

Thus high-share firms face a lower perceived elasticity and charge a higher markup, but the markup schedule remains uniformly bounded. We impose this AB markup schedule as the pricing rule in the fixed-expenditure reduced form. In the fully nested-CES model it follows from the Cournot first-order condition; fixing R_n retains the same share-dependent markup schedule while suppressing the response of sectoral expenditure to its price index.

Variable profit in market n is revenue times the operating margin:

$$VP_{nf} = \frac{R_n s_{nf}}{\mathcal{E}(s_{nf})} = R_n s_{nf} \frac{\mu(s_{nf}) - 1}{\mu(s_{nf})}. \quad (3.4)$$

Together, the cost, share, markup, and profit equations map every joint portfolio into market shares, prices, markups, and operating profit. In the Korea–Japan example, this map is what makes $v_{ab} - v_b$ differ from $v_a - v_\emptyset$: adding the Korean site lowers delivered costs by less when the Japanese site already contributes to the cost aggregator, and the profit gained from a given cost reduction depends on the share the firm already holds through $\mu(s_{nf})$.

Firm f 's ex-post profit, conditional on an activation profile and its private cost draws at the noise scale σ_ε of Section 2, is

$$\pi_f(\mathcal{I}_f, \mathcal{I}_{-f}; \varepsilon_f) = \sum_n VP_{nf}(\mathcal{I}_f, \mathcal{I}_{-f}) - \sum_{i \in \mathcal{I}_f} (\phi_{fi} + \sigma_\varepsilon \varepsilon_{fi}) w_i.$$

The deterministic cost component ϕ_{fi} may vary by firm and site; the calibration in Sec-

tion 4 restricts it to a market-level cost scaled by a provider shifter. The private fixed-cost shocks ε_{fi} are observed by firm f before entry decisions are made.

3.2 The pricing block

Fix an activation profile and a market with at least two active firms. Using the quality-adjusted price power in (3.2), write

$$x_{nf} = \log y_{nf}, \quad D_{nf} = \sum_{g \neq f} y_{ng},$$

so x_{nf} is the firm's own price power in logs and D_{nf} aggregates its rivals' price power in market n . Rivals affect firm f 's response only through D_{nf} . The markup rule defines the scalar response $y_{nf} = \mathcal{Y}_{nf}(D_{nf}; c_{nf}, \psi_{fn})$. Let Φ be the corresponding update in log price power,

$$\Phi_{nf}(x) = \log \mathcal{Y}_{nf} \left(\sum_{g \neq f} e^{x_{ng}}; c_{nf}, \psi_{fn} \right).$$

Define

$$\begin{aligned} \beta_{nf} &= \frac{(\sigma - 1)(\sigma - \eta)}{\sigma \eta} \mu(s_{nf}), \\ \tilde{\kappa}_{nf} &= \left| \frac{\partial \log y_{nf}}{\partial \log D_{nf}} \right| = \frac{s_{nf}(1 - s_{nf})\beta_{nf}}{1 + s_{nf}(1 - s_{nf})\beta_{nf}}. \end{aligned} \quad (3.5)$$

The coefficient β_{nf} measures the strength of the markup channel. It equals $(\sigma - 1)$ times the slope of the log-markup schedule at the firm's share and vanishes when $\sigma = \eta$. The elasticity $\tilde{\kappa}_{nf}$ is the firm's equilibrium pricing response: the percentage change in its own price power induced by a one-percent change in the rival aggregate. It is zero under constant markups and is damped at extreme shares through the factor $s_{nf}(1 - s_{nf})$. The pointwise interaction matrix is

$$\Lambda(x)_{(n,f),(n,g)} = \tilde{\kappa}_{nf}(x) \frac{y_{ng}(x)}{D_{nf}(x)}, \quad g \neq f, \quad (3.6)$$

with zero diagonal and cross-market entries. Thus $J_{\Phi}(x) = \Lambda(x)$, and each active row sums to $\tilde{\kappa}_{nf}(x)$. Each nonzero entry combines firm f 's price-response sensitivity with firm g 's

weight in f 's competitor aggregate D_{nf} . It is large when f 's markup is share-sensitive and g accounts for much of f 's competition. Cross-market entries are zero because the fixed-expenditure pricing problem separates market by market. When $\sigma = \eta$, markups are constant and Λ is identically zero.

The AB primitives give a uniform bound. Since

$$\beta_{nf} \leq \bar{\beta} \equiv \frac{(\sigma - 1)(\sigma - \eta)}{\sigma(\eta - 1)}$$

and $s(1 - s) \leq 1/4$,

$$\|\Lambda(x)\|_\infty \leq \bar{\lambda}_{AB} \equiv \frac{\bar{\beta}/4}{1 + \bar{\beta}/4} < 1 \quad \text{for every } x. \quad (3.7)$$

At the elasticities used in the application, $(\sigma, \eta) = (5, 2)$, this primitive bound is 0.375; direct maximization of (3.5) gives 0.3187. The maximizing share is near $s = 0.5585$. Section 4.2 explains the parameter choice. An empty market has no pricing variables. If a market has one active firm, its price follows directly from the monopoly markup and contributes no interaction row.

Proposition 4 (Pricing uniqueness). *For $\sigma > \eta > 1$, every fixed activation profile has a unique AB pricing equilibrium. In markets with at least two active firms, the log-price-power update Φ is a global sup-norm contraction with modulus $\bar{\lambda}_{AB}$ in (3.7). Equilibrium prices vary smoothly with strictly positive marginal costs on each fixed active set.*

Appendix D.1 derives (3.5) and proves the proposition. The primitive modulus also bounds the response of equilibrium prices to costs:

$$\|(I - \Lambda(x))^{-1}\|_\infty \leq \frac{1}{1 - \bar{\lambda}_{AB}}.$$

The factor $1/(1 - \bar{\lambda}_{AB})$ bounds the amplification generated by repeated repricing. A portfolio perturbation changes price power, which moves rivals' shares and price powers and then feeds back across firms. The contraction bound limits the k th round by $\bar{\lambda}_{AB}^k$, so the geometric sum bounds the total response. At $(\sigma, \eta) = (5, 2)$, the multiplier is at most $1/(1 - 0.375) = 1.6$: equilibrium repricing can add at most 60 percent to the direct response bounded in this norm.

At a solved equilibrium, $\rho(\Lambda(x^*))$ is a local pricing-interaction diagnostic; the uniform uniqueness result follows from $\bar{\lambda}_{AB}$.⁶

The unique pricing selection assigns a reduced payoff to every active profile. The next subsection bounds how portfolio changes move this fully repriced payoff through costs, accessibility, shares, and markups.

3.3 Bounding portfolio interactions under AB pricing

The large-noise theorem requires an upper bound on the portfolio interaction matrix K . For $f \neq g$, the entry K_{fg} measures how much a change in firm g 's portfolio can alter firm f 's gain from including a candidate link. Under AB pricing, this interaction includes changes in service costs and accessibility as well as the induced adjustment in market shares and markups. This subsection constructs three computable upper bounds on K (derivative, finite difference, and hybrid), using the pricing contraction of Section 3.2 to control the repricing feedback. The bounds complete the finite-noise theory; the application instead verifies portfolio responses directly, relying on the pricing block's uniqueness and error control.

To isolate the cross-firm effect, fix two portfolios $S_f, T_f \in \mathcal{X}_f$ for firm f , two portfolios $A, B \in \mathcal{X}_g$ for firm g , and the remaining firms' portfolios U . Consider the four-point payoff difference

$$\Delta_{fg}(S_f, T_f; A, B, U) = \underbrace{\left[r_f(S_f, A, U) - r_f(T_f, A, U) \right]}_{f\text{'s gain from } S_f \text{ over } T_f \text{ when } g \text{ plays } A} - \underbrace{\left[r_f(S_f, B, U) - r_f(T_f, B, U) \right]}_{\text{the same gain when } g \text{ plays } B}.$$

Their difference shows how the rival's portfolio changes firm f 's portfolio incentive after equilibrium prices have adjusted. The entry K_{fg} takes the largest such movements across the include-versus-exclude boundary of each candidate link and applies the conditional-density bounds and link-cost loadings from Theorem 1.

Each portfolio maps into service-cost and accessibility aggregators. The four joint portfolio profiles above form the corners of a rectangle in the aggregators of firms f and g . One direction changes firm f 's portfolio; the other changes firm g 's portfolio. When

⁶At the calibrated equilibrium of Section 4, Online Appendix [OA.4](#) reports $\rho(\Lambda(x^*)) = 0.236$ and $\|\Lambda(x^*)\|_\infty = 0.305$, both below the uniform bound of 0.375.

the set of active sellers remains fixed, fully repriced profit is smooth along this rectangle, and the four-point difference can be bounded by a mixed payoff derivative. This derivative includes the direct effect of the portfolio aggregators and the indirect effect through equilibrium repricing. Proposition 4 bounds the latter feedback by the pricing multiplier $1/(1 - \bar{\lambda}_{AB})$.

Entry into or exit from a market changes the set of active sellers, so the derivative calculation no longer applies throughout. The four fully repriced payoffs remain well defined. This gives three related calculations. When all four portfolio combinations keep the active seller set unchanged, smoothness bounds the four-point difference using derivatives of fully repriced profit. When a combination induces entry or exit, we instead compare the four repriced payoffs directly; in the fixed-expenditure model, a market's contribution is at most R_n/η . This quantity is monopoly variable profit in market n : perceived demand elasticity is never below η , so no firm can earn more under any portfolio profile. After ordering the firm's two portfolios by competitiveness in a market, the corresponding profit increment lies in $[0, R_n/\eta]$ under either rival portfolio, so the difference of the two increments is at most R_n/η rather than twice that amount. The hybrid bound uses derivatives when the active seller set is unchanged and direct payoff comparisons otherwise.

Proposition 5 (AB portfolio-interaction bounds). *Let $\bar{K}^{AB}$, $\bar{K}^{FD}$, and $\bar{K}^{HYB}$ denote the derivative, finite-difference, and hybrid matrices defined in Online Appendix OA.3.1. If every relevant portfolio rectangle satisfies the regularity and derivative-envelope conditions in that appendix, then $K \leq \bar{K}^{AB}$. In the fixed-expenditure AB model, $K \leq \bar{K}^{FD}$ and $K \leq \bar{K}^{HYB}$, and*

$$\bar{K}_{fg}^{FD} \leq \left(\sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} \right) \sum_n \frac{R_n}{\eta}, \quad f \neq g. \quad (3.8)$$

All matrix inequalities hold component by component, and all three upper-bound matrices have zero diagonal. For any valid upper-bound matrix $\bar{K}$, the condition $\sigma_\varepsilon > \rho(\bar{K})$ implies uniqueness of the exact finite-noise portfolio BNE and convergence of simultaneous distributional best-response iteration from every initial distribution.

Online Appendix OA.3.1 derives the three bounds, and Online Appendix OA.3.2 gives the exact AB share Jacobian used in the hybrid calculation. The finite-noise model uses these bounds to establish portfolio-equilibrium uniqueness. The cloud application takes

the complete-information limit and uses pricing uniqueness at each deployment profile: prices and shares are recomputed after every candidate portfolio deviation.

Section 4.2 defines the calibration and evaluates this bound in the calibrated model. The resulting bound is too conservative to provide a quantitative finite-noise uniqueness region for the complete-information application. We therefore use it as a sufficient theoretical condition, while the application verifies global firm responses directly and reports their numerical error.

3.4 Extension to other demand systems

Beyond the maintained assumptions on private costs and the interaction bound, the exact large-noise theorem requires from the downstream problem only a well-defined operating payoff for every joint portfolio. In smooth demand systems, a unique interior pricing equilibrium with a uniformly nonsingular first-order-condition Jacobian makes prices differentiable in costs. Bounded responses then yield the four-point interaction bound summarized in Proposition 5; Online Appendix [OA.3.1](#) gives the derivative refinement. Online Appendix [OA.2](#) states these conditions for single-product Bertrand pricing under nested-logit demand. The conditions in Nocke and Schutz (2018) provide one route to pricing uniqueness. The cloud application uses the AB pricing block above with a public-data gravity function.

4. DIGITAL-SERVICES INFRASTRUCTURE APPLICATION

Five providers—Amazon Web Services, Microsoft Azure, Google Cloud, Oracle Cloud, and Alibaba Cloud—choose cloud regions across countries. A region is a country-level infrastructure link that affects latency, serviceability, compliance, and the value of the provider’s other regions. Related work documents large content providers’ growing investment in global cable infrastructure, connects that investment to data-center siting, and studies dynamic entry by undersea-cable operators at an adjacent infrastructure layer (Caoui and Steck, 2025a,b). The application uses a single benchmark calibration for its main AB–CES comparison. All demand, accessibility, service-cost, and fixed-cost primitives are held fixed across the two pricing systems. Because the calibration targets a single cross-section of deployments, we implement the complete-information limit and set

private-cost shocks to zero. Each provider then computes a global portfolio response while rivals' portfolios are fixed, with equilibrium prices and shares recomputed for every portfolio evaluated. Iterating these responses yields the exact complete-information profiles reported below.

4.1 Data and sample

The unit of observation is a provider-country pair. We code a pair as active when the provider operates at least one cloud region in that country. The underlying deployment units are provider regions, each of which contains one or more availability zones where compute, storage, and networking services are offered from local infrastructure. We collect deployment status from each provider's official region and infrastructure pages (listed in Online Appendix [OA.5](#)), measure it as of April 7, 2026, and aggregate named regions to provider-country presence. Datacenter Map does not determine deployment status; its facility coordinates enter only the country-centroid construction used for distance. Market covariates come from public sources: World Bank output and internet use, Penn World Table compensation, and OECD digital-policy indicators.

The panel contains five firms, 42 countries, and 210 provider-country pairs. Operational deployments are observed for 129 pairs: 25 for AWS, 34 for Microsoft, 29 for Google, 27 for Oracle, and 14 for Alibaba. Five announced AWS links enter the candidate set as inactive until the deployment becomes operational, giving candidate sets of 30, 34, 29, 27, and 14 links and 134 candidate links in total. The benchmark calibration and exact AB response iteration produce a 124-link equilibrium profile. It matches deployment status for 125 of the 134 candidate links. The same primitive and fixed-cost vector is used for the exact CES equilibrium.

Candidate links are countries with an operational or announced deployment. Across providers, this produces 134 links. The calibration and main counterfactual therefore describe portfolio choice within this candidate universe; they do not model entry into countries outside it.

AWS, Azure, and Google Cloud have broad global networks. Oracle and Alibaba add smaller and more geographically concentrated portfolios. Candidate sets range from 14 countries for Alibaba to 34 for Microsoft, producing cross-provider variation in portfolio size and proximity to deployment thresholds.

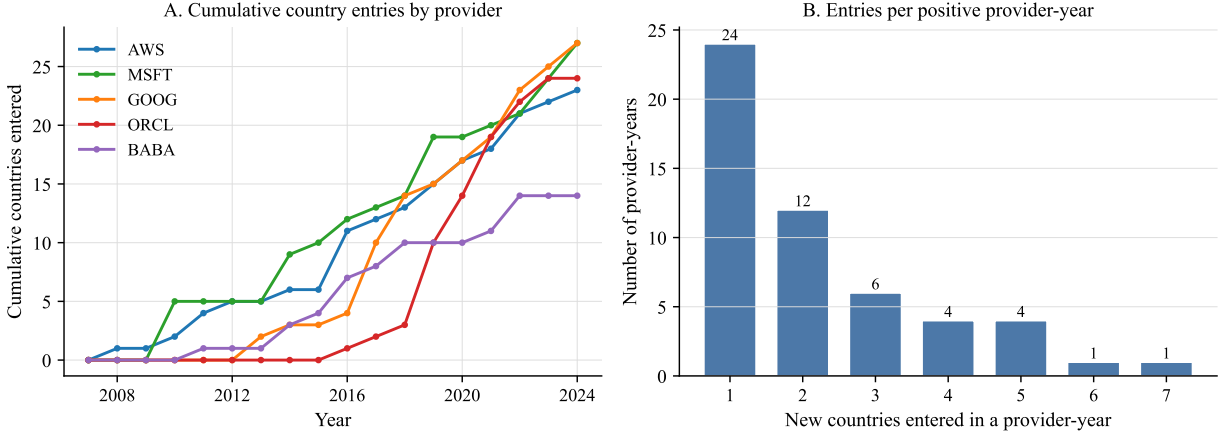


Figure 1: Multinational cloud expansion arrives in waves

Notes: Panel A: cumulative number of countries entered by each provider, 2007–2024, from the launch-date panel of official provider announcements under the main eligibility rule. Panel B: number of new countries entered per positive provider-year. See the text of this section and Online Appendix OA.5.2 for construction, caveats, and the compute-IaaS robustness.

The deployment history itself motivates modeling entry at the level of complete portfolios. From provider launch documentation we date the first region in each provider-country pair and restrict attention to years in which the provider already offered a generally available paid public-cloud service; Online Appendix OA.5.2 documents the sources and the eligibility rule. This yields 115 dated country entries over 2007–2024. Panel A of Figure 1 plots each provider’s cumulative country count. Entry is clustered within firm-years: the 115 entries occur in 52 provider-year episodes, and while 24 episodes add a single country, the remaining 28 add between two and seven (Panel B), so 91 entries (79 percent) fall in multi-country years.⁷ Because these counts aggregate launches to the calendar year, they show that expansion arrives in multi-country waves, not that the underlying decisions were simultaneous. We use them as descriptive motivation for modeling complete portfolios and do not target them in the calibration. The launch-date panel is built from providers’ launch announcements, whereas the calibration uses the April 7, 2026 deployment snapshot.⁸

⁷A stricter eligibility rule that starts each provider at compute-IaaS general availability leaves the pattern essentially unchanged (53 percent of episodes and 78 percent of entries).

⁸Online Appendix OA.5.2 reconciles the panel’s 2026 endpoint with that snapshot.

4.2 Calibration strategy

Demand and technology elasticities are fixed before matching cloud deployment. Public data determine market size, site costs, distance, regional eligibility, and digital policy. Minimum distance calibrates the market and provider fixed-cost shifters. The remaining accessibility coefficients and scale choices are reported as benchmark mappings and explicit normalizations.

Cloud serviceability depends on latency, data-residency requirements, and the ability to use one region as a hub for nearby markets. Cloud providers also follow a more concentrated location pattern than third-party data centers (Fang and Greenstein, 2025). The gravity specification separates market size from this provider-specific serviceability.

Market demand is captured by a provider-invariant public-data proxy

$$\Omega_n \propto (\text{GDP}_n \times \text{internet penetration}_n)^\beta.$$

The U.S.-normalized proxy sets fixed sectoral expenditure, $R_n = \Omega_n$. Provider-market serviceability is constructed from candidate sites. Let D_{in}^{km} be the great-circle distance between site i and market n , and define $\tilde{d}_{in} = D_{in}^{\text{km}}/1000$. The gravity score is

$$q_{in} = -\beta_d \log(1 + \tilde{d}_{in}) + \beta_{\text{same}} \mathbf{1}\{i = n\} + \beta_{\text{local}} \ell_{in} + \beta_{\text{policy}} z_n, \quad (4.1)$$

where ℓ_{in} indicates local or regional eligibility and z_n is a policy-openness weight built from 2025 OECD Digital STRI regulatory measures, with larger values for more open cross-border data regimes (construction and vintage in Online Appendix [OA.5](#)). In words, a high q_{in} means that site i is well placed to serve market n : it is nearby or in-country, locally eligible, and faces a relatively open cross-border data regime. Provider f 's accessibility index aggregates active sites through a smooth log-sum,

$$A_{fn}(\mathcal{I}_f) = \lambda_A^{-1} \log \left(1 + \sum_{i \in \mathcal{I}_f} \exp(\lambda_A q_{in}) \right). \quad (4.2)$$

We normalize accessibility against a fixed market-level frontier computed over the com-

mon candidate universe:

$$\psi_{fn}(\mathcal{I}_f) = \exp\left(\chi[A_{fn}(\mathcal{I}_f) - A_n^{\text{frontier}}]\right) \leq 1. \quad (4.3)$$

The quality shifter ψ_{fn} measures how well provider f can serve market n from its deployment portfolio. It equals one when the portfolio reaches the market-level frontier and falls below one with the accessibility gap; χ governs how quickly it falls.

Each portfolio enters the pricing block through two channels. It determines $c_{nf}(\mathcal{I}_f)$ through the Fréchet service-cost aggregator in (3.1), and it determines $\psi_{fn}(\mathcal{I}_f)$ through (4.3). The latter is the demand shifter absorbed into y_{nf} in Section 3.2, so higher ψ_{fn} raises firm f 's market share at given prices.

Literature-anchored elasticities. The implementation uses $(\sigma, \eta, \theta) = (5, 2, 3)$. As reference points, Antràs et al. (2017) estimate $\sigma = 3.85$ and $\theta = 1.789$, with an alternative estimate of $\theta = 3.175$, while Atkeson and Burstein (2008) use within- and across-sector elasticities of 10 and 1.01. Our benchmark lies between the lower elasticities in AFT and the stronger markup feedback in the AB demand parameterization.

Observed shifters and benchmark mappings. We set $\Gamma_\theta = 1$, absorbing the common Fréchet scale into a_f . Rounded cloud-revenue scale proxies of (100, 65, 40, 20, 17) billion dollars are mapped log-linearly to

$$(a_{\text{AWS}}, a_{\text{MSFT}}, a_{\text{GOOG}}, a_{\text{ORCL}}, a_{\text{BABA}}) = (2.000, 1.757, 1.483, 1.092, 1.000).$$

These proxies discipline provider rank and relative scale, while the common $[1, 2]$ mapping supplies the normalization. The two provider terms enter distinct blocks: a_f shifts variable profitability in every served market, whereas g_f scales the fixed cost of each active site and is calibrated to provider portfolio size. The site scale w_i is human-capital-adjusted compensation per worker, normalized to the United States. The service-cost matrix uses $\tau_{in} = 1$ within country, 1.15 for the same continent or bilateral distance below 5,000 kilometers, and 1.5 otherwise. Thus τ_{in} places coarse geography in marginal cost, while q_{in} uses continuous distance, local eligibility, and digital policy to shift service quality. A quality-only sensitivity specification sets $\tau_{in} = 1$ and recalibrates the fixed-cost block, iso-

lating the accessibility channel.

Minimum-distance deployment costs. Following the extensive-margin calibration logic used in quantitative sourcing and location models such as Antràs et al. (2017), we calibrate the fixed-cost block to market and provider deployment counts. The fixed cost is

$$\text{FC}_{fi} = \phi_{fi} w_i, \quad \phi_{fi} = \phi_i g_f,$$

so the general deterministic cost of Section 3 is restricted to a market-level deployment cost ϕ_i scaled by a provider-specific fixed-cost shifter g_f , with w_i the site cost scale. Distances, eligibility, and policy shifters are constructed from public data. Their score loadings, the log-sum parameter, and the market-size exponent enter as pre-specified benchmark mappings.

Table 1 summarizes what comes from the literature, public data, internal calibration, and normalization after we describe the target moments.

4.3 Minimum-distance targets and parameter choice

Conditional on the demand, technology, and accessibility inputs, the calibration chooses the market and provider shifters (ϕ_i, g_f) to minimize

$$\mathcal{L} = \sum_i (m_i^{\text{data}} - m_i^{\text{model}})^2 + \sum_f (m_f^{\text{data}} - m_f^{\text{model}})^2, \quad (4.4)$$

where m_i counts the providers in market i and m_f the markets served by provider f : the market terms discipline where providers locate, and the provider terms discipline portfolio size. We normalize $g_{\text{AWS}} = 1$ and report the parameter vector with the lowest criterion on the calibration path. Online Appendix [OA.5.3](#) gives the update equations, parameter bounds, stopping rules, and numerical tolerances.

The benchmark specification adds local eligibility and digital policy to the distance and same-country terms, and selects the accessibility curvature from a seven-point grid over $\chi \in [0.24, 0.50]$ documented in Online Appendix [OA.4](#): $\chi = 0.40$ attains the grid minimum $\mathcal{L} = 36$, and the closest competitor, $\chi = 0.45$, has $\mathcal{L} = 40$. Holding $\chi = 0.40$ and the income-group starting values for fixed costs fixed, the benchmark fixed-cost recalibration

Table 1: Calibration Discipline

Object	Baseline or normalization	Discipline
(σ, η, θ)	$(5, 2, 3)$	Reference points in Antràs et al. (2017) and Atkeson and Burstein (2008)
Ω_n	$\beta = 0.70$; United States = 1	2023 World Bank GDP and internet use; benchmark market-size curvature
q_{in} and ψ_{fn}	Score weights in (4.1) $(\beta_d, \beta_{\text{same}}, \beta_{\text{local}}, \beta_{\text{policy}}) = (0.25, 0.20, 0.15, 0.20)$; $\lambda_A = 1.50$	Public distance, regional eligibility, and 2025 OECD policy inputs held fixed; benchmark score mapping
χ	0.40	Selected from the reported grid by the entry-count criterion
a_f and Γ_θ	Revenue-ranked $a_f \in [1, 2]$; $\Gamma_\theta = 1$	Public provider-scale proxies and explicit scale normalizations
w_i and τ_{in}	U.S. wage = 1; $\tau_{in} \in \{1, 1.15, 1.5\}$	PWT compensation and a benchmark three-tier marginal-cost mapping
(ϕ_i, g_f)	$g_{\text{AWS}} = 1$	Minimum distance to market-level provider counts and provider portfolio sizes
Accessibility level	$A_{fn} - A_n^{\text{frontier}}$	Difference normalization shared across providers and markets

Notes: Public data determine the listed shifters; score coefficients and bilateral cost tiers are benchmark mappings. Sensitivity exercises recalibrate the fixed-cost block under each elasticity and geography alternative.

reaches $\mathcal{L} = 26$, selects a 128-link initial profile, and matches 127 of the 134 candidate-link indicators.

Starting from the 128-link candidate profile, iterating profitable single-link additions and deletions with equilibrium prices re-solved after each change produces a 124-link AB profile that matches 125 of the 134 candidate indicators. Unrestricted response iteration then replaces AWS Singapore with AWS Ireland while preserving both the link count and the 125-of-134 fit. These are different stages of the benchmark calibration: 127/134 describes the initial candidate profile, whereas 125/134 describes the selected and exact-equilibrium AB profiles.

We use the cross section to calibrate the deterministic deployment profile. Repeated entry histories, together with a model of entry timing, would inform activation probabilities and σ_ε ; provider-country revenues and prices would provide additional information on demand and service costs. Online Appendix OA.4 gives the accessibility and pricing calculations, Appendix E reports the benchmark calibration, and Online Appendix OA.5 documents the numerical procedure.

The application sets $\sigma_\varepsilon = 0$ and establishes equilibrium through unrestricted global-

response verification. The global interaction envelope evaluates to $\rho(\bar{K}) = 229.3369$, so it is too conservative for the large-noise uniqueness condition to apply quantitatively.

4.4 Equilibrium computation and verification

We apply the branch-and-bound method in Section 2.5 to each provider’s complete candidate set. Every evaluated portfolio is repriced. CES has a closed-form continuation; under AB, Proposition 4 gives a unique pricing equilibrium for every active portfolio, which we solve numerically. The cardinality-conditioned bound closes every provider search even though the calibrated Jia and AES order conditions fail. Online Appendix [OA.6](#) reports the algorithm, node counts, and performance comparison.

Starting from the calibrated AB profile, four providers retain their portfolios and AWS replaces Singapore with Ireland.⁹ After accepting this swap, a second round verifies that every provider’s portfolio is a global best response up to the numerical error bound.¹⁰ The verification establishes that the terminal profiles are complete-information equilibria; it does not establish that they are the unique equilibria of the application.¹¹ The terminal AB profile has 124 links. Under CES, the same response iteration reaches a 105-link profile at which every provider is a global best response. Both are therefore complete-information portfolio equilibria up to the reported numerical tolerances. Online Appendix [OA.6](#) gives the response paths and the separate CES and AB error accounting.

4.5 Exact AB–CES comparison

The fixed-cost vector is calibrated once under AB pricing and held fixed across both pricing rules, so the comparison changes only the markup rule at common primitives.

Table 2 compares the two exact portfolio equilibria using the same demand, accessibility, service-cost, and fixed-cost primitives. The first row is the 124-link AB equilibrium. The second row keeps that portfolio fixed and changes only the pricing rule. Replacing

⁹The profit gain from the swap is 1.46×10^{-6} against aggregate profit of about 1.2, so this link sits essentially at the deployment margin. In the terms of Example 1, $V_{fj}^{\text{in}} \approx V_{fj}^{\text{out}}$ for this link, the deterministic counterpart of an inclusion cutoff at zero.

¹⁰Combining the pricing and tree tolerances, the numerical optimality gap is below 1.6×10^{-10} for every reported AB response; the CES responses use the same discrete search but have no numerical pricing layer.

¹¹Selection is at least locally robust: in the conditional exercise of Section 4.7, the exact response map returns the same fixed point from three initial profiles and both provider orders under each pricing rule.

Table 2: Exact AB–CES Portfolio Decomposition

Pricing and portfolio	Active links	Net profit (normalized)	Change from AB
AB pricing and exact AB portfolio	124	1.2059	0.0%
CES pricing at the exact AB portfolio	124	0.7014	–41.8%
CES pricing and exact CES portfolio	105	0.7763	–35.6%

Notes: The first and third rows are complete-information portfolio equilibria up to the numerical tolerances reported in Section 4.4. The middle row changes only the pricing rule at the exact AB portfolio. All nonpricing primitives remain fixed. Profit is measured in the model’s expenditure units, with U.S. market expenditure normalized to one.

AB pricing with constant CES pricing lowers aggregate provider profit net of fixed costs by 41.8 percent. The third row allows every provider to reoptimize its complete portfolio under CES. Nineteen links exit and the deployment response recovers 6.2 percentage points of the pricing-only decline. Aggregate provider profit remains 35.6 percent below its AB level.

The CES portfolio is a strict subset of the AB portfolio. AWS and Microsoft each remove four links, Google and Oracle remove three each, and Alibaba removes five. Re-optimizing deployment therefore offsets part of the pricing loss by avoiding fixed costs at marginal locations; it does not reverse the finding that variable markups support a broader network. The exact comparison is the paper’s quantitative benchmark.¹² The pattern survives adding a nonstrategic residual supplier: with the calibrated costs held fixed and an expenditure-weighted residual share of 10 (20) percent, the AB network keeps 118 (110) links against 101 (95) under CES, and CES profit remains 32 (27) percent lower.

4.6 Strategic repricing and portfolio choice

To isolate strategic repricing from complete-portfolio optimization, we construct a fixed-aggregate benchmark. During a provider’s portfolio deviation, rivals’ market-level price-power indices are held at their current values and only the deviating provider reoptimizes its prices. After all providers update their portfolios, the joint network is fully repriced, the rival indices are updated, and the process repeats. Portfolio choice remains global;

¹²Each provider’s candidate set contains only countries with an operational or announced region, so these comparisons quantify deployment reallocation within the observed candidate universe. Entry into countries outside it is studied through the China access experiment of Section 4.8.

Table 3: Strategic and Fixed-Aggregate Portfolio Choice

Pricing	Strategic links	Fixed-aggregate links	Changed links	Repriced profit change (%)
AB	124	128	4	-2.04
CES	105	105	0	0.00

Notes: The fixed-aggregate iteration starts from the exact AB profile. Changed links are provider-country entries on which its fixed point differs from the strategic equilibrium. Profit changes evaluate both portfolios under a common full pricing equilibrium, relative to the strategic equilibrium.

only rivals' repricing within a provider's deviation is removed.¹³

Under AB pricing, the fixed-aggregate iteration selects four additional links (Table 3) and its fully repriced aggregate profit is 2.04 percent below the strategic equilibrium. Allowing rivals to reprice within each portfolio deviation removes those links. Under CES, the fixed-aggregate fixed point and the strategic equilibrium coincide. With constant markups, one provider's portfolio does not change its rivals' prices, so updating the aggregates between response rounds is sufficient in this calibration. The AB discrepancy therefore isolates the strategic pricing feedback generated by market-share-dependent markups. All complete-portfolio searches in this comparison close at the tolerances of Section 4.4, with the AB side using the numerical pricing continuation. Online Appendix [OA.6.5](#) reports the iteration paths and provider-country changes.

The next subsection holds the same benchmark environment fixed and asks whether country-by-country entry rules recover the exact networks.

4.7 Country-by-country response diagnostics

Section 4.6 preserves exact whole-portfolio choice and changes the treatment of rivals' repricing. This subsection instead holds the strategic pricing environment fixed and asks what changes when providers apply a country-by-country entry rule built from an additive representation of their portfolio payoffs. The exact portfolio response is the benchmark throughout. The comparison uses the same benchmark calibration and the unrestricted exact AB and CES equilibria from Section 4.4. For each provider, we make the 12 links closest to the deployment margin endogenous and hold its remaining links at the corresponding exact profile. This size keeps complete enumeration tractable: every up-

¹³This is an aggregate-taking analogue, not an implementation of the models in Antràs et al. (2017) or Arkolakis et al. (2023).

Table 4: Exact and Country-by-Country Portfolio Choice on Common Candidate Sets

Pricing	Rule	Exact / rule links	Changed links	Exact-response gain	Max. share change (pp)	Max. price-index change (%)
AB	Shapley	124 / 127	3	2.48×10^{-4}	1.04	0.43
AB	Chebyshev	124 / 126	4	9.61×10^{-4}	2.64	0.21
CES	Shapley	105 / 104	5	1.28×10^{-3}	5.28	1.03
CES	Chebyshev	105 / 104	5	1.28×10^{-3}	5.28	1.03

Notes: Changed links are provider-country entries on which the two fixed points differ. The exact-response gain sums, across providers, the profit gain from replacing the country-by-country response with the exact response at the fixed point. Share change is the largest within-market sum of absolute provider-share changes.

date evaluates all $2^{12} = 4,096$ portfolios and fully reprices the downstream equilibrium. The exact, Shapley, and Chebyshev response maps therefore use identical payoff tables without sampling error.

The Shapley rule is the additive anchor defined in Appendix B.2: it assigns each link its average marginal contribution over the complete restricted choice set. The Chebyshev rule chooses the additive payoff table with the smallest maximum approximation error. The two rules play complementary roles: Shapley is the attribution rule a practitioner would compute, while Chebyshev attains the smallest maximum error available to any additive table, so its errors cannot be attributed to a poorly fitted additive rule. Each rule then activates links with positive fitted values. Example 1 shows why this step can select the wrong links: an additive table reproduces the exact two-link choice only in the separable case $v_{ab} - v_a = v_b - v_{\emptyset}$, so when operating in Japan changes the marginal value of a Korean region, any single fitted value for that link misstates at least one portfolio comparison. For each pricing system and response rule, we run the five-provider map from the exact reference profile and from the two extreme restricted-choice-set profiles, using both forward and reverse provider orders. All 36 computations reach a fixed point, and each pricing-rule combination reaches the same fixed point along its six paths. The exact map returns the unrestricted reference profile in every case, which provides a consistency check on the exact conditional calculation.

The country-by-country rules change both network size and composition (Table 4). Under AB, Shapley adds three links relative to the exact profile, while Chebyshev adds two links and changes four link identities. Under CES, both rules remove one link in net but differ from exact choice in five provider-country links. They omit AWS deployments

Table 5: Market Effects of the CES Country-by-Country Rule

Market	Exact / rule local providers	Strategic HHI change	Provider-share change (pp)	CES price-index change (%)
Australia	4 / 3	-127.4	5.28	1.03
United States	4 / 3	-108.1	4.59	0.86
Sweden	4 / 3	-82.0	4.57	0.59
Hong Kong	2 / 3	-68.9	2.22	0.29
Norway	0 / 1	-66.5	4.27	0.29

Notes: Changes are country-by-country outcomes minus exact outcomes. The Shapley and Chebyshev rules reach the same CES fixed point. HHI uses the five strategic providers' market shares. Provider-share change is the sum of absolute share changes across the five providers.

in Australia, Sweden, and the United States and add Microsoft in Norway and Alibaba in Hong Kong. Because locations serve foreign markets, these five local changes move provider shares in all 42 markets; Table 5 reports the markets whose local provider sets change.

The preservation condition in Proposition B.2 is sufficient rather than necessary. No provider's decision margin Δ_f , the payoff gap between its best and second-best portfolios, exceeds twice its additive-approximation error γ_f in these four comparisons, and the largest $2\gamma_f/\Delta_f$ ratio ranges from 123 under CES Shapley to 6,760 under AB Chebyshev. The uniform residual bound is therefore too conservative to establish the calibrated choices. Realized gains from exact reoptimization are much smaller, yet the selected networks differ. Small optimization losses do not guarantee portfolio preservation when several links lie close to indifference. For CES, a check of the set of Chebyshev-optimal additive payoff tables (those attaining the minimax approximation error) classifies all 60 coefficient signs, so all five link differences persist for every table in that set.

4.7.1 Unrestricted country-by-country diagnostic

The conditional comparisons isolate representation error on common payoff tables. We also apply the country-by-country rule to each provider's full candidate set of 14–34 links. The exact reference profiles are the complete-information equilibria verified in Section 4.4; Online Appendix [OA.6](#) reports their full response paths and numerical diagnostics.

We next construct an unrestricted country-by-country comparison. For every provider response, random permutation paths estimate each link's Shapley value, with the full pric-

ing equilibrium recomputed at every prefix portfolio, in closed form under CES and via the numerical continuation and its error bound under AB. The number of sampled permutations increases adaptively from 128 to 1,024. Iterating the estimated response maps reaches a fixed point after three passes under both pricing systems.

The sampled maps select 110 links under AB and 79 under CES, against 124 and 105 at the exact equilibria, changing 16 and 26 provider-country links; six sign estimates under AB and three under CES remain unclassified. Online Appendix Table [OA.6.4](#) reports the full comparison, including the largest provider-level gains from exact complete-portfolio reoptimization at the sampled profiles.

Alibaba’s 14-link candidate set permits a complete check of whether sampling drives the unrestricted discrepancy. We enumerate all 2^{14} portfolios and compute its population Shapley coefficients after fully repricing each portfolio. At the reported sampled fixed points, the sampled and population Shapley signs agree for all 14 links under both pricing systems. At the exact-equilibrium rival profiles, however, the population Shapley rule selects 7 of Alibaba’s 13 exact-equilibrium links under AB and 6 of its 8 under CES. The difference for Alibaba therefore reflects the additive Shapley representation rather than Monte Carlo sign error.

The sampled fixed points are not exact portfolio equilibria. Four providers have profitable complete-portfolio deviations under AB pricing, and all five do under CES. The exact equilibrium predicts 19 exits and a 35.629 percent profit decline when AB is replaced by CES; the sampled map predicts 31 exits and a 33.686 percent decline. Because several Shapley signs remain unclassified and can affect later firm responses, we treat the sampled endpoints as diagnostics rather than population Shapley fixed points.

The three exercises have distinct roles. The unrestricted exact calculation verifies portfolio equilibrium on the full provider candidate sets. The unrestricted sampled map measures how an additive rule behaves at the same scale. The conditional comparisons enumerate every portfolio on common candidate sets, allowing exact and country-by-country response maps to be compared without sampling error.

4.8 Policy experiments on the exact benchmark

The first policy experiment captures the compliance costs that European data-protection and digital-sovereignty rules impose on operating local data centers (Ferracane et al., 2020;

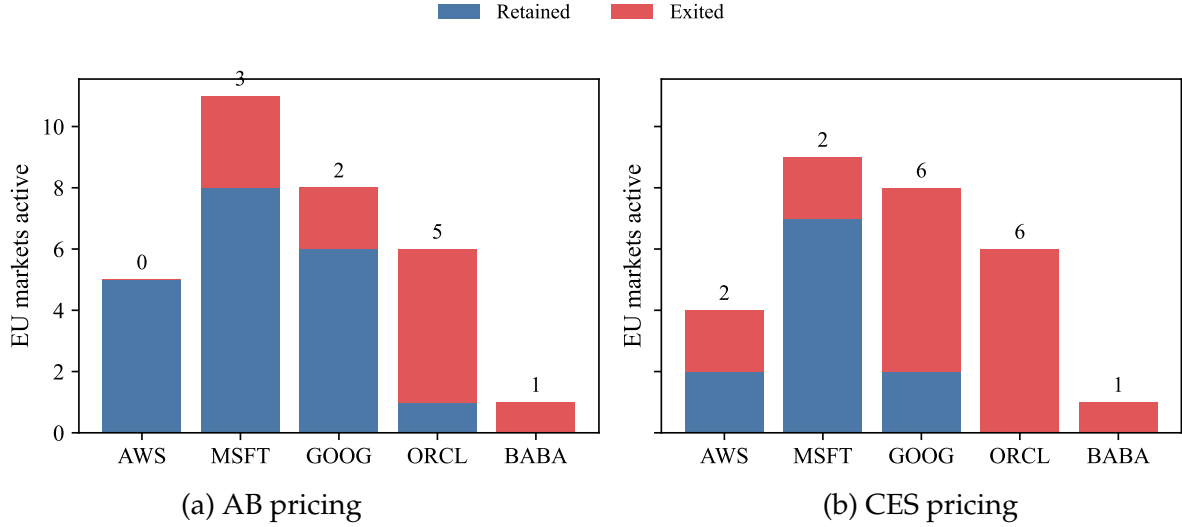


Figure 2: EU Compliance-Cost Increase at $\Delta\phi = 0.001$ on the Exact Benchmark

Notes: Within each pricing regime, blue bars show baseline EU markets retained under the shock and red segments show exited EU markets; labels report the number of exits by provider. Gross EU exits are 11 under AB and 17 under CES; the net changes in Table 6 also include offsetting non-EU entries, one under AB and four under CES.

Demirer et al., 2024; Manganelli and Schnurr, 2024): it raises the fixed cost of operating a location in each EU country and re-solves the exact portfolio equilibrium at every shock level under both pricing rules. The two pricing rules give different answers, and the ordering reverses as the shock grows: at a moderate increase the AB network sheds fewer locations than the CES network (-10 against -13), while at a large increase it sheds more (-26 against -20 ; Table 6). The reversal is compositional. At the moderate increase, endogenous markups keep more of the AB network’s near-marginal EU deployments profitable. Large increases push both networks toward complete local exit—at $\Delta\phi = 0.005$ the AB network has shed 28 of its 31 baseline EU deployments and the CES network 27 of 28—so the AB network loses more links simply because it operates more EU locations at baseline. Provider profit net of fixed costs rises under both rules as marginal locations exit and competition softens among the remaining providers, the equilibrium counterpart of the product exit and market concentration documented after the EU’s General Data Protection Regulation (Janssen et al., 2022; Johnson et al., 2023). The response saturates beyond $\Delta\phi = 0.010$, where no local EU deployment remains under either rule. Figure 2 shows which providers exit at the moderate shock: exits concentrate in providers with thin EU positions, with Oracle shedding five of its six EU locations under AB pricing and all six

Table 6: EU Compliance-Cost Ladder on the Exact Benchmark

$\Delta\phi$	Increase relative to median fixed cost	Active		Δn		$\Delta\pi$ (%)	
		AB	CES	AB	CES	AB	CES
0 (baseline)	0.0%	124	105	—	—	—	—
0.001	11.1%	114	92	-10	-13	+0.3	+2.1
0.005	55.4%	98	85	-26	-20	+4.1	+4.5
0.010	110.8%	95	84	-29	-21	+5.6	+5.4
0.030	332.5%	95	84	-29	-21	+5.6	+5.4

Notes: Each row reports the exact portfolio equilibrium under the benchmark calibration with the fixed-cost shock $\Delta\phi$ added for every provider in each EU market. Every AB equilibrium is search-certified with the bounded pricing residual and every CES equilibrium is fully certified; each rung is re-verified from the reverse provider order. Changes are measured relative to the corresponding baseline. Profit is aggregate provider profit net of fixed costs; the median market fixed cost is 0.009023.

under CES, while AWS’s EU network is unchanged under AB.

The shock adds the same amount to every provider’s fixed-cost parameter in each EU market and leaves the cost of serving EU users from locations outside the EU unchanged.¹⁴ The main channel is therefore relocation rather than exit from EU demand: a provider that closes an EU location keeps serving those users from sites elsewhere, giving up the accessibility advantage of local presence. EU markets keep cloud service but lose locally operating providers, and it is the largest providers that remain: after the moderate shock, the three largest providers operate 19 of the 20 remaining EU locations under AB pricing and all 11 under CES.¹⁵

The second experiment separates costs from access in China, where market access for foreign firms has long been granted only against costly conditions (Holmes et al., 2015) and digital services face the Great Firewall and app restrictions (Sun, 2024; Sun and Trefler, 2023). Three cost scenarios move China-related parameters in both directions: reducing China’s market fixed cost by 90 percent, adding a service-cost friction to China-related provision, and liberalizing China-related service costs. Each leaves the exact network unchanged apart from a single AWS relocation from Singapore to Ireland (Table 7): no

¹⁴European data rules also restrict cross-border transfers. That instrument raises the cost of serving EU users from locations outside the EU and pushes in the opposite direction, toward local presence, closing the relocation margin that the fixed-cost shock activates. The China experiment below studies its extreme form, a change in feasible market access.

¹⁵Every reported equilibrium is certified by the whole-portfolio search of Section 2.5; re-solving from the reverse provider order and one-link-stable adjustment on the same calibration reproduce the same profiles at every rung.

Table 7: China Cost and Access Scenarios on the Exact Benchmark

Scenario	Active		Changed links		Alibaba entries		$\Delta\pi$ (%)	
	AB	CES	AB	CES	AB	CES	AB	CES
Baseline	124	105	—	—	—	—	—	—
Fixed cost -90%	124	105	2	0	0	0	+0.8	+1.4
Service-cost friction ($\Delta\tau = +0.15$)	124	105	2	0	0	0	-0.1	0.0
Service-cost liberalization ($\tau = 1.15$)	124	105	2	0	0	0	+0.3	0.0
China closed to non-Chinese providers	124	105	4	4	0	1	-0.5	+0.6
Liberalization + Alibaba access	143	120	21	21	20	18	-5.1	-2.0

Notes: Each row re-solves the exact portfolio equilibrium under the benchmark calibration with the stated change to China-related parameters. Changed links counts firm–market cells that differ from the baseline profile, entries plus exits. Alibaba entries counts Alibaba’s added markets. Profit is aggregate provider profit net of fixed costs. Every AB equilibrium is search-certified with the bounded pricing residual and every CES equilibrium is fully certified.

provider enters or leaves China, and Alibaba’s portfolio does not move. The providers with feasible access to China already operate there at baseline, and cheaper or costlier operation changes none of their entry choices. Closing China to non-Chinese providers also leaves the total unchanged: AWS and Microsoft are forced out of China, and two offsetting entries elsewhere restore the count under each pricing rule. Profit changes in all of these scenarios are small under both pricing rules.

Broad entry occurs only when Alibaba’s candidate set expands to all markets. With liberalized service costs, Alibaba adds 20 markets, its rivals shed one link in net, and aggregate provider profit falls by 5.1 percent as competition intensifies. The added markets span Europe, the Americas, Africa, and the Asia–Pacific rather than clustering near China. The dichotomy is not specific to the pricing rule. Under CES the cost scenarios reproduce the baseline exactly, and the access expansion adds 15 links, with Alibaba entering 18 markets and rivals shedding three European links; the strategic AB response is larger, with more entry and a profit decline of 5.1 against 2.0 percent. Feasible access, not cost, is the binding margin.

5. CONCLUSION

This paper develops an exact framework for strategic portfolio equilibrium with endogenous downstream pricing. Contingent link cutoffs represent complete-portfolio responses

without restricting interaction signs, and an interaction matrix gives sufficient conditions for equilibrium uniqueness and convergence. When Jia's or AES's portfolio ordering is unavailable, a cardinality-conditioned branch-and-bound method supplies global responses. Under AB, a unique pricing continuation and explicit numerical-error accounting make these responses comparable to closed-form CES calculations.

In the cloud application, exact equilibrium contains 124 links under AB pricing and 105 under CES, and aggregate provider profit net of fixed costs is 35.6 percent lower under CES. A fixed-aggregate benchmark reproduces the strategic CES profile but selects four additional AB links and yields 2.0 percent lower fully repriced profit, locating the discrepancy in rivals' endogenous repricing. For unrestricted subset choice, additivity of operating value is the sharp condition for uniform country-by-country entry; in the calibrated comparisons the simpler rule selects a different international network even when the gain from exact reoptimization is small. The exact portfolio solution provides a quantitative test of when market-by-market entry rules preserve the international network and its response to pricing or policy changes.

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APPENDIX

Appendix organization. The equilibrium proofs are in Appendix A. Appendix B gives the formal modularity, decision-margin, and finite-noise approximation results summarized at the end of Section 2. Appendix C proves the cardinality-conditioned search bound, gives the CES and AB specializations, and adds node-count bounds and an NP-hardness result for the search problem. Appendix D proves AB pricing uniqueness; the online appendix gives the full interaction-bound calculations, exact share Jacobian, and portfolio-ordering check. Appendix E documents the cloud sample and calibration. The online appendix covers nested logit, gravity-based access, cloud data and calibration, the portfolio algorithm, and replication sources; Online Appendix [OA.1](#) maps these parts.

A. PROOFS FOR EXACT PORTFOLIO EQUILIBRIUM

A.1 Proof of Proposition 1

The state space $\mathcal{Q} = \prod_f \Delta(\mathcal{X}_f)$ is nonempty, compact, and convex. For each portfolio, $V_f(S_f; Q_{-f})$ is continuous in Q . Two distinct portfolios have payoff difference $V_f(S_f; Q_{-f}) - V_f(T_f; Q_{-f}) - \sigma_\varepsilon \sum_j w_{fj} (\mathbf{1}\{j \in S_f\} - \mathbf{1}\{j \in T_f\}) \varepsilon_{fj}$, whose stochastic loading is nonzero, and absolute continuity assigns zero probability to the associated affine indifference hyperplane. If $Q^k \rightarrow Q$, the best-response indicator therefore converges for almost every type. Dominated convergence implies $\mathcal{B}_f(Q_{-f}^k) \rightarrow \mathcal{B}_f(Q_{-f})$. The joint map $\mathcal{B} : \mathcal{Q} \rightarrow \mathcal{Q}$ is continuous, so Brouwer's theorem supplies a fixed point.

A.2 Proof of Theorem 1

Fix firm f , link j , and the other own shocks. When its deterministic portfolio values change from v_f to $v_f + h$, the difference-of-maxima formula in (2.6) gives

$$|\bar{\varepsilon}_{fj}(v_f + h) - \bar{\varepsilon}_{fj}(v_f)| \leq \frac{1}{\sigma_\varepsilon w_{fj}} \max_{\substack{S_f \ni j \\ T_f \not\ni j}} |h(S_f) - h(T_f)|.$$

Couple the two problems using the same private-cost realization. Link j can change status only when its shock lies between the two cutoffs. The conditional-density bound and a union over links therefore imply

$$d_{\text{TV}}(\mathcal{L}(b_f(v_f)), \mathcal{L}(b_f(v_f + h))) \leq \sigma_\varepsilon^{-1} \|h\|_{\partial f}.$$

Replace rival distributions one firm at a time. Decompose the signed measure $Q'_g - Q_g$ into its positive and negative parts and couple those parts. The definition in (2.10), followed by telescoping over rivals, gives (2.11).

Choose λ with $\rho(K) < \lambda < \sigma_\varepsilon$ and set $a = (I - K/\lambda)^{-1}\mathbf{1} \gg 0$. Then $Ka < \lambda a$. Equation (2.11) is a contraction in the complete metric $\max_f d_{\text{TV}}(Q_f, Q'_f)/a_f$, with modulus below $\lambda/\sigma_\varepsilon$. Banach's theorem proves uniqueness and global convergence.

Proposition A.1 (Perron minimality within firm reweightings). *Let K be the interaction matrix of Theorem 1. For any positive firm weights u , the comparison inequality (2.11) shows that simultaneous best response is Lipschitz in the weighted metric $\max_f d_{\text{TV}}(Q_f, Q'_f)/u_f$ with constant $(1/\sigma_\varepsilon) \max_f (Ku)_f/u_f$, hence a contraction whenever this constant is below one. No choice of weights certifies a constant below $\rho(K)/\sigma_\varepsilon$: $\inf_{u \gg 0} \max_f (Ku)_f/u_f = \rho(K)$, with the infimum attained by the right Perron vector when K is irreducible. Moreover, rescaling the firm-level metrics by positive constants c_f replaces K by DKD^{-1} with $D = \text{diag}(c_1, \dots, c_F)$ and leaves $\rho(K)$ unchanged.*

Proof. The first claim repeats the weighted-norm step of the preceding proof with weights u in place of the vector $a = (I - K/\lambda)^{-1}\mathbf{1}$. The variational identity is the Collatz–Wielandt characterization of the spectral radius of a nonnegative matrix (Horn and Johnson, 2013, Chapter 8), and irreducibility gives attainment at the right Perron vector. The rescaling claim follows from rewriting (2.11) in the rescaled distances $c_f d_{\text{TV}}(Q_f, Q'_f)$, and similar matrices share eigenvalues. $\square$

A.3 Exact small-noise uniqueness

The small-noise argument bounds the probability of near ties around the strict complete-information equilibrium.

For a deterministic value vector v_f and a private type, let $\Gamma_f(v_f, \varepsilon_f; \sigma_\varepsilon)$ be the gap between the largest and second-largest values in (2.4), with $\Gamma_f = +\infty$ when $|X_f| = 1$. Suppose

that, for every firm and $m > 0$, there are $\bar{t}_f(m) > 0$ and coefficients $a_f^{\text{loc}}(\sigma_\varepsilon, m) \rightarrow 0$ as $\sigma_\varepsilon \downarrow 0$ such that

$$\Pr\{\Gamma_f(v_f, \varepsilon_f; \sigma_\varepsilon) \leq t\} \leq a_f^{\text{loc}}(\sigma_\varepsilon, m)t, \quad 0 \leq t \leq \bar{t}_f(m), \quad (\text{A.1})$$

uniformly over value vectors with a portfolio S_f^* satisfying $v_f(S_f^*) - v_f(S_f) \geq m$ for every $S_f \neq S_f^*$.

Lemma A.1 (Primitive local boundary condition). *Suppose the link shocks within firm f are independent, their marginal densities satisfy $\bar{g}_{fj} \equiv \|g_{fj}\|_\infty < \infty$, and*

$$d_{fj}(x) = \text{ess sup}_{|z| \geq x} g_{fj}(z)$$

obeys

$$x d_{fj}(x) \rightarrow 0, \quad x \Pr\{|\varepsilon_{fj}| \geq x\} \rightarrow 0 \quad \text{as } x \rightarrow \infty. \quad (\text{A.2})$$

Then (A.1) holds. In particular, independent standard normal or logistic link shocks satisfy the condition.

Proof. Fix f and $m > 0$, and let S_f^* have deterministic value at least m above every other portfolio. The claim is immediate when $|\mathcal{X}_f| = 1$. Otherwise, after omitting links that are constant across feasible portfolios, write

$$W_f^\Sigma = \sum_{j \in \mathcal{A}_f} w_{fj}, \quad \delta_f(m) = \frac{m}{8W_f^\Sigma}, \quad \bar{t}_f(m) = \frac{m}{4}.$$

Fix a link j and condition on $\varepsilon_{f,-j}$. Let V_{fj}^{in} and V_{fj}^{out} be the best conditional values among portfolios that include and exclude j , as in Lemma 1, and define $\bar{\varepsilon}_{fj} = (V_{fj}^{\text{in}} - V_{fj}^{\text{out}})/(\sigma_\varepsilon w_{fj})$. On the event $E_{f,-j} = \{|\varepsilon_{fk}| < \delta_f(m)/\sigma_\varepsilon \text{ for every } k \neq j\}$, the difference between the other-shock components of any two conditional portfolio values is at most $m/8$. The group containing S_f^* therefore beats the other group by at least $7m/8$: $|V_{fj}^{\text{in}} - V_{fj}^{\text{out}}| \geq 7m/8$.

If the winner and runner-up differ in link j and their payoff gap is at most t , then ε_{fj} lies in one of the two intervals adjacent to $\bar{\varepsilon}_{fj}$, each of length $t/(\sigma_\varepsilon w_{fj})$. For $t \leq m/4$, every point in those intervals has absolute value at least $m/(2\sigma_\varepsilon w_{fj})$ on $E_{f,-j}$. Independence and

a union bound therefore give

$$\Pr\{\Gamma_f \leq t\} \leq \frac{2t}{\sigma_\varepsilon} \sum_{j \in \mathcal{A}_f} \frac{1}{w_{fj}} \left[d_{fj} \left(\frac{m}{2\sigma_\varepsilon w_{fj}} \right) + \bar{g}_{fj} \sum_{k \neq j} \Pr \left\{ |\varepsilon_{fk}| \geq \frac{\delta_f(m)}{\sigma_\varepsilon} \right\} \right].$$

Every distinct winner–runner-up pair differs in at least one nonconstant link, so the union over j covers the event. Define $a_f^{\text{loc}}(\sigma_\varepsilon, m)$ as the coefficient multiplying t on the right. Each density term divided by σ_ε converges to zero by the first condition in (A.2), and each tail probability divided by σ_ε converges to zero by the second. Thus $a_f^{\text{loc}}(\sigma_\varepsilon, m) \rightarrow 0$, proving (A.1). $\square$

Proposition 2 can assume (A.1) directly, allowing dependent within-firm shocks.

Proof of Proposition 2. For a mixed portfolio profile Q , define complete-information regret

$$\mathcal{R}(Q) = \max_f \left\{ \max_{S_f} V_f(S_f; Q_{-f}) - \sum_{S_f} Q_f(S_f) V_f(S_f; Q_{-f}) \right\}. \quad (\text{A.3})$$

The zero set of this continuous function is exactly the set of mixed Nash equilibria. That set is the singleton $Q^* = (\delta_{S_1^*}, \dots, \delta_{S_F^*})$. Compactness implies that for every neighborhood N of Q^* ,

$$\inf_{Q \notin N} \mathcal{R}(Q) > 0. \quad (\text{A.4})$$

Let $Z_f(S_f) = -\sum_{j \in S_f} w_{fj} \varepsilon_{fj}$ and $D_f^\varepsilon = \max_{S_f} Z_f(S_f) - \min_{S_f} Z_f(S_f) < \infty$ a.s. Because the reduced payoff table is finite, there is a finite constant $\bar{R}_f$ bounding the deterministic payoff difference between any two portfolios. A noisy best response has complete-information regret at most $\min\{\bar{R}_f, \sigma_\varepsilon D_f^\varepsilon\}$. At any BNE,

$$\mathcal{R}(Q^{\sigma_\varepsilon}) \leq \max_f \mathbb{E} \left[\min\{\bar{R}_f, \sigma_\varepsilon D_f^\varepsilon\} \right] \rightarrow 0 \quad (\text{A.5})$$

by bounded convergence. Equations (A.4) and (A.5) place every small-noise BNE in any prescribed neighborhood of Q^* .

Define the strict pure-profile gap and total interaction bound

$$\Delta_f = \min_{S_f \neq S_f^*} \{r_f(S^*) - r_f(S_f, S_{-f}^*)\} > 0, \quad B_f = \sum_{g \neq f} \beta_{fg},$$

where

$$\beta_{fg} = \max |r_f(S_f, S_g, R) - r_f(T_f, S_g, R) - r_f(S_f, T_g, R) + r_f(T_f, T_g, R)|$$

is the maximum absolute cross-firm cross-difference in firm f 's reduced payoff. Choose $r > 0$ so that, on $\bar{N}_r = \{Q : \max_g d_{\text{TV}}(Q_g, Q_g^*) \leq r\}$, the deterministic best portfolio remains S_f^* with gap at least $m_f = \Delta_f/2$, and $2rB_f \leq \bar{t}_f(m_f)$. Such an r exists because the payoff table is finite, S^* is strict, and r can be reduced without changing the deterministic gap conclusion.

Uniformly on $\bar{N}_r$, selection of another portfolio requires $\sigma_\varepsilon D_f^\varepsilon \geq m_f$. Hence

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \delta_{S_f^*}) \leq \Pr\{D_f^\varepsilon \geq m_f/\sigma_\varepsilon\} \rightarrow 0.$$

For sufficiently small noise, $\mathcal{B}$ maps $\bar{N}_r$ into itself. For $Q, Q' \in \bar{N}_r$, define the deterministic payoff perturbation $h_f(S_f) = V_f(S_f; Q'_{-f}) - V_f(S_f; Q_{-f})$. Replacing rival distributions one at a time and using the definition of β_{fg} yields

$$\text{osc}_{S_f} h_f \leq t_f(Q, Q') \equiv \sum_{g \neq f} \beta_{fg} d_{\text{TV}}(Q_g, Q'_g) \leq 2rB_f \leq \bar{t}_f(m_f).$$

Couple the two best responses using the same private type. If the selected portfolios differ, the winner–runner-up gap under Q is at most $\text{osc}_{S_f} h_f$: a larger gap could not be reversed by the perturbation. The coupling inequality and (A.1) therefore give

$$d_{\text{TV}}(\mathcal{B}_f(Q_{-f}), \mathcal{B}_f(Q'_{-f})) \leq \sum_{g \neq f} a_f^{\text{loc}}(\sigma_\varepsilon, m_f) \beta_{fg} d_{\text{TV}}(Q_g, Q'_g). \quad (\text{A.6})$$

The nonnegative matrix on the right converges entrywise to zero. For sufficiently small noise, its spectral radius is below one. The corresponding Perron–Frobenius weighted metric makes $\mathcal{B}$ a contraction on $\bar{N}_r$. Banach's theorem gives a unique fixed point there, while (A.5) excludes fixed points outside. The same argument establishes convergence to Q^* . $\square$

The full mixed-equilibrium condition gives global containment. A finite game can have a strict profile as its only pure equilibrium while also having a mixed equilibrium. Smooth

payoff perturbations can then generate a second small-noise branch that converges to the mixed equilibrium. Uniqueness in the full mixed space rules out this limit in (A.4).

Corollary A.1 (Two-tail uniqueness). *Under Theorem 1 and Proposition 2, there are positive constants σ_{low} and σ_{high} such that the exact portfolio BNE is unique whenever $\sigma_\varepsilon \in (0, \sigma_{\text{low}}) \cup (\sigma_{\text{high}}, \infty)$. Intermediate noise requires a direct spectral bound or another equilibrium argument.*

B. MODULAR APPROXIMATION AND FINITE-NOISE ERROR BOUNDS

This appendix gives the formal results behind the country-by-country representation summarized in Section 2.6. It first characterizes the additive boundary, then develops modular anchors, decision-margin conditions, and finite-noise equilibrium-error bounds. Throughout, the exact distributional best-response map $\mathcal{B}$ and the cross-firm interaction matrix K are those defined in Section 2. A modular residual, the error from approximating a complete portfolio-value table by an additive one, controls the error in a firm's portfolio response, while K determines how response errors propagate through equilibrium feedback across firms.

B.1 Link-by-link cutoffs and the modularity boundary

Let z index the policy or counterfactual environment, and fix a rival portfolio distribution Q_{-f} . We suppress z when no confusion arises. For this subsection, let the feasible family be the full power set $\mathcal{X}_f = 2^{\mathcal{A}_f}$. Write $W_f(S_f; Q_{-f}, z)$ for firm f 's interim operating value before deterministic and private link costs. For a deterministic link-cost vector ϕ_f , define

$$V_f^\phi(S_f; Q_{-f}, z) = W_f(S_f; Q_{-f}, z) - \sum_{j \in S_f} \phi_{fj} w_{fj}. \quad (\text{B.1})$$

Thus $V_f^\phi(\cdot; Q_{-f}, z)$ is the Section 2 interim value in environment z . The corresponding type payoff subtracts the private-cost term in (2.2).

Definition B.1 (Separable link-by-link cutoff representation). At (Q_{-f}, z, ϕ_f) , a separable link-by-link cutoff representation is a vector $c_f = (c_{fj})_{j \in \mathcal{A}_f}$ such that, outside a null tie set,

the exact best response is

$$b_f(\varepsilon_f; Q_{-f}, z) = \{j \in \mathcal{A}_f : \varepsilon_{fj} < c_{fj}\}. \quad (\text{B.2})$$

The cutoffs may depend on beliefs, policy, and deterministic costs, but not on the firm's other realized private costs.

Here “separable” describes the coordinatewise payoff representation, not the joint distribution of link choices. The coordinates of ε_f may be correlated, in which case the induced choices and portfolio distribution are also generally correlated.

Assumption B.1 (Rich within-firm cost support). For each firm, the joint law of ε_f is absolutely continuous and assigns positive probability to every nonempty open subset of $\mathbb{R}^{|\mathcal{A}_f|}$.

Theorem B.1 (Modularity is the exact separability boundary). Fix (f, Q_{-f}, z) , suppose $\mathcal{X}_f = 2^{\mathcal{A}_f}$, and impose Assumption B.1. The following are equivalent:

(i) $W_f(\cdot; Q_{-f}, z)$ is modular, so there are numbers a_f and $(v_{fj})_{j \in \mathcal{A}_f}$ such that

$$W_f(S_f; Q_{-f}, z) = a_f + \sum_{j \in S_f} v_{fj} \quad \text{for every } S_f \subseteq \mathcal{A}_f.$$

(ii) For every deterministic link-cost vector ϕ_f , the exact best response admits a separable link-by-link cutoff representation almost surely.

The result permits correlated private costs within a firm.

Proof. If the operating value is modular, the type payoff is an additive sum over links. The exact maximizer includes j precisely when

$$\varepsilon_{fj} < \frac{v_{fj} - \phi_{fj} w_{fj}}{\sigma_\varepsilon w_{fj}},$$

which proves sufficiency.

For necessity, suppose the value table is not modular. Then there are distinct links j, k and a portfolio $T \subseteq \mathcal{A}_f \setminus \{j, k\}$ for which the two-link second difference

$$\Delta_{jk} W_f(T) = W_f(T \cup \{j, k\}) - W_f(T \cup \{j\}) - W_f(T \cup \{k\}) + W_f(T)$$

is nonzero. Deterministic costs can make every link in T strictly active and every link outside $T \cup \{j, k\}$ strictly inactive on an open set of private costs. Conditional on that open set, the remaining choice is two-dimensional. If $\Delta_{jk}W_f(T) > 0$, an open region compares T with $T \cup \{j, k\}$ across a boundary in $\varepsilon_{fj} + \varepsilon_{fk}$. If $\Delta_{jk}W_f(T) < 0$, an open region compares $T \cup \{j\}$ with $T \cup \{k\}$ across a boundary in $\varepsilon_{fj} - \varepsilon_{fk}$. Neither oblique boundary can agree on an open set with the axis-aligned regions generated by (B.2). Assumption B.1 gives the disagreement positive probability, contradicting (ii). $\square$

Theorem B.1 shows precisely when the separable threshold model in Remark 1 is valid.

B.2 Modular anchors and typewise regret

For the approximation results, let $V_f(\cdot; Q_{-f}, z)$ denote an interim portfolio-value table net of deterministic link costs, such as V_f^ϕ in (B.1). A modular anchor approximates this value by a constant plus a sum of link-specific contributions:

$$M_f(S_f; Q_{-f}, z) = \alpha_f(Q_{-f}, z) + \sum_{j \in S_f} v_{fj}(Q_{-f}, z). \quad (\text{B.3})$$

Define its residual and sup-norm error by

$$R_f^M(S_f; Q_{-f}, z) = V_f(S_f; Q_{-f}, z) - M_f(S_f; Q_{-f}, z), \quad (\text{B.4})$$

$$\gamma_f^M(Q_{-f}, z) = \max_{S_f \subseteq \mathcal{A}_f} |R_f^M(S_f; Q_{-f}, z)|. \quad (\text{B.5})$$

The best-modular residual is

$$\gamma_f^*(Q_{-f}, z) = \inf_{\alpha, v} \max_{S_f \subseteq \mathcal{A}_f} \left| V_f(S_f; Q_{-f}, z) - \alpha - \sum_{j \in S_f} v_j \right|. \quad (\text{B.6})$$

The additive coefficients of a chosen anchor induce the link-by-link rule

$$\tilde{b}_f^M(\varepsilon_f; Q_{-f}, z) = \left\{ j : \varepsilon_{fj} < \frac{v_{fj}(Q_{-f}, z)}{\sigma_\varepsilon w_{fj}} \right\}. \quad (\text{B.7})$$

Lemma B.1 (Typewise modular-anchor regret). *For every $(Q_{-f}, z, \varepsilon_f)$, the payoff loss from*

using (B.7) rather than the exact portfolio response is at most

$$2\gamma_f^M(Q_{-f}, z). \quad (\text{B.8})$$

The bound does not depend directly on the number of feasible portfolios.

Proof. The anchor rule maximizes the modular objective. Replacing the true value by the anchor can lower the value assigned to the anchor maximizer by at most γ_f^M and raise the value assigned to an exact maximizer by at most the same amount. $\square$

Minimizing over all modular anchors yields $2\gamma_f^*$, the tightest regret bound available from this sup-norm argument at a fixed belief profile. Constructing an equilibrium response map, however, requires a particular anchor whose coefficients vary continuously with beliefs. The Shapley anchor provides such a map by averaging each link's marginal contribution over all orders in which the other links can be added:

$$\alpha_f^{\text{Sh}}(Q_{-f}, z) = V_f(\emptyset; Q_{-f}, z), \quad (\text{B.9})$$

$$v_{fj}^{\text{Sh}}(Q_{-f}, z) = \sum_{T \subseteq \mathcal{A}_f \setminus \{j\}} \frac{|T|!(|\mathcal{A}_f| - |T| - 1)!}{|\mathcal{A}_f|!} [V_f(T \cup \{j\}; Q_{-f}, z) - V_f(T; Q_{-f}, z)]. \quad (\text{B.10})$$

Because interim values are continuous, and typically multilinear, in rival portfolio distributions, the Shapley coefficients are continuous in Q_{-f} .

Proposition B.1 (Existence of a modular-anchor fixed point). *Suppose $\mathcal{X}_f = 2^{\mathcal{A}_f}$ for every firm, the selected anchor coefficients are continuous in rival portfolio distributions, and each within-firm type law is absolutely continuous. Then the distributional anchor-response map $\tilde{\mathcal{B}}^M$ induced by (B.7) has a fixed point. In particular, the result applies to the Shapley anchor.*

Proof. The product of finite portfolio simplices is compact and convex. Away from the finite union of threshold hyperplanes, the anchor choice is locally constant in the coefficients. Absolute continuity makes those hyperplanes null. Dominated convergence therefore makes $\tilde{\mathcal{B}}^M$ continuous, and Brouwer's theorem applies. $\square$

At any fixed point $\tilde{Q}^M(z)$ of the anchor-response map, the portfolio prescribed by firm f 's anchor rule loses at most $2\gamma_f^M(\tilde{Q}_{-f}^M(z), z)$ relative to its exact best response, for every type. Thus the fixed point is an approximate BNE in payoff terms.

B.3 Decision margins and complete-information preservation

A small payoff loss need not imply the same discrete choice. The relevant comparison is between the modular-approximation error and the exact objective's gap between its best and second-best portfolios. For a type with a unique exact response S_f^* , define

$$\Delta_f(\varepsilon_f; Q_{-f}, z) = \min_{T_f \neq S_f^*} \left\{ U_f(S_f^*; \varepsilon_f, Q_{-f}, z) - U_f(T_f; \varepsilon_f, Q_{-f}, z) \right\}, \quad (\text{B.11})$$

where

$$U_f(S_f; \varepsilon_f, Q_{-f}, z) = V_f(S_f; Q_{-f}, z) - \sigma_\varepsilon \sum_{j \in S_f} w_{fj} \varepsilon_{fj}.$$

Proposition B.2 (Portfolio-preservation condition). *If*

$$\Delta_f(\varepsilon_f; Q_{-f}, z) > 2\gamma_f^M(Q_{-f}, z), \quad (\text{B.12})$$

then the exact and modular-anchor responses select the same portfolio for that type and belief profile.

Proof. Let S_f^* maximize the exact type objective, and let $\tilde{U}_f^M$ denote the modular objective after subtracting the same private costs. Since the private-cost term is common to both objectives, $|U_f - \tilde{U}_f^M| \leq \gamma_f^M$ pointwise; hence for every $T_f \neq S_f^*$,

$$\tilde{U}_f^M(S_f^*) - \tilde{U}_f^M(T_f) \geq U_f(S_f^*) - U_f(T_f) - 2\gamma_f^M > 0.$$

Thus S_f^* is also the unique modular-anchor maximizer. $\square$

At the complete-information limit, the same comparison can be made directly at each pure rival portfolio. Write $\delta_{S_{-f}}$ for the degenerate rival distribution concentrated on S_{-f} .

Corollary B.1 (Complete-information equilibrium preservation). *Let S^* be a strict pure equilibrium of the exact complete-information game, and define its unilateral decision margins by*

$$\Delta_f^{CI}(S^*, z) = \min_{T_f \neq S_f^*} \left\{ V_f(S_f^*; \delta_{S_{-f}}, z) - V_f(T_f; \delta_{S_{-f}}, z) \right\}.$$

If $\Delta_f^{CI}(S^, z) > 2\gamma_f^M(\delta_{S_{-f}}, z)$ for every firm, then S^* is also a strict pure equilibrium of the modular-anchor game. If the same condition holds for the unique exact best response at every pure rival*

profile, the exact and modular pure best-response maps, and hence their pure equilibrium sets, coincide.

At finite noise, the next subsection controls the probability that the exact and modular-anchor responses disagree and then propagates it through strategic interaction.

B.4 From Response Disagreement to Equilibrium Distance

For a fixed policy state, let Q^* be the exact portfolio BNE and let $\tilde{Q}^M$ be a modular-anchor fixed point. Define the one-step error at the anchor beliefs by

$$e_f^M = d_{\text{TV}}\left(\mathcal{B}_f(\tilde{Q}_{-f}^M), \tilde{\mathcal{B}}_f^M(\tilde{Q}_{-f}^M)\right). \quad (\text{B.13})$$

The next results bound e_f^M by the probability that the exact and anchor problems disagree on common private-cost draws, first analytically and then by simulation.

Theorem B.2 (Equilibrium propagation of one-step portfolio error). *Suppose the conditions of Theorem 1 hold and $\sigma_\varepsilon > \rho(K)$. Let $A = K/\sigma_\varepsilon$. Then, componentwise,*

$$d_f \equiv d_{\text{TV}}(Q_f^*, \tilde{Q}_f^M) \leq \left[(I - A)^{-1} e^M\right]_f. \quad (\text{B.14})$$

The exact fixed point is unique. The bound applies to every anchor fixed point $\tilde{Q}^M$, whether or not the anchor-response map itself has a unique fixed point.

Proof. Insert the exact response to the anchor beliefs between the two fixed-point distributions. The exact-map Lipschitz bound in (2.11) and the definition of e_f^M give

$$d_f \leq e_f^M + \sum_{g \neq f} A_{fg} d_g.$$

Thus $d \leq e^M + Ad$. Since A is nonnegative and $\rho(A) < 1$, $(I - A)^{-1} = \sum_{m \geq 0} A^m$ exists and is nonnegative. Multiplication gives (B.14). Exact-equilibrium uniqueness follows from Theorem 1. $\square$

The resolvent $(I - A)^{-1} = I + A + A^2 + \dots$ is the strategic amplification factor: it adds the direct response error and every subsequent round of feedback through rivals.

Corollary B.2 (Probability bound). *Retain the conditional-density bounds from the exact model.*

Then

$$e_f^M \leq \delta_f^M \equiv \min \left\{ 1, \frac{4\gamma_f^M(\tilde{Q}_{-f}^M)}{\sigma_\varepsilon} \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} \right\}. \quad (\text{B.15})$$

Consequently, $d \leq (I - K/\sigma_\varepsilon)^{-1} \delta^M$.

Proof. Couple the exact and anchor choices using the same private-cost vector. If the portfolios differ, the modular objective loss from moving away from its exact anchor maximizer is at most $2\gamma_f^M$. Because that loss is the sum of nonnegative coordinate margins over changed links, at least one link lies within $2\gamma_f^M/(\sigma_\varepsilon w_{fj})$ of its anchor cutoff. The conditional-density bound assigns this interval probability at most $4\gamma_f^M \bar{g}_{fj}/(\sigma_\varepsilon w_{fj})$. A union bound over links and the coupling inequality give (B.15). $\square$

Proposition B.3 (Sample-based disagreement bound). *Fix the anchor beliefs $\tilde{Q}_{-f}^M$ independently of a simulation sample. For firm f , draw N_f independent private-cost vectors and solve the exact and anchor portfolio problems on each common draw. Let q_f^M be the probability that the two choices differ under this coupling, and let $\hat{q}_f^M$ be its sample analogue. For any $\alpha \in (0, 1)$, with probability at least $1 - \alpha$ simultaneously across firms,*

$$e_f^M \leq q_f^M \leq q_f^U \equiv \min \left\{ 1, \hat{q}_f^M + \sqrt{\frac{\log(F/\alpha)}{2N_f}} \right\}. \quad (\text{B.16})$$

On that event, $d \leq (I - K/\sigma_\varepsilon)^{-1} q^U$.

Proof. The coupling inequality gives $e_f^M \leq q_f^M$. Conditional on the fixed belief profile, each disagreement indicator is Bernoulli with mean q_f^M . A one-sided Hoeffding bound and a union bound over firms give (B.16). Apply Theorem B.2. $\square$

An exact one-sided binomial confidence bound can replace the Hoeffding term. The simulation draws used to construct q^U must be independent of those used to compute the fixed point; otherwise, sample splitting or an additional empirical-process argument is required.

Corollary B.3 (Exact modular nesting). *If the selected anchor has $\gamma_f^M(Q_{-f}, z) = 0$ for every firm and every relevant rival profile, then $\tilde{\mathcal{B}}^M = \mathcal{B}$. Exact and anchor fixed points coincide, and the equilibrium-distance bound is zero.*

Corollary B.4 (Counterfactual outcome error). *Consider policy states z_0, z_1 and a bounded outcome H_z on the joint portfolio space. Let*

$$r(z) = (I - K(z)/\sigma_\varepsilon)^{-1} q^U(z)$$

be a valid equilibrium-distance bound at each endpoint, and define $\text{osc}(H_z) = \max_S H_z(S) - \min_S H_z(S)$. Then the absolute error in the anchor model's predicted counterfactual change is at most

$$\text{osc}(H_{z_1}) \sum_f r_f(z_1) + \text{osc}(H_{z_0}) \sum_f r_f(z_0). \quad (\text{B.17})$$

Proof. For product distributions, joint total variation is at most the sum of firm-level total variations, and expectations of a bounded outcome differ by at most its oscillation times total variation. Apply both facts at each policy endpoint and add. $\square$

Scope and use. The exact-equilibrium results in Section 2 permit any finite feasible portfolio family. The coordinatewise link-by-link characterization and anchor rule use the full power set. With capacity, hierarchy, or other portfolio constraints, an additive score instead defines a constrained modular optimizer. Computation and approximation bounds must then be formulated for that constrained optimizer.

The Shapley anchor provides a coefficient map that is continuous in beliefs and therefore supports the fixed-point result. A Chebyshev-optimal anchor minimizes the largest absolute approximation error, but applying the general existence result also requires a continuous selection among optimal coefficients. In a finite complete-information application, a deterministic Chebyshev response rule can instead be iterated and its fixed point checked directly. We use that comparison to assess robustness across anchors.

At the complete-information limit $\sigma_\varepsilon = 0$, deterministic exact and anchor responses are compared directly. For small positive noise, Proposition 2 supplies exact-equilibrium uniqueness under its stated conditions.

C. CARDINALITY-CONDITIONED PORTFOLIO SEARCH

This appendix proves Proposition 3 and states the conditions used by the CES and AB implementations.

C.1 Validity

Fix a branch-and-bound node (S_f, U_f) and a feasible completion $T_f = S_f \cup H$, where $H \subseteq U_f$ and $|H| = k$. For every rival state r and statistic ℓ ,

$$\begin{aligned} z_{r\ell}(T_f) &= z_{r\ell}(S_f) + \sum_{j \in H} a_{r\ell j} \\ &\leq z_{r\ell}(S_f) + \sum_{h=1}^k a_{r\ell, (h)}^\downarrow(U_f) = \bar{z}_{r\ell}(S_f, U_f, k). \end{aligned}$$

The inequality holds component by component. Coordinatewise monotonicity of the continuation envelope gives

$$\pi_{fr}(T_f) \leq \bar{\pi}_{fr}(z_r(T_f)) \leq \bar{\pi}_{fr}(\bar{z}_r(S_f, U_f, k)).$$

Since any k -element subset of U_f costs at least the sum of the k smallest link costs, the total cost of T_f satisfies $\sum_{j \in T_f} c_{fj} \geq \underline{C}_f(S_f, U_f, k)$. Multiplying the state inequalities by $q_r \geq 0$, summing over r , and subtracting the cost lower bound proves that the expression inside braces in (2.15) dominates the payoff of T_f . Maximizing over k proves Proposition 3.

If the attainable completion sizes at a node are known, the maximization over k can be restricted to them. Negative link costs also require no separate treatment: the order statistic $\sum_{h=1}^k c_{f, (h)}^\uparrow$ is a lower bound for every k -link completion whether costs are positive or negative. A preprocessing rule that instead fixes every negative-cost link into the portfolio would need additional closure and monotonicity conditions; the implementation does not use that rule.

C.2 Pricing specializations

Corollary C.1 (Constant-markup CES). *Suppose $\sigma > 1$, $\theta > 0$, and the portfolio enters each market through nonnegative additive service and accessibility statistics. Suppose accessibility has a nonnegative loading. Holding a rival-portfolio state fixed, the closed-form CES variable profit satisfies the monotonicity conditions of Proposition 3.*

Proof. Write the service statistic as $H_{nf}(T_f) = H_{nf}^0 + \sum_{j \in T_f} h_{fjn}$ with $h_{fjn} \geq 0$. The Fréchet service cost is proportional to $H_{nf}^{-1/\theta}$, so own competitiveness contains the increasing term

$H_{nf}^{(\sigma-1)/\theta}$. The accessibility log-sum is increasing in its nonnegative additive weights, and a nonnegative loading preserves that order. With rival price power fixed, the constant-markup CES share is increasing in own competitiveness, and variable profit is $R_n s_{nf}/\sigma$. Applying the componentwise top- k envelopes therefore gives a valid continuation bound. $\square$

Corollary C.2 (Atkeson–Burstein pricing). *Suppose $\sigma > \eta > 1$ and the primitive service and accessibility conditions of Corollary C.1 hold. If $\bar{s}_{nfr}(z_r)$ is a valid coordinatewise increasing upper bound on the AB equilibrium share in rival state r , then*

$$\bar{\pi}_{fr}(z_r) = \sum_n R_n \left[\frac{\bar{s}_{nfr}(z_r)}{\sigma} + \left(\frac{1}{\eta} - \frac{1}{\sigma} \right) \bar{s}_{nfr}(z_r)^2 \right]$$

satisfies Proposition 3.

Proof. Under AB pricing, variable profit in market n is $R_n g(s_{nf})$ with g the bracketed function in the corollary display; because $\sigma > \eta > 1$, $g'(s) = 1/\sigma + 2(1/\eta - 1/\sigma)s > 0$ on $[0, 1]$. Substituting any valid share upper bound therefore gives a variable-profit upper bound. The cloud implementation obtains $\bar{s}_{nfr}$ from the markup interval $\sigma/(\sigma-1) \leq \mu \leq \eta/(\eta-1)$ and the AB equilibrium-share envelope documented in Online Appendix [OA.6](#). $\square$

These corollaries distinguish the mathematical node bound from the payoff evaluation used at complete portfolios. CES prices and shares have a closed-form continuation in the implementation. AB leaves are evaluated by numerical fixed point iteration. Exhausting the discrete tree in the AB case therefore proves global optimality of the portfolio response conditional on that numerical continuation.

C.3 Relation to the maximal-portfolio envelope

For comparison, construct a pure maximal-portfolio bound using the same continuation envelopes, and write $a_{rj} = (a_{r1j}, \dots, a_{rLj})$:

$$B_f^{\max}(S_f, U_f) = \sum_r q_r \bar{\pi}_{fr} \left(z_r(S_f) + \sum_{j \in U_f} a_{rj} \right) - \sum_{j \in S_f} c_{fj} - \sum_{j \in U_f} \min\{c_{fj}, 0\}. \quad (\text{C.1})$$

Corollary C.3 (Envelope dominance). *When (2.15) and (C.1) use the same componentwise continuation envelopes, $B_f^{\text{card}}(S_f, U_f) \leq B_f^{\text{max}}(S_f, U_f)$.*

Proof. For every k , each top- k statistic is no larger than the corresponding sum over all undecided links. Moreover $\sum_{h=1}^k c_{f,(h)}^\uparrow(U_f) \geq \sum_{j \in U_f} \min\{c_{fj}, 0\}$. Monotonicity of the continuation envelope gives the result for each k , and maximizing over k preserves it. $\square$

The dominance statement compares only the two bounds built from the same market envelopes. The baseline implementation also contains specification-specific caps that can occasionally be tighter. The reported algorithm therefore uses $\min\{B_f^{\text{max}}, B_f^{\text{card}}\}$ rather than discarding the baseline bound.

C.4 Node-count bounds

Fix a firm, a private-cost type, and a target cardinality k , and write $\tilde{V}(T_f)$ for the search objective (2.13) over feasible k -link portfolios and $\tilde{V}^*$ for its maximum. In this subsection $n = |\mathcal{A}_f|$ denotes the number of candidate links, the symbols ω , Γ , and γ are local, and maxima and minima over $|T_f| = k$ range over feasible k -link portfolios. For $\epsilon \geq 0$, let N_ϵ count the feasible k -link portfolios with $\tilde{V}(T_f) \geq \tilde{V}^* - \epsilon$. For a node (S_f, U_f) , let $C_{f,k}(S_f, U_f)$ denote its feasible cardinality- k completions, the members of $C_f(S_f, U_f)$ with exactly k links.

Proposition C.1 (Relaxation-gap node bound). *Suppose the search discards any node with an empty cardinality- k completion set, that every node bound at a node with a nonempty cardinality- k completion set satisfies*

$$\max_{T_f \in C_{f,k}(S_f, U_f)} \tilde{V}(T_f) \leq B_f(S_f, U_f) \leq \max_{T_f \in C_{f,k}(S_f, U_f)} \tilde{V}(T_f) + \Gamma$$

for some $\Gamma \geq 0$, that a node is expanded only when its bound strictly exceeds the best payoff already found (all comparisons in exact arithmetic; numerical tolerances are treated separately), and that the search starts from a feasible portfolio with payoff at least $\tilde{V}^ - \omega$. Then the search expands at most $(n + 1)N_{\omega+\Gamma}$ nodes. If every expanded node generates at most two children, the total number of nodes whose bounds are evaluated is at most $2(n + 1)N_{\omega+\Gamma} + 1$.*

Proof. The best payoff already found never falls below $\tilde{V}^* - \omega$. For an expanded node, $\max_{T_f \in C_{f,k}(S_f, U_f)} \tilde{V}(T_f) \geq B_f(S_f, U_f) - \Gamma > \tilde{V}^* - (\omega + \Gamma)$: the node admits an $(\omega + \Gamma)$ -near-

optimal cardinality- k completion. Assign each expanded node to one of these completions. Because each branching step includes or excludes a single undecided link, at most one child of any node keeps a fixed portfolio in its completion set, so the nodes whose completion sets contain a given portfolio form a single root-to-leaf path deciding one link per step. The path contains at most $n + 1$ nodes, so each near-optimal portfolio is assigned at most $n + 1$ expanded nodes. Every evaluated node is the root or a child of an expanded node, which gives the second count. $\square$

The bound isolates the two objects that govern search effort: the relaxation gap of the node bound and the number of near-optimal portfolios. The worst case remains exponential because $N_{\omega+\Gamma}$ can be exponential; tractability is a property of instances, not of the algorithm.

Corollary C.4 (Near-modular payoffs). *Suppose $\tilde{V}(T_f) = M(T_f) + R(T_f)$ with $M(T_f) = c + \sum_{j \in T_f} a_j$ modular, and let $\gamma = \max_{|T_f|=k} R(T_f) - \min_{|T_f|=k} R(T_f)$ be the range of the residual. Start the search from a feasible k -link portfolio S^M that maximizes M over feasible k -link portfolios and use the node bound $B_f(S_f, U_f) = \max_{T_f \in C_{f,k}(S_f, U_f)} M(T_f) + \max_{|T_f|=k} R(T_f)$. Then $\Gamma \leq \gamma$ and $\omega \leq \gamma$, so the search expands at most $(n + 1)N_{2\gamma}$ nodes. If in addition the decision margin Δ , the payoff gap between the best and second-best feasible k -link portfolios, exceeds γ , then S^M is the unique optimum and the search expands at most $n + 1$ nodes. In the cardinality-only feasibility class used by the solver, the modular maximizations in the bound and the starting portfolio are obtained by sorting.*

Proof. The stated bound overstates a node's best completion by at most the range of the residual, so $\Gamma \leq \gamma$. For the starting portfolio, with T^* optimal and both S^M and T^* feasible k -link portfolios, $M(S^M) \geq M(T^*)$, so $\tilde{V}(S^M) \geq M(T^*) + R(S^M) \geq \tilde{V}^* - [R(T^*) - R(S^M)] \geq \tilde{V}^* - \gamma$ and $\omega \leq \gamma$; Proposition C.1 gives the first claim. If $\Delta > \gamma$, every γ -near-optimal portfolio is the optimum itself, so $S^M = T^*$ and $\omega = 0$; portfolios within $\omega + \Gamma \leq \gamma < \Delta$ of the optimum reduce to T^* , and Proposition C.1 applies with $N_{\omega+\Gamma} = 1$. $\square$

Corollary C.5 (Ambiguous band). *In the setting of Corollary C.4, suppose in addition that every k -link subset of $\mathcal{A}_f$ is feasible with $1 \leq k \leq n - 1$, and order the modular scores $a_{(1)} \geq \dots \geq a_{(n)}$. Every optimal portfolio contains link i whenever $a_i - a_{(k+1)} > \gamma$ and excludes link j whenever $a_{(k)} - a_j > \gamma$. Exact search therefore reduces to the links whose scores lie within γ of the cutoff pair*

$(a_{(k)}, a_{(k+1)})$; if b links remain ambiguous and r of the k slots are not filled by forced links, at most $\binom{b}{r} \leq 2^b$ configurations of the band need to be compared.

Proof. Suppose $a_i - a_{(k+1)} > \gamma$ and an optimal T_f excludes i . The top- k score set contains i , so the k elements of T_f include some j with $a_j \leq a_{(k+1)}$. The swapped set $T_f \setminus \{j\} \cup \{i\}$ is again a k -link subset and hence feasible, and the swap changes the payoff by $(a_i - a_j) + [R(T_f \setminus \{j\} \cup \{i\}) - R(T_f)] \geq (a_i - a_{(k+1)}) - \gamma > 0$, contradicting optimality. The exclusion claim is symmetric. $\square$

Two remarks connect these bounds to the paper’s other objects. First, an additive payoff table with maximum error γ_f (Section 2.6) has a residual with range at most $2\gamma_f$ over any set of portfolios; since every conclusion of Corollaries C.4 and C.5 only weakens as γ grows, they apply with the range replaced by $2\gamma_f$. Second, the diagnostics of Section 4.7 use unrestricted best-versus-second-best margins—largest $2\gamma_f/\Delta_f$ ratios range from 123 (CES Shapley) to 6,760 (AB Chebyshev) across the four comparisons—and therefore do not directly test the fixed-cardinality condition of Corollary C.4; they show that the available additive approximations are too coarse to certify the unrestricted exact choices. In the calibrated searches, low node counts coincide with the tightness of the cardinality bound of Proposition 3: the reported searches discard nearly the entire tree against the best payoff found (Online Appendix OA.6), the behavior Proposition C.1 associates with a tight relaxation and a sparse near-optimal set. Failure of a sufficient condition leaves room for the observed disagreement between country-by-country and exact choices; it does not characterize it.

C.5 Hardness of the cardinality-conditioned response

Tight node bounds are worth developing because no generally fast rule is available.

Proposition C.2 (NP-hardness). *Deciding whether some cardinality- k portfolio attains payoff at least a given rational threshold is NP-hard, already for one firm best responding to a single fixed rival portfolio under the constant-markup continuation with fixed expenditure, zero fixed costs, and rational demand and trade-cost parameters.*

Proof. Reduce from maximum coverage, which is NP-hard because set cover (Karp, 1972) is its special case with target $t = m$: given sets $F_1, \dots, F_n$ over elements $1, \dots, m$, an integer

$k \leq n$, and an integer target t , decide whether some k sets cover at least t elements. The at-most- k and exactly- k versions are equivalent because a solution using fewer than k sets can be padded without reducing coverage. Build one market per element with expenditure $R_e = 1$ and one candidate location per set, and take $\theta = 1$, $\sigma = 2$, unit wages, and unit demand shifters, so the constant markup is $\sigma/(\sigma - 1) = 2$. Give the optimizing firm productivity $a_f = 2A$ with $A = 8m(k + 1)$ and trade costs $\tau_{ie} = 1$ if $e \in F_i$ and $\tau_{ie} = A^2$ otherwise; every trade cost is rational and at least one. Fix the rival at one mandatory location with productivity 2 and unit trade costs: its constant-markup price is one, so its price power is exactly one in every market. With the Fréchet normalization absorbed into productivity, the optimizing firm's delivered cost and price power reduce to

$$c_e(T) = \frac{1}{a_f} \left(\sum_{i \in T} \tau_{ie}^{-1} \right)^{-1}, \quad z_e(T) = p_e(T)^{1-\sigma} = \frac{a_f}{2} \sum_{i \in T} \tau_{ie}^{-1},$$

so each location contributes $a_f/(2\tau_{ie})$ to price power— A for $e \in F_i$ and $1/A$ otherwise—and the firm's share is $z_e/(z_e + 1)$, with payoff $\sum_e z_e/(z_e + 1)$ divided by σ . If T covers e , the market's share contribution is at least $1 - 1/(A + 1)$; if not, it is at most k/A . Summing over markets, σ times the payoff differs from the number of covered elements by at most $m/(A + 1) + mk/A \leq m(k + 1)/A = 1/8$. Because coverage counts are integers, some k sets cover at least t elements if and only if some cardinality- k portfolio attains payoff at least $(t - 1/4)/\sigma$. All parameters are rationals whose digit counts are polynomial in the instance size, and each payoff evaluates exactly in polynomial time, so the reduction is polynomial. $\square$

The construction uses only nonnegative access contributions and a monotone objective: hardness requires neither signed substitution effects nor strategic rivals, and adding either feature preserves it because the constructed instances embed in the larger model. Variable markups are not the source of difficulty either: hardness arises already in the simplest constant-markup case.

C.6 Complexity and implementation checks

Let $M = |U_f|$ and let D be the number of state-statistic pairs. Constructing all top- k sums and the cost order statistics takes $O((D + 1)M \log M)$ operations, followed by one

continuation-envelope evaluation for each k . This is polynomial work per visited node. The branch-and-bound tree can still contain $2^{|\mathcal{A}_f|}$ leaves in the worst case.

The validity proof above is separate from the numerical implementation checks. Eighteen such checks pass. In synthetic five-link CES and AB economies, we enumerate all 3^5 branch-node states and verify that the cardinality bound dominates every feasible completion at every node. The calibrated verification then establishes the 14–34-link responses. These checks validate the implementation but do not constitute a directed-rounding proof for floating-point arithmetic.

D. AB PRICING AND PORTFOLIO-INTERACTION PROOFS

This appendix proves Proposition 4; the AB interaction-bound derivations are in Online Appendix [OA.3](#) and the entry-game contraction in Theorem 1.

D.1 Proof of Proposition 4

Fix an activation profile. Pricing separates across markets, so consider one market and suppress n . An empty market has no pricing variables. A market with one active firm has $s_f = 1$ and price $p_f = \mu(1)c_f$. Now suppose at least two firms are active. Let $\psi_f > 0$ denote firm f 's fixed quality shifter in this market and define

$$y_f = \psi_f p_f^{1-\sigma}, \quad D_f = \sum_{g \neq f} y_g, \quad s_f = \frac{y_f}{y_f + D_f}.$$

For fixed $D_f > 0$, the markup rule gives $y_f = \xi_f \mu(s_f)^{1-\sigma}$, where $\xi_f = \psi_f c_f^{1-\sigma} > 0$.

This scalar equation has one solution. To see this, define

$$G_f(y; D_f) = \log y - \log \xi_f - (1 - \sigma) \log \mu \left(\frac{y}{y + D_f} \right).$$

Its derivative with respect to $\log y$ is

$$\frac{\partial G_f}{\partial \log y} = 1 + s_f(1 - s_f)\beta_f > 0, \quad \beta_f = \frac{(\sigma - 1)(\sigma - \eta)}{\sigma\eta} \mu(s_f) > 0.$$

Moreover, $G_f(y; D_f)$ tends to $-\infty$ as $y \downarrow 0$ and to $+\infty$ as $y \uparrow \infty$, because $\mu(s)$ is bounded.

The solution $y_f = \mathcal{Y}_f(D_f; c_f, \psi_f)$ is therefore unique and smooth.

Logarithmic differentiation of the scalar response gives $d \log y_f = -\beta_f ds_f$ and $ds_f = s_f(1 - s_f)(d \log y_f - d \log D_f)$; solving these two equations yields $\partial \log y_f / \partial \log D_f = \tilde{\kappa}_f$, which is (3.5). Set $x_f = \log y_f$; the solved scalar response makes the update Φ of Section 3.2 well defined. Its Jacobian has entries $\partial \Phi_f / \partial x_g = \tilde{\kappa}_f y_g / D_f$ for $g \neq f$ and zero diagonal; this is the matrix in (3.6), and its row sum is $\tilde{\kappa}_f$. If B denotes the update in levels, then at a fixed point

$$J_\Phi(x^*) = \text{diag}(y^*)^{-1} J_B(y^*) \text{diag}(y^*).$$

The log- and level-coordinate Jacobians therefore have the same eigenvalues.

The markup bound $\mu(s) \leq \eta/(\eta - 1)$ gives $\beta_f \leq \bar{\beta}$; since $s_f(1 - s_f) \leq 1/4$ and $q \mapsto q/(1 + q)$ is increasing, every row sum of $J_\Phi(x)$ is at most $\bar{\lambda}_{AB} < 1$ for every $x \in \mathbb{R}^m$, which is (3.7). Thus Φ is a contraction of the complete space $(\mathbb{R}^m, \|\cdot\|_\infty)$. Banach's theorem gives a unique fixed point in each market and hence a unique pricing equilibrium for the activation profile.

Finally, write $F(x, c) = x - \Phi(x, c)$, so $F(x^*(c), c) = 0$ and $F_x = I - \Lambda(x^*)$. Since $\|\Lambda(x^*)\|_\infty \leq \bar{\lambda}_{AB} < 1$, the Neumann series gives invertibility with $\|(I - \Lambda(x^*))^{-1}\|_\infty \leq (1 - \bar{\lambda}_{AB})^{-1}$, and the implicit-function theorem yields a continuously differentiable $x^*(c)$, hence continuously differentiable prices, for strictly positive active-firm costs. This proves Proposition 4. $\square$

E. CLOUD APPLICATION: SAMPLE AND CALIBRATION

This appendix documents coding by provider and country, the shared candidate universe, and the gravity inputs used in the cloud application. The exact benchmark and the supplementary calibration use the same 134 candidate links but fit separate fixed-cost vectors. Online Appendices [OA.4](#) and [OA.5.3](#) document the supplementary calibration and the numerical procedure.

The unit of observation is a provider-country pair. A pair is active when the April 7, 2026 provider-region snapshot records an operational region for that provider and country. Announced status is recorded separately for future regions. Online Appendix [OA.5](#) lists the canonical official pages and the replication archive maps each input row to its provider source. Datacenter Map enters only the separate construction of facility-based country centroids used for distance.

Table 8: Candidate-Set Accounting across the Two Calibrations

Calibration and profile	Active links	Operational	Announced
Benchmark initial profile	128	125	3
Benchmark exact AB equilibrium	124	122	2
Supplementary initial profile	124	122	2
Supplementary one-link profile	121	119	2

Notes: The common candidate universe consists of 129 operational deployments and five announced links. The benchmark initial and exact AB profiles match 127 and 125 of the 134 observed candidate-link indicators, respectively. The supplementary initial and one-link profiles match 125 and 122 indicators, respectively. The two calibrations use different fitted fixed-cost vectors.

Candidate sets consist of operational deployments and announced links. Both calibrations therefore condition on observed and announced deployments as the candidate universe.

Table 8 separates the benchmark profiles from the supplementary-calibration profiles. The provider-country panel has 210 pairs, of which 134 are candidate links: 129 operational deployments and five announced links. The benchmark recalibration produces a 128-link initial profile, and unrestricted response iteration selects the 124-link exact AB equilibrium reported in Section 4.5. The supplementary calibration produces a different 124-link initial profile; fully repriced one-link improvement selects a 121-link AB profile.

Both calibrations and their AB–CES comparisons condition on this candidate universe.

The demand proxy uses 2023 World Bank WDI GDP and internet use, with an IMF GDP estimate and an internet-use imputation for Taiwan. The distance term is great-circle distance between country facility centroids, with capital-city fallbacks, and the site cost scale uses human-capital-adjusted Penn World Table 11.0 compensation per worker for 2023. The fixed-cost prior uses the World Bank’s July 2024 income-classification vintage. Online Appendix [OA.5](#) reports the transformations and data provenance in one table.

Given these inputs, we choose (ϕ_i, g_f) to minimize squared market- and provider-entry-count errors, normalizing $g_{\text{AWS}} = 1$ (Section 4.3). Online Appendix [OA.5.3](#) gives the update equations, bounds, stopping rules, and tie-breaking.

Online Appendix

Strategic Multinational Location Choice with Endogenous Pricing

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OA.1. GUIDE TO THE ONLINE APPENDIX

This appendix collects demand extensions, numerical methods, and source files. Section [OA.2](#) gives sufficient conditions for extending the pricing argument to nested-logit demand. Section [OA.3](#) gives the full AB portfolio-interaction derivations, exact share Jacobian, and ordering check. Section [OA.4](#) defines the cloud accessibility and pricing blocks, distinguishes the two calibrated specifications used in the numerical exercises, and documents the fully repriced one-link procedure of the supplementary calibration. Section [OA.5](#) reports the inputs and calibration procedure. Section [OA.6](#) describes the unrestricted whole-portfolio search, the exact-equilibrium verification, the fixed-aggregate portfolio benchmark, and the unrestricted sampled-Shapley extension.

OA.2. NESTED LOGIT WITH BERTRAND PRICING

This section states sufficient conditions for replacing the CES demand block with nested-logit demand. The share derivatives are explicit. Smooth price and payoff responses require additional regularity of the pricing first-order conditions.

OA.2.1 Share derivatives

Fix a market n . Let s_{nf} denote firm f 's market share, $s_{nf|k(f)}$ its within-nest share, and S_{nk} the total share of nest k . Thus $s_{nf} = S_{n,k(f)}s_{nf|k(f)}$. The price coefficient is $\gamma_n > 0$, and the nesting parameter is $\lambda_{n,k} \in (0, 1]$.

Differentiating the nested-logit share system with respect to firm f 's own price gives

$$\frac{\partial s_{nf}}{\partial p_{nf}} = -\gamma_n s_{nf} \left[\lambda_{n,k(f)}^{-1} (1 - s_{nf|k(f)}) + s_{nf|k(f)} - s_{nf} \right]. \quad (\text{OA.2.1})$$

For a rival $g \neq f$ in the same nest, $k(g) = k(f)$,

$$\frac{\partial s_{nf}}{\partial p_{ng}} = \gamma_n s_{nf} s_{ng|k(f)} \left[\lambda_{n,k(f)}^{-1} - 1 + S_{n,k(f)} \right], \quad (\text{OA.2.2})$$

and for a rival in a different nest,

$$\frac{\partial s_{nf}}{\partial p_{ng}} = \gamma_n s_{nf} s_{ng}. \quad (\text{OA.2.3})$$

These formulas reduce to the standard logit cross-price derivative when $\lambda_{n,k} = 1$. They are uniformly bounded when $\gamma_n \leq \bar{\gamma} < \infty$ and $\lambda_{n,k} \geq \underline{\lambda} > 0$.

Lemma OA.2.1 (Bounded nested-logit share derivatives). *Suppose $\gamma_n \leq \bar{\gamma} < \infty$ and $\lambda_{n,k} \geq \underline{\lambda} > 0$ for all markets and nests. On a compact interior region of the nested-logit share space, there exists a finite constant L_s such that*

$$\max_{n,f,g} \left| \frac{\partial s_{nf}}{\partial p_{ng}} \right| \leq L_s.$$

Proof. The formulas above express each derivative as γ_n times bounded shares and a term bounded by the inverse nesting parameter. Since $\gamma_n \leq \bar{\gamma}$ and $\lambda_{n,k}^{-1} \leq \underline{\lambda}^{-1}$, the derivatives are uniformly bounded. $\square$

OA.2.2 Conditions on Bertrand pricing

For a single-product firm, the Bertrand first-order condition is

$$s_{nf} + (p_{nf} - c_{nf}) \frac{\partial s_{nf}}{\partial p_{nf}} = 0.$$

Substituting (OA.2.1) yields the nested-logit markup equation

$$p_{nf} - c_{nf} = \frac{1}{\gamma_n \left[\lambda_{n,k(f)}^{-1} (1 - s_{nf|k(f)}) + s_{nf|k(f)} - s_{nf} \right]}. \quad (\text{OA.2.4})$$

The denominator is positive on the interior of the nested-logit demand system.

Proposition OA.2.2 (Conditional nested-logit pricing response). *Let z collect marginal costs and other payoff-relevant states, and write the pricing first-order conditions as $F(p, z) = 0$. Let $\mathcal{Z}$ be a compact state set, and let $\mathcal{U}$ be a compact neighborhood of the equilibrium graph $\{(p^*(z), z) : z \in \mathcal{Z}\}$ contained in the interior of the domain of F . Suppose that:*

- (i) *each active-product configuration has a unique interior pricing equilibrium $p^*(z)$ for every $z \in \mathcal{Z}$;*
- (ii) *F is continuously differentiable on an open neighborhood of $\mathcal{U}$; and*
- (iii) *$F_p(p^*(z), z)$ is uniformly nonsingular, so $\sup_{z \in \mathcal{Z}} \|F_p(p^*(z), z)^{-1}\|_\infty < \infty$.*

Then $p^(z)$ is continuously differentiable and*

$$\left\| \frac{\partial p^*}{\partial z} \right\|_\infty \leq L_p$$

for a finite constant L_p . If F is $C^{1,1}$ or C^2 with uniformly bounded derivatives on a neighborhood of $\mathcal{U}$, the price response is respectively $C^{1,1}$ or C^2 .

Proof. The implicit-function theorem gives

$$\frac{\partial p^*(z)}{\partial z} = -F_p(p^*(z), z)^{-1} F_z(p^*(z), z).$$

The inverse in condition (iii) and F_z are uniformly bounded on the compact equilibrium graph. Local implicit-function selections agree on overlaps by uniqueness and therefore

form the global selection p^* on $\mathcal{Z}$. Standard implicit-function results give the stated $C^{1,1}$ or C^2 regularity under the stronger assumptions. $\square$

Results such as [Nocke and Schutz \(2018\)](#) can establish pricing uniqueness for particular generalized extreme-value demand systems. Proposition [OA.2.2](#) also requires the nonsingularity and smoothness conditions used by the implicit-function argument.

OA.2.3 Implications for the portfolio interaction bound

The exact portfolio theorem defines K from four-point differences in fully repriced payoffs. Nested-logit demand can be used in that construction whenever the pricing selection and the resulting reduced payoffs satisfy the regularity conditions in Proposition [OA.2.2](#). The finite four-point definition remains valid without differentiability. A derivative-based bound is available under stronger smoothness.

Proposition OA.2.3 (Nested-logit derivative bound). *Let $z = (z_1, \dots, z_F)$ collect the portfolio aggregators. Suppose that the conditions of Lemma [OA.2.1](#) and Proposition [OA.2.2](#) hold on every relevant regular portfolio rectangle. If the primitive payoff and pricing equations are $C^{1,1}$ with uniformly bounded derivatives, then each fully repriced payoff $\tilde{r}_f(z) = r_f(z, p^*(z))$ is $C^{1,1}$ on those rectangles. Its cross-firm block derivatives are uniformly bounded, so the rectangle argument in Proposition [OA.3.1](#) supplies a finite entrywise bound for K .*

Proof. Proposition [OA.2.2](#) gives a uniformly bounded derivative of $p^*(z)$. The $C^{1,1}$ implicit-function argument gives the corresponding Lipschitz derivative. Applying the chain rule to $r_f(z, p^*(z))$ bounds the cross-firm block derivatives of the reduced payoff. For a $C^{1,1}$ reduced payoff, the mixed second derivatives exist almost everywhere, and the rectangle integral is bounded by their essential supremum. Integrating those bounds over each regular portfolio rectangle gives the claimed bound. Rectangles that cross an entry face are evaluated from their four fully repriced corner payoffs. $\square$

OA.3. AB PORTFOLIO-INTERACTION BOUND DERIVATIONS

This section supplies the full derivations summarized in Appendix [D](#) of the main paper. It first constructs the derivative, finite-difference, and hybrid bounds on the exact portfolio

interaction matrix. It then derives the active-face AB share Jacobian and reports the theoretical and calibrated ordering checks used to determine whether Jia or AES elimination applies.

OA.3.1 Derivation of the AB portfolio-interaction bounds

Under Assumption 1, Theorem 1 defines K from the finite payoff table. In the AB benchmark, demand and pricing primitives bound its entries.

The remainder of this subsection constructs derivative, finite-difference, and hybrid versions of the AB interaction bound. To bound K_{fg} , compare firm f 's payoff difference between two own portfolios before and after firm g changes portfolios. On a rectangle contained in one fixed active set, the fully repriced payoff is smooth and its cross-difference has a mixed-derivative representation. A rectangle that reaches an entry face is instead bounded by four fully repriced corner payoffs.

Let $z = (z_1, \dots, z_F)$ collect portfolio aggregators. In the benchmark, one component is the Fréchet cost sum of (3.1), $z_{nf}^c(S_f) = \sum_{i \in S_f} (w_i \tau_{in})^{-\theta}$, so, absorbing Γ_θ into a_f , $c_{nf}(S_f) = a_f^{-1} \{z_{nf}^c(S_f)\}^{-1/\theta}$. Additional smooth accessibility or quality indices can be appended to z_f . Let x collect the log price-power state, with $x_{nf} = \log y_{nf}$, and write its equilibrium as

$$x^*(z) = \Psi(x^*(z), z). \quad (\text{OA.3.1})$$

Firm f 's fully repriced variable payoff is $\tilde{r}_f(z) = r_f(z, x^*(z))$. Modular fixed costs are omitted in this subsection because they cancel from every cross-firm cross-difference.

Fix portfolios $S_f, T_f \in \mathcal{X}_f$, $A, B \in \mathcal{X}_g$, and $U = S_{-(f,g)}$. Set

$$\Delta_f = z_f(S_f) - z_f(T_f), \quad \Delta_g = z_g(A) - z_g(B),$$

and let z_{st} , $s, t \in [0, 1]$, be the rectangle that holds U fixed and moves from (T_f, B) in directions (Δ_f, Δ_g) . Call this rectangle regular if it lies in the relative interior of a single fixed-active-set region and $x^*(z)$ and $\tilde{r}_f(z)$ extend as twice continuously differentiable functions to a neighborhood of the rectangle. Its interior aggregator values provide an analytic interpolation and need not correspond to feasible discrete portfolios. On every

regular rectangle,

$$\begin{aligned} & r_f(S_f, A, U) - r_f(T_f, A, U) - r_f(S_f, B, U) + r_f(T_f, B, U) \\ &= \int_0^1 \int_0^1 D_{z_f z_g}^2 \tilde{r}_f(z_{st}) [\Delta_f, \Delta_g] ds dt. \end{aligned} \quad (\text{OA.3.2})$$

Applying the fundamental theorem of calculus to $\tilde{r}_f(z_{st})$ first in s and then in t gives (OA.3.2). Inserting the absolute value of this integral into the maxima and weighted sum in (2.10) gives a derivative bound for K_{fg} . The bound retains firm-pair, link, direction, and rectangle heterogeneity.

For an analytic envelope, use the sup norm for the pricing state and compatible operator norms for the portfolio blocks. Suppose on the relevant regular rectangles

$$\begin{aligned} \|\Psi_x\|_\infty &\leq \lambda < 1, & \|\Psi_{z_h}\| &\leq A_h, & \|\Psi_{z_h z_k}\| &\leq A_{hk}, \\ \|\Psi_{z_h x}\| &\leq A_{hx}, & \|\Psi_{xz_k}\| &\leq A_{xk}, & \|\Psi_{xx}\| &\leq A_{xx}. \end{aligned} \quad (\text{OA.3.3})$$

The quantities L_h and L_{hk} below translate those primitive derivatives into equilibrium price responses, including feedback through the pricing fixed point. Define

$$L_h = \frac{A_h}{1 - \lambda}, \quad L_{hk} = \frac{A_{hk} + A_{hx}L_k + A_{xk}L_h + A_{xx}L_hL_k}{1 - \lambda}. \quad (\text{OA.3.4})$$

For the primitive payoff $r_f(z, x)$, let

$$\begin{aligned} \|r_{f,z_f z_g}\| &\leq P_{f;fg}, & \|r_{f,z_f x}\| &\leq P_{f;fx}, & \|r_{f,x z_g}\| &\leq P_{f;xg}, \\ \|r_{f,xx}\| &\leq P_{f;xx}, & \|r_{f,x}\| &\leq P_{f;x}, \end{aligned}$$

and set

$$H_{fg}^{AB} = P_{f;fg} + P_{f;fx}L_g + P_{f;xg}L_f + P_{f;xx}L_fL_g + P_{f;x}L_{fg}. \quad (\text{OA.3.5})$$

The constant H_{fg}^{AB} combines direct payoff derivatives with the equilibrium price responses and bounds the fully repriced mixed payoff derivative. Define the rival portfolio diameter and the own-link boundary diameter

$$D_g = \max_{A, B \in X_g} \|z_g(A) - z_g(B)\|, \quad D_{fj}^\partial = \max_{\substack{S_f \ni j \\ T_f \not\ni j}} \|z_f(S_f) - z_f(T_f)\|. \quad (\text{OA.3.6})$$

Proposition OA.3.1 (Derivative bound for K). *Suppose every relevant portfolio rectangle is regular and (OA.3.1)–(OA.3.3) hold on each. Then, for $g \neq f$,*

$$K_{fg} \leq \bar{K}_{fg}^{AB} \equiv H_{fg}^{AB} D_g \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} D_{fj}^\partial, \quad \bar{K}_{ff}^{AB} = 0. \quad (\text{OA.3.7})$$

Any entrywise tighter value obtained from the rectangle identity can replace the corresponding entry of $\bar{K}^{AB}$. Hence

$$\sigma_\varepsilon > \rho(\bar{K}^{AB}) \quad (\text{OA.3.8})$$

is a derivative-based sufficient condition for uniqueness of the exact portfolio BNE.

Proof. Implicit differentiation of (OA.3.1) gives

$$x_{z_h}^* = (I - \Psi_x)^{-1} \Psi_{z_h}, \quad \|x_{z_h}^*\| \leq L_h.$$

Differentiating in block z_k once more gives

$$(I - \Psi_x) x_{z_h z_k}^* = \Psi_{z_h z_k} + \Psi_{z_h x} x_{z_k}^* + \Psi_{x z_k} x_{z_h}^* + \Psi_{xx} [x_{z_h}^*, x_{z_k}^*],$$

so $\|x_{z_h z_k}^*\| \leq L_{hk}$. The mixed block Hessian of the reduced payoff is

$$\begin{aligned} D_{z_f z_g}^2 \tilde{r}_f &= r_{f, z_f z_g} + r_{f, z_f x} x_{z_g}^* + (x_{z_f}^*)' r_{f, x z_g} \\ &\quad + (x_{z_f}^*)' r_{f, xx} x_{z_g}^* + r_{f, x} x_{z_f z_g}^*. \end{aligned}$$

Its block norm is bounded by (OA.3.5). Combining the rectangle identity with the two portfolio diameters bounds the four-corner payoff change. Summing these bounds with the link-boundary weights in (2.10) proves (OA.3.7). Entrywise matrix domination and Theorem 1 then give (OA.3.8). $\square$

The AB pricing derivatives entering this bound are explicit. Write

$$\begin{aligned} a &= \frac{1}{\sigma}, \quad b = \frac{1}{\eta} - \frac{1}{\sigma} > 0, \quad \varrho(s) = a + bs, \quad \mu(s) = \frac{1}{1 - \varrho(s)}, \\ \mu'(s) &= b\mu(s)^2, \quad \mu''(s) = 2b^2\mu(s)^3, \quad h(s) = as + bs^2. \end{aligned}$$

The log elasticity of the price-power response is

$$\begin{aligned}\beta(s) &= \frac{(\sigma-1)(\sigma-\eta)}{\sigma\eta}\mu(s), & \tilde{\kappa}(s) &= \frac{s(1-s)\beta(s)}{1+s(1-s)\beta(s)}, \\ \beta(s) \leq \bar{\beta} &= \frac{(\sigma-1)(\sigma-\eta)}{\sigma(\eta-1)}, & \tilde{\kappa}(s) \leq \bar{\lambda}_{AB} &= \frac{\bar{\beta}/4}{1+\bar{\beta}/4} < 1.\end{aligned}\tag{OA.3.9}$$

Thus the log-price-power map supplies the uniform choice $\lambda = \bar{\lambda}_{AB}$ in (OA.3.3). Together with derivatives of the Fréchet cost sum, these expressions bound the blocks in (OA.3.5). Evaluating the block constants still requires suprema over the relevant portfolio rectangles.

The next bound uses finite market-level payoff differences and avoids calibrated Hessian suprema. Let $v_{fn}(S) = R_n h(s_{fn}(S))$. Because h is increasing on $[0, 1]$, $0 \leq v_{fn}(S) \leq R_n/\eta$. For own portfolios S, T , rival- g portfolios A, B , and fixed portfolios $U = S_{-(f,g)}$, define

$$\begin{aligned}\Delta_{fg}^n(S, T; A, B, U) &= v_{fn}(S, A, U) - v_{fn}(T, A, U) \\ &\quad - v_{fn}(S, B, U) + v_{fn}(T, B, U),\end{aligned}$$

and

$$\Omega_{fgn}^j = \max_{\substack{S \ni j, T \not\ni j \\ A, B, U}} |\Delta_{fg}^n(S, T; A, B, U)|.\tag{OA.3.10}$$

Proposition OA.3.2 (Market-by-market AB bound). *In the fixed-expenditure AB model,*

$$K_{fg} \leq \bar{K}_{fg}^{FD} \equiv \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} \sum_n \Omega_{fgn}^j, \quad g \neq f.\tag{OA.3.11}$$

Set $\bar{K}_{ff}^{FD} = 0$. Moreover, $\Omega_{fgn}^j \leq R_n/\eta$, so

$$K_{fg} \leq \left(\sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} \right) \sum_n \frac{R_n}{\eta}.\tag{OA.3.12}$$

The finite-difference spectral condition $\sigma_\varepsilon > \rho(\bar{K}^{FD})$ is therefore sufficient for uniqueness of the exact portfolio BNE.

Proof. Apply the triangle inequality across markets inside (2.10) and then maximize each market term separately to obtain (OA.3.11). In a given market, order each pair of own competitiveness levels and each pair of rival competitiveness levels. AB share and $h(s)$ are increasing in own competitiveness and decreasing in rival competitiveness. Each one-direction payoff difference therefore lies in $[0, R_n/\eta]$, so the difference between two such increments has absolute value at most R_n/η . This proves (OA.3.12). The spectral conclusion follows from Theorem 1 and entrywise matrix domination. $\square$

The finite-difference bound covers entry and exit but discards the calibrated share response. A hybrid bound uses exact derivatives where the active set is fixed and fully repriced corner payoffs where it changes. Define log effective competitiveness by

$$t_{hn}(S_h) = \log \psi_{hn}(S_h) + (1 - \sigma) \log c_{hn}(S_h).$$

Let H_{fgn} be a verified upper bound on the absolute mixed derivative of firm f 's market- n variable profit with respect to t_{fn} and t_{gn} on the relevant fixed-active-set share region. Define the active-face diameters

$$d_{fn}^j = \max_{\substack{S_f \ni j, T_f \not\ni j \\ S_f, T_f \neq \emptyset}} |t_{fn}(S_f) - t_{fn}(T_f)|,$$

$$d_{gn} = \max_{A, B \neq \emptyset} |t_{gn}(A) - t_{gn}(B)|.$$

Either diameter is set to zero when its maximizing class is empty. Finally, let $\Omega_{fgn}^{j, \partial}$ be the fully repriced four-corner maximum over portfolio rectangles that reach an entry or exit face, with value zero when no such rectangle exists.

Corollary OA.3.3 (Hybrid AB interaction bound). *In the fixed-expenditure AB model,*

$$K_{fg} \leq \bar{K}_{fg}^{HYB} \equiv \sum_{j \in \mathcal{A}_f} \frac{\bar{g}_{fj}}{w_{fj}} \sum_n \max \left\{ \Omega_{fgn}^{j, \partial}, \min \left(\frac{R_n}{\eta}, H_{fgn} d_{fn}^j d_{gn} \right) \right\}. \quad (\text{OA.3.13})$$

Set $\bar{K}_{ff}^{HYB} = 0$. Hence $\sigma_\varepsilon > \rho(\bar{K}^{HYB})$ is sufficient for uniqueness of the exact portfolio BNE.

Proof. Partition the portfolio rectangles in (OA.3.10) into those that remain on a fixed active face and those that reach entry or exit. On a fixed active face, the fundamental theorem of calculus gives the product $H_{fgn} d_{fn}^j d_{gn}$, capped by the global market envelope R_n/η .

The second class is bounded directly by its fully repriced four-corner maximum. Taking the larger class bound and substituting it into (OA.3.11) proves (OA.3.13). The spectral conclusion follows from Theorem 1. $\square$

The next subsection derives the exact AB share Jacobian and the mixed payoff derivative used to construct H_{fgn} .

OA.3.2 Exact AB share derivatives and a hybrid interaction bound

Recall $a = 1/\sigma$, $b = 1/\eta - 1/\sigma$, and $\mu(s) = \{1 - a - bs\}^{-1}$. The closed-form envelope in (OA.3.12) discards the calibrated share response. On each face with a fixed set of active firms, it can be tightened using log effective competitiveness

$$t_{hn} = \log \psi_{hn} + (1 - \sigma) \log c_{hn}.$$

Let

$$m(s) = 1 - a - bs, \quad \beta_s(s) = \frac{(\sigma - 1)b}{m(s)}, \quad \omega(s) = \frac{s}{1 + s\beta_s(s)}.$$

For the active firms in a market, put $W = \sum_h \omega(s_h)$ and $\omega_h = \omega(s_h)/W$. On a fixed active face, the AB share system is

$$s_h = \frac{\exp(t_h)\mu(s_h)^{1-\sigma}}{\sum_\ell \exp(t_\ell)\mu(s_\ell)^{1-\sigma}}.$$

Writing the denominator as $\mathcal{Z}$, logarithmic differentiation gives

$$\frac{ds_h}{s_h} = dt_h - \beta_s(s_h)ds_h - d \log \mathcal{Z}, \quad ds_h = \omega(s_h)\{dt_h - d \log \mathcal{Z}\}.$$

Because shares sum to one, $\sum_h ds_h = 0$, and hence

$$d \log \mathcal{Z} = \sum_k \omega_k dt_k.$$

Substitution gives the exact share Jacobian

$$J_{hk} \equiv \frac{\partial s_h}{\partial t_k} = \omega(s_h)(\mathbf{1}\{h = k\} - \omega_k). \quad (\text{OA.3.14})$$

It satisfies both required adding-up checks:

$$\sum_h J_{hk} = \omega(s_k) - \omega_k W = 0, \quad \sum_k J_{hk} = \omega(s_h) \left(1 - \sum_k \omega_k\right) = 0.$$

The remaining scalar derivative is

$$\omega'(s) = \frac{1 - s^2(\sigma - 1)b^2/m(s)^2}{\{1 + s\beta_s(s)\}^2}.$$

Define

$$P_{fg} \equiv \frac{\partial \omega_f}{\partial t_g} = \frac{\omega'(s_f)J_{fg} - \omega_f \sum_h \omega'(s_h)J_{hg}}{W},$$

$$Q_{ffg} \equiv \frac{\partial J_{ff}}{\partial t_g} = \omega'(s_f)J_{fg}(1 - \omega_f) - \omega(s_f)P_{fg}.$$

The exact mixed payoff derivative on an active face is

$$\mathcal{H}_{fgn}(s) = R_n \{2bJ_{ff}J_{fg} + (a + 2bs_f)Q_{ffg}\}. \quad (\text{OA.3.15})$$

Thus any verified upper bound H_{fgn} on $\sup_s |\mathcal{H}_{fgn}(s)|$ over the relevant active-share region gives

$$\Omega_{fgn}^{j,\text{act}} \leq \min \left\{ \frac{R_n}{\eta}, H_{fgn} d_{fn}^j d_{gn} \right\}. \quad (\text{OA.3.16})$$

The diameters and entry-face corner term are defined in Corollary [OA.3.3](#). Combining [\(OA.3.16\)](#) with those corner payoffs yields [\(OA.3.13\)](#): finite differences cover changes in the active set, while the implicit AB Hessian covers fixed-active-set rectangles.

Repriced link interactions may change sign across active sets; the bound remains valid.

OA.3.3 Why order-based portfolio elimination need not apply

The contingent threshold in Lemma [1](#) exists for every finite portfolio-value table. Order-based portfolio algorithms require more. The lattice bounds in [Jia \(2008\)](#) use supermodularity of the firm's own portfolio payoff. The elimination procedure in [Arkolakis et al. \(2023\)](#) allows weaker conditions, but still requires the sign of a link's marginal value to be ordered in one direction across nested portfolios. The following example shows that a

unique AB pricing equilibrium supplies neither property.

Proposition OA.3.4 (AB repricing need not preserve portfolio order). *There is a two-firm AB market with a unique pricing equilibrium at every portfolio and a three-link portfolio problem whose fully repriced payoff is neither supermodular nor submodular and satisfies neither direction of the AES choice-monotonicity condition.*

Proof. Set $(\sigma, \eta, \theta) = (5, 2, 3)$ and normalize market expenditure to one. Firm 2 has cost $c_2 = 1$. Firm 1 has productivity aggregator

$$\Sigma(S) = 0.001 + \sum_{j \in S} d_j, \quad c_1(S) = \Sigma(S)^{-1/3},$$

with link increments $d_1 = d_2 = 0.02$ and $d_3 = 0.099$. For every S , both firms' prices are recomputed from

$$p_k = \mu(s_k) c_k, \quad s_k = \frac{p_k^{1-\sigma}}{p_1^{1-\sigma} + p_2^{1-\sigma}}.$$

Proposition 4 gives a unique solution. Firm 1's gross variable profit is

$$h(S) = s_1(S) \varrho(s_1(S)).$$

Assign link 1 the modular cost 0.007; the other two link costs are zero. For the resulting net payoff $\tilde{h}$, the marginal value of link 1 along $\emptyset \subset \{2\} \subset \{2, 3\}$ is

A	$\tilde{h}(A \cup \{1\}) - \tilde{h}(A)$
$\emptyset$	-0.000337164214
$\{2\}$	0.000897247933
$\{2, 3\}$	-0.001046642228

Thus link 1 is unprofitable, then profitable, then unprofitable as the other portfolio expands. The first change in its marginal value is positive and the second is negative, so the payoff is neither submodular nor supermodular. The positive marginal value at $\{2\}$ fails to remain positive when the portfolio is expanded to $\{2, 3\}$ or contracted to $\emptyset$, which violates both directions of AES choice monotonicity. $\square$

The example separates pricing uniqueness from portfolio order. Jia or AES can still be used as an exact inner procedure when its respective order condition is verified over every

Table OA.3.1: Fully repriced AES ordering check at the calibrated cloud profiles

Notes: Rivals' portfolios are fixed at the selected profile for the indicated pricing rule. "Fails" means that the check found a nested-portfolio sign reversal that rejects the corresponding direction of SCD-C. Comparisons are the number of nested pairs evaluated before witnesses for both directions were found.

Provider	Pricing	Candidate links	Comparisons	From below	From above
Oracle	CES	27	1,454	Fails	Fails
AWS	AB	30	159	Fails	Fails
AWS	CES	30	1,776	Fails	Fails
Microsoft	AB	34	2,354	Fails	Fails
Microsoft	CES	34	2,331	Fails	Fails
Google	AB	29	125	Fails	Fails
Google	CES	29	1,639	Fails	Fails

relevant off-path portfolio. Without such verification, their elimination steps can discard the globally optimal response. The calculation is reproduced by `scripts/verify_ab_aes_counterexample.py`.

Calibrated ordering check. We also test the AES condition in seven large-candidate-set cloud responses. Fix rivals at the selected AB or CES profile. For a candidate link j and nested own portfolios $A \subset B$, we fully reprice the two marginal values

$$\Delta_j \pi_f(A) = \pi_f(A \cup \{j\}) - \pi_f(A), \quad \Delta_j \pi_f(B) = \pi_f(B \cup \{j\}) - \pi_f(B).$$

A positive-to-negative reversal rejects SCD-C from below; a negative-to-positive reversal rejects SCD-C from above. Table OA.3.1 reports the number of nested comparisons made before both witnesses were found. A single witness rejects the corresponding global order condition, so this falsification does not require enumerating every portfolio.

The check uses a 10^{-6} sign tolerance. For AB, portfolio profits are enclosed in numerical intervals before marginal signs are classified; the largest profit-interval width across these responses is 1.95×10^{-8} . All four CES responses and all three AB responses reject both directions. These are calibrated-instance results rather than a claim that CES or AB pricing always violates SCD-C. They show that neither constant markups nor a unique pricing continuation is sufficient to validate AES elimination in the cloud specification.

OA.4. GRAVITY-ACCESS IMPLEMENTATION

The cloud application combines the benchmark Fréchet service-cost aggregator, a smooth accessibility shifter, and quality-adjusted AB pricing. The reported counterfactuals follow selected deployment profiles. This section defines these blocks, establishes their regularity, and reports calibration fit within the provider-specific candidate sets.

OA.4.1 Service-cost primitives

The implemented service-cost block is

$$c_{nf}(I_f) = \frac{1}{a_f} \left[\sum_{i \in I_f} (w_i \tau_{in})^{-3} \right]^{-1/3}. \quad (\text{OA.4.1})$$

Thus $\theta = 3$, and the common Fréchet factor is normalized to $\Gamma_\theta = 1$ and absorbed into provider productivity. Let Rev_f denote the rounded cloud-revenue scale proxy. The calibration uses (100, 65, 40, 20, 17) billion dollars for AWS, Microsoft, Google, Oracle, and Alibaba, respectively. We set

$$a_f = 1 + \frac{\log \text{Rev}_f - \min_g \log \text{Rev}_g}{\max_g \log \text{Rev}_g - \min_g \log \text{Rev}_g},$$

which gives

$$(a_{\text{AWS}}, a_{\text{MSFT}}, a_{\text{GOOG}}, a_{\text{ORCL}}, a_{\text{BABA}}) = (2.000, 1.757, 1.483, 1.092, 1.000).$$

The site scale w_i is Penn World Table human-capital-adjusted compensation per worker, normalized to one in the United States. The baseline bilateral service-cost matrix is

$$\tau_{in} = \begin{cases} 1, & i = n, \\ 1.15, & i \neq n \text{ and the pair is on the same continent or within 5,000 km,} \\ 1.5, & \text{otherwise.} \end{cases}$$

Country coordinates determine this three-tier classification. Equation (OA.4.1) places it in marginal cost. The accessibility score below uses facility-centroid distance and policy

measures to shift service quality. The quality-only sensitivity specification sets $\tau_{in} = 1$ for all pairs, retains the accessibility score, and recalibrates (ϕ_i, g_f) .

OA.4.2 Accessibility construction and frontier normalization

Let D_{in}^{km} denote great-circle distance and define $\tilde{d}_{in} = D_{in}^{\text{km}}/1000$. For candidate site i and market n , the gravity implementation uses the score

$$q_{in} = -\beta_d \log(1 + \tilde{d}_{in}) + \beta_{\text{same}} \mathbf{1}\{i = n\} + \beta_{\text{local}} \ell_{in} + \beta_{\text{policy}} z_n, \quad (\text{OA.4.2})$$

where ℓ_{in} indicates local or regional eligibility and z_n is a researcher-constructed policy weight based on OECD Digital STRI regulatory measures for market n . Specifically, $z_n = u_n - 1$, where the tier weights u_n are 1.00 for open transfer, 0.95 for open safeguards, 0.85 for pre-authorized safeguards, 0.70 for case-by-case approval, 0.60 for a commercial-presence requirement, 0.55 for a local-presence requirement, 0.35 for local storage, and 0.10 for prohibited cross-border transfer. An OECD row without a classified measure receives 0.75. When several measures apply, the most restrictive tier determines the weight. The five markets outside OECD coverage receive the neutral value $u_n = 1$. Hence $z_n \in [-0.90, 0]$, and a larger value raises serviceability. For a realized activation set I_f , provider f 's accessibility index is

$$A_{fn}(I_f) = \lambda_A^{-1} \log \left(1 + \sum_{i \in I_f} \exp(\lambda_A q_{in}) \right). \quad (\text{OA.4.3})$$

The market-level frontier is computed over the common candidate universe $\mathcal{S}$:

$$A_n^{\text{frontier}} = \lambda_A^{-1} \log \left(1 + \sum_{i \in \mathcal{S}} \exp(\lambda_A q_{in}) \right). \quad (\text{OA.4.4})$$

The frontier-normalized quality shifter is

$$\psi_{fn}(I_f) = \exp \left(\chi [A_{fn}(I_f) - A_n^{\text{frontier}}] \right) \leq 1. \quad (\text{OA.4.5})$$

For derivative calculations, embed the binary portfolio in $z_f \in [0, 1]^{|S|}$, with $z_{fi} = 0$

outside provider f 's candidate set, and define

$$A_{fn}(z_f) = \lambda_A^{-1} \log \left(1 + \sum_{i \in S} z_{fi} \exp(\lambda_A q_{in}) \right), \quad (\text{OA.4.6})$$

and

$$\psi_{fn}(z_f) = \exp \left(\chi [A_{fn}(z_f) - A_n^{\text{frontier}}] \right). \quad (\text{OA.4.7})$$

The embedding agrees with the set-based definition at binary vectors and is used only to bound payoff changes between portfolio corners.

Proposition OA.4.1 (Smoothness of frontier-normalized accessibility). *Fix finite site and market sets. Suppose $|q_{in}| \leq \bar{q}_{\text{grav}} < \infty$ and $\lambda_A \in (0, \infty)$. For each provider-market pair (f, n) , the frontier-normalized accessibility shifter $\psi_{fn}(z_f)$ is continuously differentiable on $[0, 1]^{|S|}$, with derivatives uniformly bounded on that domain. It depends only on firm f 's portfolio coordinates.*

Proof. The map inside the logarithm in (OA.4.6) is a finite sum of exponentials plus one, so it is strictly positive and smooth on $[0, 1]^{|S|}$. The frontier term in (OA.4.4) is fixed and therefore does not affect derivatives. Boundedness follows from the finite site universe and the uniform bound on q_{in} . $\square$

OA.4.3 Quality-adjusted AB pricing

Conditional on an active set, the gravity specification shifts provider quality and service costs. The AB pricing bound is unchanged. Let $y_{nf} > 0$ denote the quality-adjusted price-power term, set $x_{nf} = \log y_{nf}$, and define $D_{nf} = \sum_{g \neq f} y_{ng}$. For $g \neq f$, the log- y price-response Jacobian in market n is

$$(J_n^x)_{fg} = \tilde{\kappa}_{nf} \frac{y_{ng}}{D_{nf}}, \quad \tilde{\kappa}_{nf} = \frac{s_{nf}(1 - s_{nf})\beta_{nf}}{1 + s_{nf}(1 - s_{nf})\beta_{nf}}, \quad (\text{OA.4.8})$$

with diagonal entries equal to zero and

$$\beta_{nf} = \frac{(\sigma - 1)\mu_{nf}(\sigma - \eta)}{\sigma\eta}.$$

We also write $\Lambda_n \equiv J_n^x$. Each row of J_n^x sums to $\tilde{\kappa}_{nf}$. The AB markup bound implies

$$\beta_{nf} \leq \bar{\beta} \equiv \frac{(\sigma - 1)(\sigma - \eta)}{\sigma(\eta - 1)}.$$

Since $s(1 - s) \leq 1/4$,

$$\|J_n^x\|_\infty \leq \bar{\lambda}_{AB} \equiv \frac{\bar{\beta}/4}{1 + \bar{\beta}/4} < 1. \quad (\text{OA.4.9})$$

This bound depends only on the demand elasticities and applies to every active set. A market with one active firm has no cross-firm pricing response.

Proposition OA.4.2 (Quality-adjusted pricing regularity). *Suppose $\sigma > \eta > 1$. For every active set, the quality-adjusted AB pricing map is a contraction in $x = \log y$ with modulus at most $\bar{\lambda}_{AB}$ and therefore has a unique equilibrium. If quality and cost shifters are C^1 in z with bounded derivatives on a compact set, the equilibrium price map $p(z)$ is C^1 with a uniformly bounded Jacobian.*

Proof. Equation (OA.4.9) gives a global sup-norm contraction, so Banach's theorem gives the unique pricing fixed point. The fixed-point Jacobian has inverse norm bounded by $(1 - \bar{\lambda}_{AB})^{-1}$. The implicit-function theorem and the bounded derivatives of the shifters then give the last claim. $\square$

At the supplementary calibration's selected AB deployment profile, the local pricing row sum is 0.306 and the local spectral radius is 0.238. Both lie below the share-specific envelope 0.3187 and the uniform modulus 0.375. These local measures concern the pricing subgame and are distinct from the entry-interaction radius $\rho(K)$.

At the exact benchmark profile, the corresponding row sum is 0.305 and the local spectral radius is 0.236.

At the exact benchmark calibration, the global finite-difference and hybrid portfolio bounds both give $\rho(\bar{K}) = 229.3369$. The implied matrix is

$$\begin{pmatrix} 0 & 64.700 & 64.700 & 64.700 & 64.700 \\ 67.593 & 0 & 67.593 & 67.593 & 67.593 \\ 60.973 & 60.973 & 0 & 60.973 & 60.973 \\ 61.318 & 61.318 & 61.318 & 0 & 61.318 \\ 34.672 & 34.672 & 34.672 & 34.672 & 0 \end{pmatrix},$$

where rows index the affected provider and columns the rival whose portfolio changes, in the order AWS, Microsoft, Google, Oracle, and Alibaba. Every market-level term reaches the R_n/η cap, so the active-face derivative calculation cannot tighten the full-family finite-difference bound. The value is too large to be informative for the calibrated application and is not used to claim finite-noise uniqueness. It is reported to distinguish an evaluated but conservative global portfolio bound from the much smaller local pricing matrix Λ .

OA.4.4 Numerical specifications and reported portfolio objects

The paper reports results from two calibrated specifications. They share the same demand elasticities, candidate universe, and benchmark accessibility and service-cost mappings, but they use different estimates of the fixed-cost block. The supplementary-calibration archive is the basis for the archived one-link, regional, and residual exercises. The benchmark recalibration supplies the fixed primitives for the main unrestricted and conditional comparisons. Table [OA.4.1](#) summarizes the distinction.

OA.4.5 Supplementary calibration: one-link selection and update-order checks

The supplementary fixed-cost calibration uses a conditional active-set routine to construct its 124-link candidate profile. For a fixed vector of rival price powers, that routine generates lower and upper sequences of link sets. Increasing differences would order these sequences and support a global conditional response, but AB repricing can reverse the ordering. We therefore treat the calibration output as a candidate profile and apply a separate, fully repriced selection step before reporting the application results.

Starting from a supplied profile, the selection routine evaluates every feasible one-link addition and deletion. It solves the full AB or CES pricing equilibrium for all firms after every proposed change, using a pricing residual tolerance of 10^{-10} . The reported rule takes the change with the largest positive gain and repeats. A final check evaluates all 134 possible toggles and stops only when the maximum gain is at most 10^{-10} . The result is one-link stable, conditional on the pricing calculation and numerical tolerances. Section [OA.6](#) describes the separate branch-and-bound method for verifying global whole-portfolio optimality.

We vary both initialization and update order. The six initial profiles are the two cal-

Table OA.4.1: Calibrated Specifications and Reported Portfolio Objects

Specification	Calibration output	Repriced portfolios	Role in the paper
Supplementary calibration	$\mathcal{L} = 36$; 124 candidate links; 125/134 candidate-profile fit	One-link AB: 121 links and 122/134 fit; one-link CES: 102 links; AB-to-CES profit change: -36.616%	Documentation of a retired supplementary calibration; not used in the main results
Benchmark recalibration	$\mathcal{L} = 26$; 128 candidate links; 127/134 candidate-profile fit	One-link AB: 124 links and 125/134 fit; one-link CES: 105 links; AB-to-CES profit change: -35.665%	Fixed costs and initial profiles for the unrestricted exact comparison
Unrestricted exact verification	benchmark primitives; no further recalibration	Terminal AB: 124 links after AWS replaces Singapore with Ireland; terminal CES: 105 links; AB-to-CES profit change: -35.629%	Main exact AB–CES decomposition with explicit AB numerical-error accounting
Sampled-Shapley extension	benchmark primitives; fully repriced permutation paths	Sampled AB: 110 links; sampled CES: 79 links; exact-profile Hamming distances: 16 and 26	Unrestricted separable diagnostic; six AB and three CES signs remain unclassified
Conditional comparison	benchmark primitives and exact reference profiles	Shapley: 127 AB and 104 CES links; Chebyshev: 126 AB and 104 CES links; exact-profile differences: 3–5 links	Main conditional comparison on common 12-link candidate sets
Fixed-aggregate benchmark	benchmark primitives and exact reference profiles	Fixed-aggregate fixed point: 128 AB and 105 CES links; strategic equilibrium: 124 AB and 105 CES links	Isolates rivals' repricing while preserving global whole-portfolio choice

Notes: Profit changes are aggregate provider profit net of fixed costs. The first two rows differ in calibrated fixed costs, not only in the portfolio algorithm. The main exact comparison uses the benchmark primitives for both AB and CES; the supplementary-calibration results are separately labeled as a supplementary calibration. The sampled-Shapley fixed points are fixed points of estimated response maps; they do not establish a fixed point of the population Shapley map.

ibration branches, the stored CES profile, the observed profile, all candidate links, and the empty profile. For each pricing rule, cyclic updates use forward, reverse, and three random orders. We also run largest-gain improvement from the two calibration branches. Table OA.4.2 summarizes the 64 runs.

OA.4.6 Supplementary-calibration gravity fit

The supplementary fixed-cost calibration activates 124 of 134 candidate links, recovers 122 of 129 operational deployments, and activates two of five announced links. The reported largest-gain calculation removes three links and selects a 121-link AB profile. It matches

Table OA.4.2: Fully Repriced One-Link Sweep under the Supplementary Calibration

Pricing	Runs	Initial profiles	Final active	Distinct profiles	Profit range
AB	32	6	121	3	1.2140–1.2147
CES	32	6	102	2	0.7686–0.7695

Notes: Each pricing rule is run from six initial profiles with forward, reverse, and three random cyclic orders. Largest-gain improvement is also run from the two calibration branches. Every run is one-link stable under full repricing. Across all pairings of the terminal AB and CES runs, the aggregate profit change ranges from -36.73% to -36.62% .

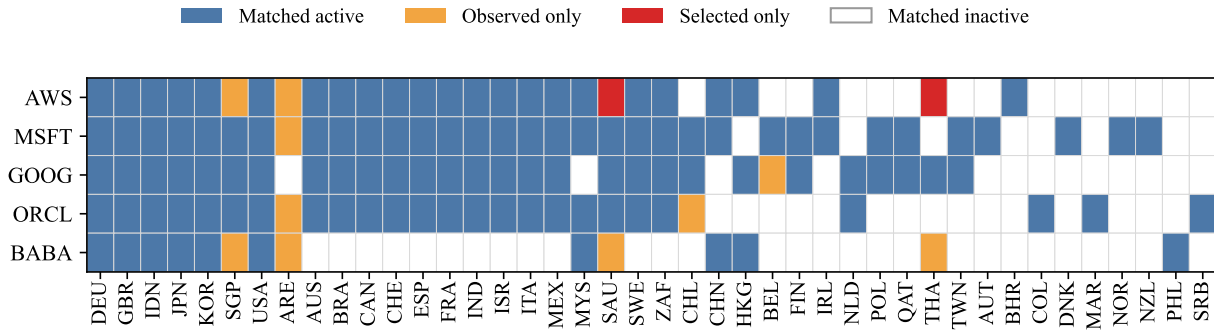


Figure OA.4.1: Observed and Selected Deployment Status by Provider and Country

Notes: Colors classify each provider-country pair as matched active, matched inactive, observed only, or selected only. Statistics condition on the 134 provider-specific candidate links.

deployment status for 122 of 134 candidate links, retains 119 operational deployments, and activates two announced links.

A conditional diagnostic used in calibration held the converged rival-demand vector fixed and assigned AWS–Chile a gain of 2.10×10^{-4} . At the selected profile, the gain from adding AWS–Chile after re-solving all firms' prices is -1.60×10^{-3} . More generally, every deletion from the 121 active links and every addition among the 13 inactive links lowers the deviating firm's profit after full repricing. The smallest deletion loss is 3.41×10^{-6} , and the largest inactive-link gain is -5.26×10^{-5} .

The sweep also separates robustness of aggregate outcomes from robustness of composition. Every run reaches 121 AB links and 102 CES links, while three AB profiles and two CES profiles appear across update paths. The selected largest-gain rule reaches the same AB profile from both calibration branches. Starting CES from that selected AB profile yields 102 links and the upper end of the CES profit range in Table OA.4.2.

The broader pattern of mismatches points to peering, network design, and regulatory hub value omitted from the public-data accessibility index.

Table OA.4.3: AB–CES One-Link Decomposition under the Supplementary Calibration

Scenario	Pricing	Deployment profile	Active	Δn	$\Delta\pi$ (%)
AB one-link-stable profile	AB	recomputed	121	—	—
CES pricing, AB profile	CES	AB fixed	121	—	-42.3
CES one-link-stable profile	CES	recomputed	102	-19	-36.6

Notes: The fixed-profile row changes only the pricing rule. The final row recomputes the deployment profile under CES. Δn and $\Delta\pi$ are measured relative to the selected AB deployment profile, and π is aggregate provider profit net of fixed costs. Percentage-point differences are computed from unrounded profit values.

OA.4.7 AB–CES calculation

The AB–CES exercise holds demand and cost primitives and the accessibility mapping fixed while changing the pricing rule. Table OA.4.3 decomposes the comparison into a pricing change that holds the selected AB deployment profile fixed and the additional change from recomputing deployment under CES.

OA.5. CLOUD DATA, CALIBRATION, AND REPRODUCIBILITY

OA.5.1 Input construction

Table OA.5.1 lists the inputs used in the benchmark gravity calibration. The market-size proxy is

$$\Omega_n = \left[\frac{\text{GDP}_n \cdot \text{InternetUse}_n}{\text{GDP}_{US} \cdot \text{InternetUse}_{US}} \right]^{0.70}.$$

The fixed-expenditure demand system sets $R_n = \Omega_n$. GDP is measured in current U.S. dollars and InternetUse is the share of the population using the internet. Both series use 2023 World Bank WDI values, except that Taiwan uses a 2023 IMF GDP estimate and a 93 percent internet-use imputation.

The distance term is a geographic proxy for latency. We average facility coordinates within each country and use a capital-city coordinate when no facility coordinate is available. The resulting great-circle distance is measured in kilometers. With $\tilde{d}_{in} = D_{in}^{\text{km}}/1000$, it enters the gravity score as $\log(1 + \tilde{d}_{in})$. The local-region indicator equals one for the same country and for bilateral pairs assigned a discrete service-cost value no greater than 1.15. The latter category includes countries on the same continent or less than 5,000 kilometers

apart.

The canonical infrastructure pages are those of [AWS](#), [Microsoft Azure](#), [Google Cloud](#), [Oracle Cloud](#), and [Alibaba Cloud](#). The provider-region data were assembled and held fixed as of April 7, 2026. The file has SHA-256 prefix 822f3dcf3e51. The replication archive lists the complete hash, provider-level URLs, source-check dates, page hashes, and a source mapping for each of its 191 rows.

The replication archive contains the dated OECD policy file. Its SHA-256 prefix is b10c3581299e. The neutral policy fallback applies to ARE, BHR, HKG, QAT, and TWN, which are outside OECD coverage in this vintage.

OA.5.2 Launch-date panel and expansion-wave counts

The expansion-wave figure in the main text uses a launch-date panel constructed and checked separately from the archived deployment vintage. For each provider-country pair we date the general availability of the first customer-facing cloud region from provider documentation: release notes, official product blogs, newsroom releases, and, where the original pages are no longer served, Internet Archive copies of those pages, with the original and archive URLs retained together. Announcements of intent, previews, restricted releases, and datacenter-campus openings are not coded as launches. Six dates that previously rested on secondary press sources were re-adjudicated against primary evidence; all six years were confirmed, and day-level fields were corrected or withdrawn where the provider evidence did not support them. The verified panel dates 132 provider-country first entries. Its 2026 cross section has three more active links in aggregate than the archived calibration file and differs in nine provider-country pairs: six additions and three removals. The calibration and all equilibrium objects in the paper use the archived vintage, and the launch-date panel is used only for the descriptive evidence reported here.

A provider enters the risk set in the year its public-cloud service first became generally available as a paid offering, dated from primary evidence for each provider: AWS 2006, Microsoft 2010, Google 2011, Alibaba 2011, and Oracle 2016. Under this rule, 115 of the 132 dated entries fall in 2007–2024 with a one-year lag of the network state available. These 115 entries occur in 52 positive provider-years; 28 episodes (54 percent) add two or more countries, and 91 entries (79 percent) fall in such multi-country years. A stricter rule that starts each provider at compute-IaaS general availability (AWS 2008, Microsoft 2013, Google 2013, Alibaba 2011, Oracle 2016) yields 110 entries in 51 episodes, of which

27 (53 percent) are multi-country, containing 86 entries (78 percent). The counts aggregate launches to the calendar year; they measure clustering of realized expansion, not simultaneity of the underlying decisions, and they are not calibration targets.

OA.5.3 Calibration procedure

Conditional on the gravity coefficients, the calibration chooses market fixed costs ϕ_i and provider shifters g_f to reduce discrepancies in market and provider entry counts. Let

$$e_i = m_i^{data} - m_i^{model}, \quad e_f = m_f^{data} - m_f^{model},$$

and use

$$\mathcal{L} = \sum_i e_i^2 + \sum_f e_f^2 \quad (\text{OA.5.1})$$

to compare the discrete profiles generated during calibration. The routine stores the parameter vector with the smallest value of (OA.5.1).

The benchmark recalibration performs the market-cost update in logs:

$$\log \phi_i^{new} = \log \phi_i - \alpha_\phi e_i + \lambda_\phi (\log \phi_i^0 - \log \phi_i), \quad (\text{OA.5.2})$$

with $\alpha_\phi = 0.10$ and $\lambda_\phi = 0.05$. The last term implements shrinkage toward the initial income-group prior, which remains fixed throughout the benchmark update sequence. The archived earlier implementation reset the shrinkage center to the current iterate before each pass. Its last term was therefore zero at the point of updating. The two calibrated specifications differ in this feature of the fixed-cost estimator, as well as in their saved fixed-cost vectors. Each routine selects among its visited profiles using (OA.5.1). Provider costs are updated as

$$\log g_f^{new} = \log g_f - \alpha_g e_f, \quad \alpha_g = 0.03. \quad (\text{OA.5.3})$$

AWS is normalized to $g_{AWS} = 1$. The other provider shifters start at one. Market costs start at ϕ_i^0 . The bounds are

$$\phi_i \in [0.005, 0.50], \quad g_f \in [0.10, 5.0].$$

The saved supplementary-calibration run allows eight outer provider-cost updates. The first market-cost pass allows 18 iterations, and subsequent passes use 15. The market-cost routine stops at exact market-count fit, after eight consecutive stalled iterations, or at its iteration cap. The outer routine stops at exact fit, after three stalled iterations, or at its cap. For every firm update, pricing is solved with damping 0.3 to a residual below 10^{-9} , with a 1,200-iteration cap. The sequential active-set routine accepts a proposed portfolio when its profit gain exceeds 10^{-6} and stops after a pass with no accepted changes, with a 50-pass cap. We run this calibration step from the full and low-cost initializations. After fixing the selected parameter vector, the fully repriced one-link routine uses a 10^{-10} improvement tolerance, a 10^{-10} pricing-residual tolerance, and a 5,000-iteration pricing cap.

The benchmark run holds the income-group vector ϕ^0 fixed and uses the supplementary calibration’s selected costs and portfolio only as an initial value. It selects that path with $\mathcal{L} = 26$ over two alternative initializations with $\mathcal{L} = 104$. Its one-link profiles and the subsequent unrestricted verification are reported separately in Table [OA.4.1](#); the benchmark and retired supplementary outputs are not combined.

OA.5.4 Specification choice and saved results

The comparison set contains G1 (distance), G2 (distance and same-country), and G3 (local access and policy) specifications. Comparable G3 runs using the benchmark cloud-demand proxy evaluate

$$\chi \in \{0.24, 0.27, 0.30, 0.33, 0.40, 0.45, 0.50\}.$$

The entry-count criterion in [\(OA.5.1\)](#) reaches its minimum of 36 at $\chi = 0.40$ and matches deployment status for 201 of 210 provider-country pairs. The adjacent value $\chi = 0.45$ matches 200 pairs but has $\mathcal{L} = 40$. We therefore use $\chi = 0.40$ as the central value and report $\chi = 0.45$ as a sensitivity check. The archived supplementary-calibration run matches deployment status for 125 of 134 candidate links and has 124 active links. Within each calibration run, the routine stores the first visited profile attaining the lowest value of $\mathcal{L}$, updating the saved state only after a strict improvement. Holding that run’s parameter vector fixed, largest-gain fully repriced improvement selects 121 AB links and matches deployment status for 122 candidate links. The archived AB-CES and regional exercises

start from this selected profile.

OA.6. WHOLE-PORTFOLIO SOLUTION METHOD FOR THE EXACT MODEL

This section documents the branch-and-bound algorithm for unrestricted portfolio responses. The same routine handles the finite-noise model and its complete-information limit. Each private-cost type compares complete portfolios, with prices recomputed after every deviation. In the complete-information application, the rival distribution is a point mass and the private-cost scale is zero.

Jia's lattice bounds are exact under own-portfolio supermodularity, and AES elimination is exact under its weaker choice-monotonicity condition. These are distinct from the own-cost monotonicity behind the inclusion cutoff. The calibrated check in Appendix Table [OA.3.1](#) finds witnesses against both AES orderings in each large-candidate-set response examined. The branch-and-bound calculation therefore uses node bounds that remain valid without either portfolio-order restriction. Appendix [C](#) proves the cardinality-conditioned bound; this section describes its implementation and numerical use.

The ordering check covers four CES and three AB responses, uses a 10^{-6} marginal-value sign tolerance, and finds both directions of reversal within the nested-pair comparisons alone. AB marginal values are classified from payoff intervals; their largest width in these tasks is 1.95×10^{-8} . A witness falsifies an ordering for the corresponding calibrated response. The check does not claim that CES or AB pricing violates the ordering in every environment.

OA.6.1 Portfolio search

Fix firm f , a private-cost realization ε_f , and a finite distribution Q_{-f} over rival portfolios. For a complete own portfolio S_f , the type-specific objective is

$$\tilde{V}_f(S_f; \varepsilon_f, Q_{-f}) = V_f(S_f; Q_{-f}) - \sigma_\varepsilon \sum_{j \in S_f} w_{fj} \varepsilon_{fj} = \sum_{S_{-f}} Q_{-f}(S_{-f}) r_f(S_f, S_{-f}) - \sigma_\varepsilon \sum_{j \in S_f} w_{fj} \varepsilon_{fj},$$

where $r_f(S) = \Pi_f(S; p^\dagger(S)) - \sum_{j \in S_f} \phi_{fj} w_{fj}$, as in (2.1). Thus the total realized cost attached to link j is

$$C_{fj}(\varepsilon_{fj}) = w_{fj}(\phi_{fj} + \sigma_\varepsilon \varepsilon_{fj}).$$

We write C_{fj} for this realized cost below. The pricing calculation and reduced payoff are cached by complete portfolio profile.

At a search node, let S_f denote selected links and U_f undecided links. In the cloud candidate-set problem, every subset of $\mathcal{A}_f$ is feasible. Adding a link weakly raises accessibility and weakly lowers the Fréchet service-cost aggregator. The all-in completion $S_f \cup U_f$ therefore gives the optimistic own shifters

$$\bar{\psi}_{nf} = \psi_{nf}(S_f \cup U_f), \quad \underline{c}_{nf} = c_{nf}(S_f \cup U_f).$$

For a fixed rival portfolio state S_{-f} , evaluate rival quality and cost at that state and define

$$\underline{y}_{ng}(S_{-f}) = \psi_{ng}(\mu_{\max} c_{ng})^{1-\sigma}, \quad \bar{y}_{nf} = \bar{\psi}_{nf}(\mu_{\min} \underline{c}_{nf})^{1-\sigma},$$

where

$$\mu_{\min} = \frac{\sigma}{\sigma - 1}, \quad \mu_{\max} = \frac{\eta}{\eta - 1}.$$

Because $\sigma > 1$, these expressions use the most favorable own quality and cost and the least favorable rival price power allowed by the AB markup bounds. Hence

$$\bar{s}_{nf}(S_{-f}) = \frac{\bar{y}_{nf}}{\bar{y}_{nf} + \sum_{g \neq f} \underline{y}_{ng}(S_{-f})} \quad (\text{OA.6.1})$$

is an upper bound on firm f 's market share under every completion of the node.

Let

$$\varrho(s) = \frac{1-s}{\sigma} + \frac{s}{\eta}.$$

The variable-profit term $R_n s \varrho(s)$ is increasing in s . A valid node bound is therefore

$$\sum_{S_{-f}} Q_{-f}(S_{-f}) \sum_n R_n \bar{s}_{nf}(S_{-f}) \varrho(\bar{s}_{nf}(S_{-f})) - \sum_{j \in S_f} C_{fj} - \sum_{j \in U_f} \min\{C_{fj}, 0\}, \quad (\text{OA.6.2})$$

where C_{fj} is the deterministic plus realized private fixed cost. The last term includes every undecided link with a negative cost, which keeps the bound above the payoff of every node

completion.

This maximal-portfolio envelope is the baseline node bound. The algorithm discards a node only when (OA.6.2) is below the best payoff already found. If the feasible family is not closed under taking subsets, $S_f \cup U_f$ need not be a feasible completion. The same search remains valid only after replacing the all-in completion by a node-specific upper bound over feasible completions, or by an optimistic relaxation that is known to dominate them.

Algorithm 1 Branch-and-bound complete-portfolio response

Require: Type (ε_f, Q_{-f}) , feasible family $\mathcal{X}_f$, and a node bound B_f satisfying (2.14)

- 1: Fully reprice a feasible T_f^0 and set $(T_f^*, v^*) \leftarrow (T_f^0, \tilde{V}_f(T_f^0; \varepsilon_f, Q_{-f}))$
- 2: Initialize the node list $\mathcal{L} \leftarrow \{(\emptyset, \mathcal{A}_f)\}$
- 3: **while** $\mathcal{L} \neq \emptyset$ **do**
- 4: Remove a node (S_f, U_f) from $\mathcal{L}$
- 5: **if** $C_f(S_f, U_f) = \emptyset$ **then**
- 6: Discard the infeasible node
- 7: **else if** $B_f(S_f, U_f) \leq v^*$ **then**
- 8: Prune the node
- 9: **else if** $U_f = \emptyset$ **then**
- 10: Fully reprice S_f and compute $v \leftarrow \tilde{V}_f(S_f; \varepsilon_f, Q_{-f})$
- 11: **if** $v > v^*$ **then**
- 12: $(T_f^*, v^*) \leftarrow (S_f, v)$
- 13: **end if**
- 14: **else**
- 15: Choose $j \in U_f$ and add the exclude child $(S_f, U_f \setminus \{j\})$ to $\mathcal{L}$
- 16: Add the include child $(S_f \cup \{j\}, U_f \setminus \{j\})$ to $\mathcal{L}$
- 17: **end if**
- 18: **end while**

Ensure: Response T_f^* , payoff v^* , and the reported search gap

OA.6.2 Cardinality-conditioned implementation

The implemented refinement conditions on the number k of undecided links added by a completion. For each rival state and market-level service or accessibility statistic, it adds the k largest contributions among links in U_f . It then subtracts the costs of links in S_f and the k smallest costs in U_f . The maximizing links may differ across markets, statistics, and the cost calculation, so the resulting vector need not represent a feasible common portfolio. Every feasible size- k completion is nevertheless weakly less favorable in ev-

ery component and weakly more costly. Maximizing this relaxed payoff over k gives the bound in Proposition 3.

The algorithm takes the smaller of the baseline envelope in (OA.6.2) and the cardinality envelope. The refinement therefore never discards fewer nodes than the baseline. Computing the top- k sums is polynomial at each visited node, while the worst-case search tree remains exponential in the number of candidate links. The implementation does not impose a negative-cost preprocessing rule and allows the feasible portfolio family to exclude some subset sizes.

We check every branch node in synthetic five-link CES and AB problems against exhaustive completion values. In the benchmark recalibration, the baseline envelope reaches a 10,000-node limit in each of the four large CES responses. Cardinality conditioning reduces their root gaps by 95–98 percent and establishes global optimality in 427–2,347 nodes. The corresponding AB responses require 55–889 nodes. Their numerical pricing error is accounted for below. Table OA.6.1 reports the CES comparison.

OA.6.3 Numerical error accounting

For every reported exact response, the search exhausts the tree before the node limit and certifies a global best response to within 10^{-10} under CES and below 1.6×10^{-10} under AB. CES pricing is closed form; the AB bound additionally accounts for the numerical pricing solution, as follows. The global contraction in Proposition 4 bounds the distance to the exact pricing solution: a computed log price-power vector that one further update moves by at most r_x lies within $r_x/(1 - \bar{\lambda}_{AB})$ of the fixed point in the sup norm. We convert that distance into a uniform share-error bound r . At $(\sigma, \eta) = (5, 2)$, a share-error bound r implies at most $13.20r$ error in one firm’s aggregate variable profit and $26.40r$ in a comparison of two portfolio payoffs. The largest reported share-error bound is 4.11×10^{-12} , giving bounds of 5.5×10^{-11} and 1.1×10^{-10} . Because the pricing allowance is added to every node upper bound, the returned portfolio’s own pricing error enters the final search gap only once: adding 5.5×10^{-11} to the 10^{-10} tree tolerance gives an optimality-gap bound below 1.6×10^{-10} .¹

For finite type and rival supports, the distributional algorithm sums over all supplied nodes and iterates the resulting laws over complete portfolios. It uses total variation as

¹All bounds are computed in standard floating-point arithmetic.

its fixed-point residual. With continuous private costs, quadrature or common-random-number sampling adds an integration error.

OA.6.4 Unrestricted equilibrium verification

The application calculation fixes the benchmark primitives in Table OA.4.1 and begins from the associated fully repriced AB and CES profiles. Providers update sequentially using global whole-portfolio responses. The AB path accepts one response in its first round: AWS removes Singapore and adds Ireland, increasing its profit by 1.46×10^{-6} . A second complete round leaves every provider unchanged. The terminal AB profile therefore contains 124 links. The maximum profit gain from any provider's complete-portfolio deviation is below the numerical bound just reported.

The CES path starts with 105 links and has no accepted response. Its pricing continuation is closed form. Exhausting every provider's search tree therefore establishes global optimality at the profile, up to the reported numerical tolerance. Table OA.6.2 summarizes the two paths.

OA.6.5 Fixed-aggregate portfolio benchmark

The strategic response in the main paper resolves the full pricing equilibrium after every candidate portfolio deviation. The fixed-aggregate benchmark changes only this feature. For each provider, it holds fringe price power and the current price power of all rivals fixed by market, while allowing the provider to choose its complete portfolio and reoptimize its own prices. The five simultaneous responses are then combined, the joint profile is repriced, and the rival aggregates are updated before the next round. This calculation is an aggregate-taking analogue; it is not a literal implementation of AFT or AES.

The AB path is $124 \rightarrow 128 \rightarrow 128$. At the first response, AWS adds Chile and Singapore, while Google adds Belgium and Taiwan; Microsoft, Oracle, and Alibaba retain their portfolios. The four additions remain after the aggregates are updated. Relative to the strategic AB equilibrium, the fixed-aggregate profile has 3.2 percent more links but 2.04 percent lower fully repriced aggregate profit net of fixed costs.

The CES path is $124 \rightarrow 101 \rightarrow 105 \rightarrow 105$. The one-shot response reflects aggregates inherited from the AB profile. Once the CES aggregates are updated, the fixed-aggregate fixed point coincides with the strategic CES equilibrium. With constant CES markups,

a provider's portfolio does not change its rivals' prices. Under AB pricing it changes rivals' shares and markups, which in turn changes the provider's return to its candidate locations. The contrast isolates this strategic repricing channel from both global portfolio optimization and aggregate consistency.

OA.6.6 Unrestricted sampled-Shapley comparison

For each firm and pricing system, the separable response assigns a Shapley value to every candidate link. The implementation samples random permutations of the full candidate set and fully reprices every prefix portfolio. It increases the number of permutations from 128 to at most 1,024 until all simultaneous Monte Carlo diagnostic intervals exclude zero or the sampling limit is reached. The selected response includes links with positive sample means. Sequential iteration reaches a fixed point after three passes under both pricing systems.

Alibaba's 14-link candidate set can be enumerated completely. At the reported sampled fixed points, the sampled and population Shapley signs agree for all 14 links under both pricing systems; the largest coefficient errors are 1.95×10^{-5} under AB and 3.20×10^{-5} under CES. At the exact-equilibrium rival profiles, population Shapley selects 7 links rather than the exact AB portfolio's 13 and 6 rather than the exact CES portfolio's 8. This provider-level check separates approximation error from Monte Carlo sign error.

The sampled fixed points differ from the exact equilibria in 16 AB links and 26 CES links. They also fail the exact-response verification: four of five providers have profitable complete-portfolio deviations under AB, and all five do under CES. The exact AB-to-CES comparison reduces deployment by 19 links and aggregate provider profit net of fixed costs by 35.629 percent. The sampled map reduces deployment by 31 links and profit by 33.686 percent. Six AB and three CES coefficients remain sign-unclassified. Because an unresolved decision can alter later firm responses, the exercise describes the reported sampled map and does not establish a population Shapley equilibrium.

Table OA.5.1: Provenance of the Benchmark Cloud Inputs

Input	Source and vintage	Construction
Provider-country activity	Public provider infrastructure pages; working file accessed and held fixed as of April 7, 2026	An operational region makes the provider-country link active; an announced region adds an inactive link to the candidate set.
Market size	World Bank WDI, 2023 observations	GDP times the internet-use share, normalized to the United States and raised to 0.70.
Demand and Fréchet parameters	Quantitative-trade reference points	$\sigma = 5$, $\eta = 2$, and $\theta = 3$; sensitivity specifications use AFT-centered and AB-demand values and recalibrate the fixed-cost block before comparison. The common Fréchet factor is normalized to $\Gamma_\theta = 1$.
Provider productivity a_f	Rounded public cloud-revenue scale proxies	Revenue is mapped log-linearly to $[1, 2]$; the resulting vector is (2.000, 1.757, 1.483, 1.092, 1.000).
Service cost τ_{in}	Continent classification and bilateral country coordinates	1 within country, 1.15 on the same continent or within 5,000 kilometers, and 1.5 otherwise.
Distance	Licensed Datacenter Map extract, April 2026, with capital-city fallback	Haversine distance in kilometers, scaled by 1,000 inside the logarithm.
Local eligibility	Discrete geography and service-cost coding	Same country, same continent, or bilateral distance below 5,000 kilometers.
Digital policy	OECD Digital STRI regulatory data, 2025 observation	Researcher-constructed policy tier and weight based on OECD regulatory measures, accessed and held fixed as of April 19, 2026; missing markets receive the neutral value.
Site cost scale w_i	Penn World Table 11.0, 2023	Human-capital-adjusted compensation per worker, normalized to the United States. Source fallbacks are recorded by country in the replication archive.
Fixed-cost prior ϕ_i^0	World Bank income classification, July 2024	Initial values are 0.15, 0.12, 0.10, and 0.08 for high-, upper-middle-, lower-middle-, and low-income groups.

Table OA.6.1: Cardinality Conditioning in Unrestricted CES Responses

Provider	Links	Portfolios	Root gap reduction (%)	Nodes		Seconds
				Baseline	Cardinality	Cardinality
Oracle	27	2^{27}	97.1	10,000 [†]	427	0.88
AWS	30	2^{30}	95.0	10,000 [†]	1,451	2.20
Microsoft	34	2^{34}	98.1	10,000 [†]	977	1.40
Google	29	2^{29}	96.0	10,000 [†]	2,347	3.59

Notes: The two columns use the same benchmark recalibration, initial best-known portfolio, cost-based branch order, and 10,000-node limit. Root-gap reduction compares the cardinality-conditioned root bound with the baseline maximal-portfolio bound. [†]The baseline calculation reached the node limit before exhausting the search; the cardinality calculations exhausted the remaining tree. Seconds are total computation time on the same computer.

Table OA.6.2: Unrestricted Portfolio-Equilibrium Verification

Pricing	Initial links	Terminal links	Accepted responses	Status
AB	124	124	AWS: Singapore out, Ireland in	Portfolio equilibrium up to reported numerical error
CES	105	105	None	Complete-information portfolio equilibrium up to numerical tolerance

Notes: The calculation starts from one benchmark calibrated profile per pricing rule and completes a final round with no changes after any accepted response. It does not establish uniqueness or convergence from arbitrary initial profiles. The AB error bound includes its unique downstream pricing continuation; no discrete search tree remains open.

Table OA.6.3: Fixed-Aggregate Iteration and Strategic Equilibrium

Pricing	Calculation	Links	Changed links	Repriced profit
AB	Fixed-aggregate one shot	128	4	1.181296
AB	Fixed-aggregate fixed point	128	4	1.181296
AB	Strategic equilibrium	124	0	1.205909
CES	Fixed-aggregate one shot	101	4	0.784195
CES	Fixed-aggregate fixed point	105	0	0.776259
CES	Strategic equilibrium	105	0	0.776259

Notes: Both iterations start from the 124-link exact AB profile. Changed links are provider-country entries that differ from the strategic equilibrium under the pricing system in that row. Repriced profit evaluates the combined portfolio under one common full pricing equilibrium. It is therefore comparable across rows; the sum of provider decision values computed against different fixed aggregates is not. All discrete searches attain the reported tolerance. AB results also use the numerical pricing continuation and its error accounting.

Table OA.6.4: Provider-Level Unrestricted Exact and Sampled Responses

Pricing	Provider	Exact links	Sampled links	Profile Hamming	Unclassified	Gain from exact reoptimization
AB	AWS	25	26	1	2	1.416×10^{-3}
AB	Microsoft	33	33	0	0	0
AB	Google	27	24	3	2	1.655×10^{-3}
AB	Oracle	26	20	6	2	1.840×10^{-3}
AB	Alibaba	13	7	6	0	2.089×10^{-3}
CES	AWS	21	20	1	3	3.549×10^{-4}
CES	Microsoft	29	27	2	0	9.586×10^{-4}
CES	Google	24	13	11	0	6.672×10^{-3}
CES	Oracle	23	13	10	0	3.000×10^{-3}
CES	Alibaba	8	6	2	0	5.652×10^{-4}

Notes: Exact links and profile Hamming compare the sampled fixed point with the exact unrestricted equilibrium. The last column reoptimizes the firm's complete portfolio against rivals at the sampled fixed point. The exact response need not equal the firm's portfolio at the exact equilibrium. Unclassified counts refer to simultaneous Monte Carlo diagnostic intervals that contain zero. CES searches are global up to numerical tolerance; AB searches include the numerical pricing-error accounting above.

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